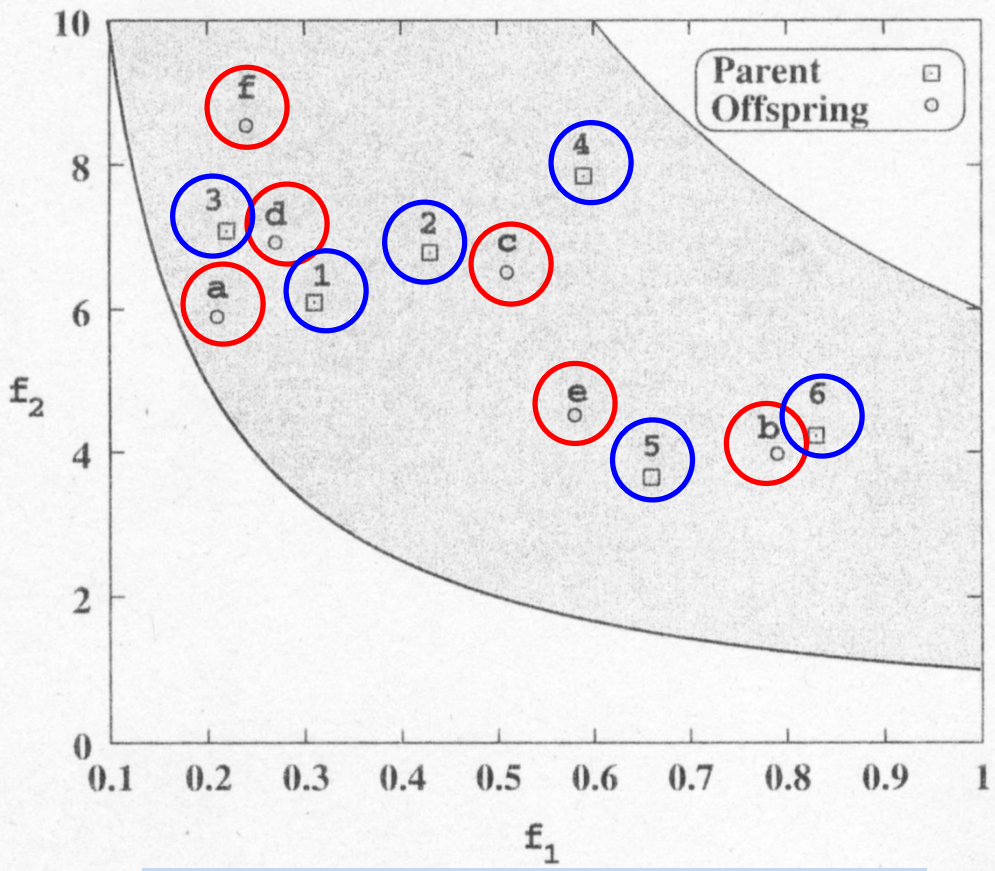
授業受講者以外への再配布禁止です
Redistribution to anyone who don't take this lecture is prohibited

2018.5.21

第5回
航空宇宙設計最適化
Design optimization in aerospace engineering

大山　聖
Akira Oyama

Select 6 solutions from 12 solution candidates



Parent population, Pt				
solution	x1	x2	f1	f2
1	0.31	0.89	0.31	6.1
2	0.43	1.92	0.43	6.79
3	0.22	0.56	0.22	7.09
4	0.59	3.63	0.59	7.85
5	0.66	1.41	0.66	3.65
6	0.83	2.51	0.83	4.23

Minimize $f_1(x) = x_1$

Minimize $f_2 = (1 + x_2) / x_1$

subject_to $0.1 \le x_1 \le 1$

$0 \le x_2 \le 5$

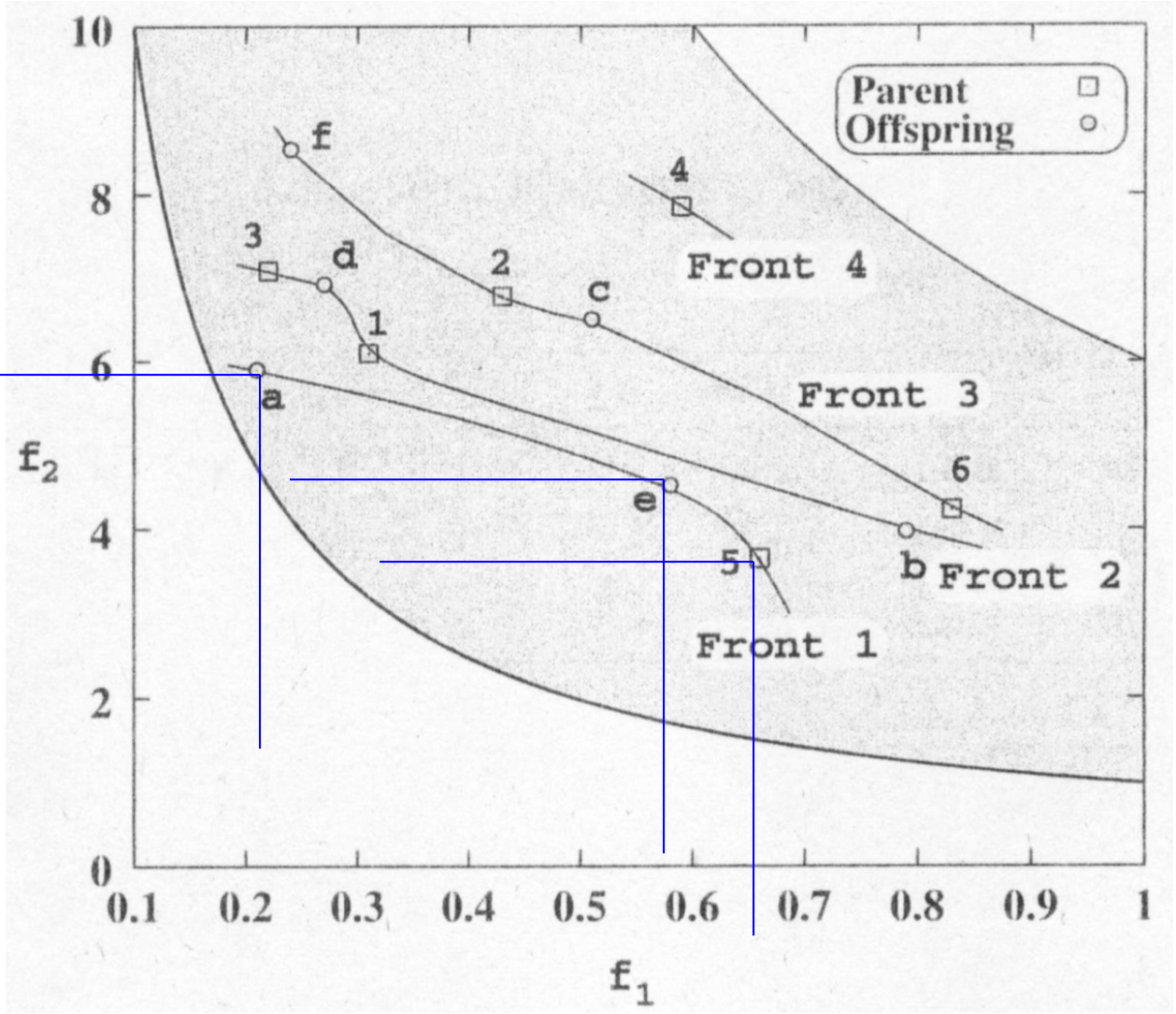
Offspring population, Qt				
solution	x1	x2	f1	f2
a	0.21	0.24	0.21	5.9
b	0.79	2.14	0.79	3.97
c	0.51	2.32	0.51	6.51
d	0.27	0.87	0.27	6.93
e	0.58	1.62	0.58	4.52
f	0.24	1.05	0.24	8.54

解答 answer

解答（非優越ランキング）
Answer (non-dominated sorting)

- ランクを計算

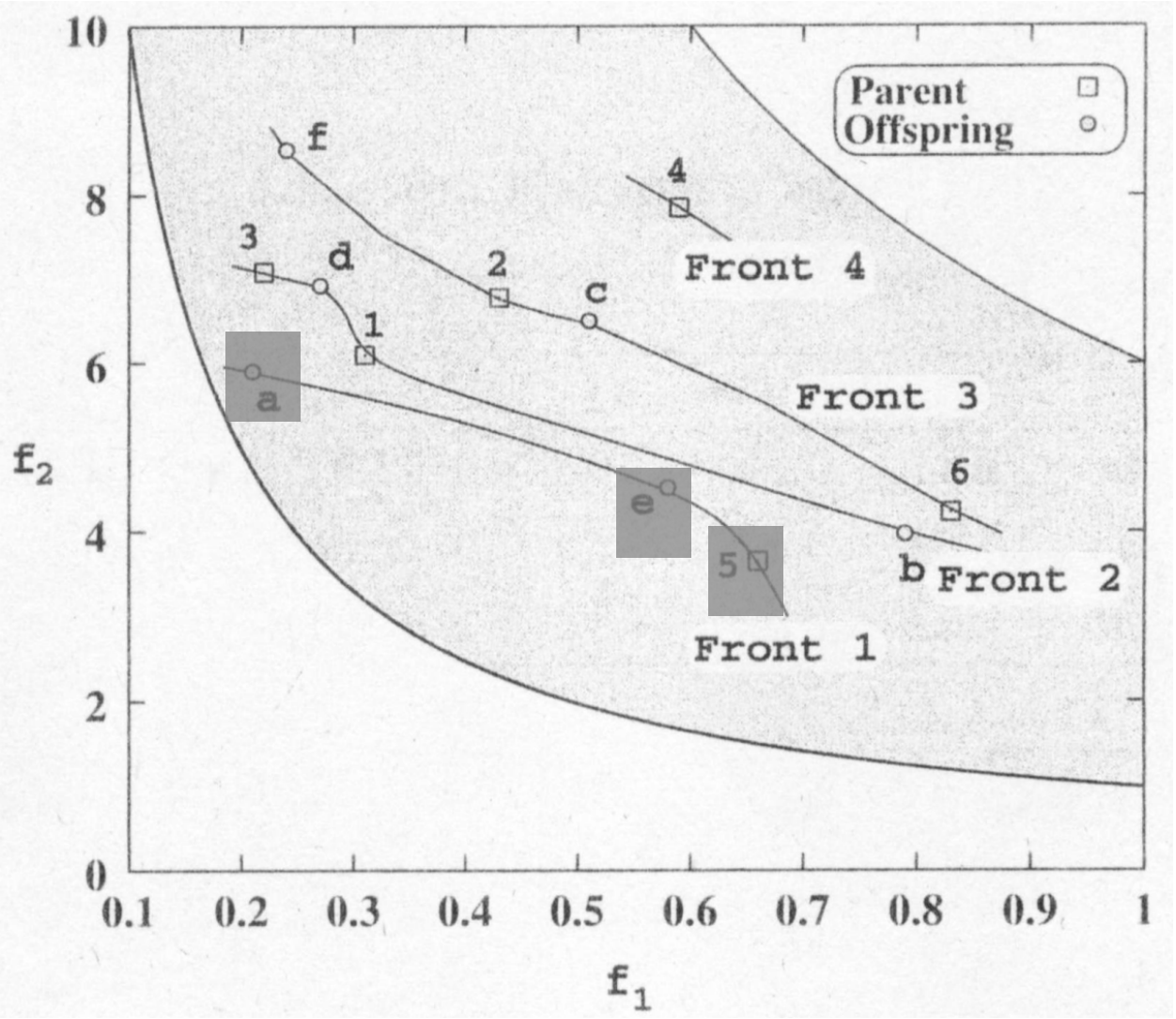
  Calculation of rank



解答（非優越ランキング）
Answer (non-dominated sorting)

- ランクを計算

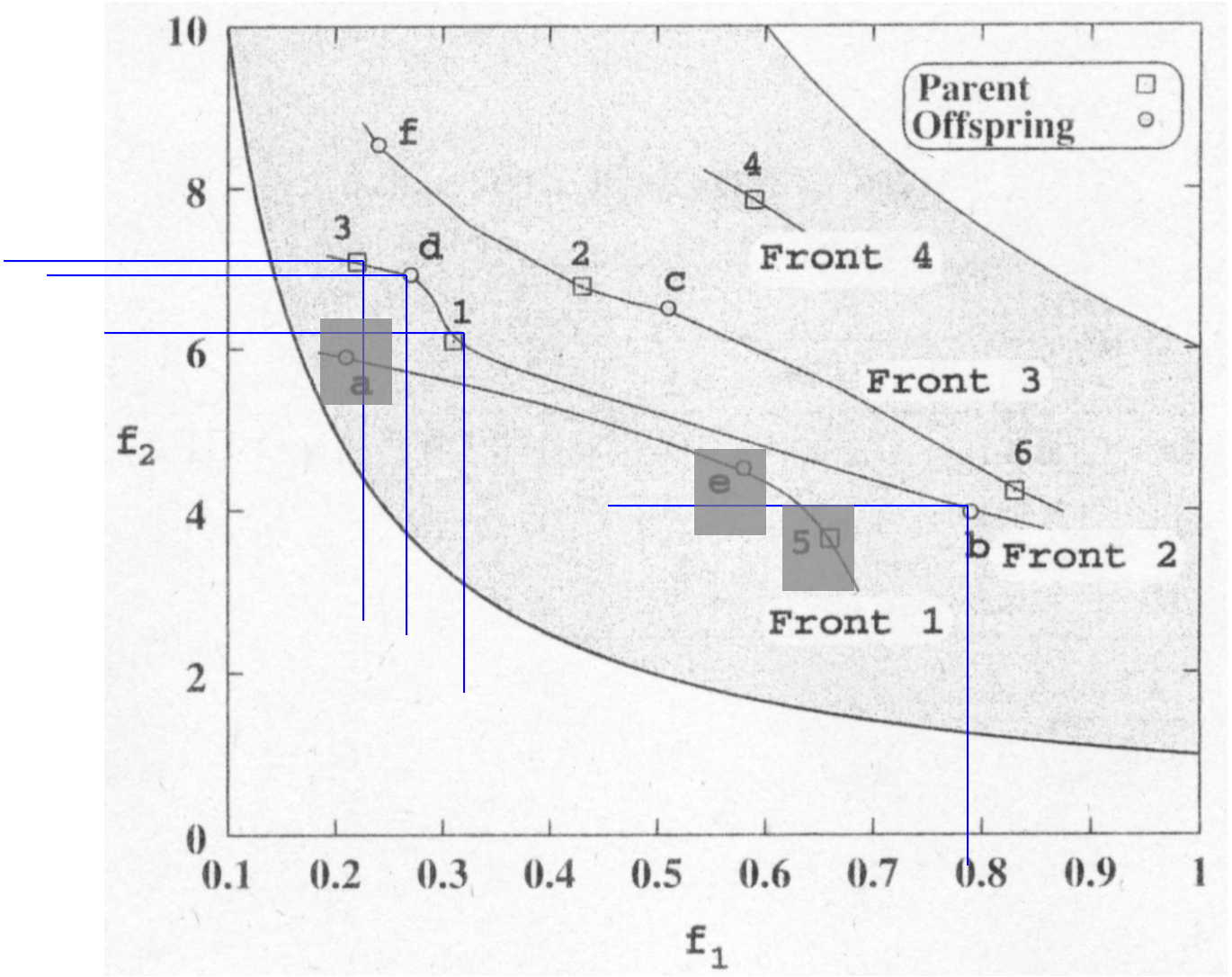
  Calculation of rank



解答（非優越ランキング）
Answer (non-dominated sorting)

- ランクを計算

  Calculation of rank



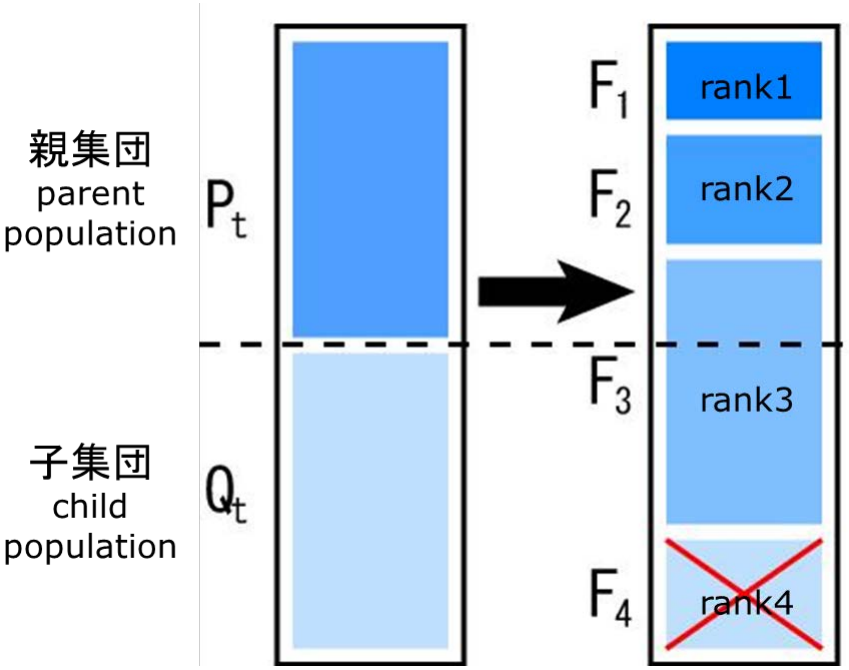
解答（非優越ランキング）
Answer (non-dominated sorting)

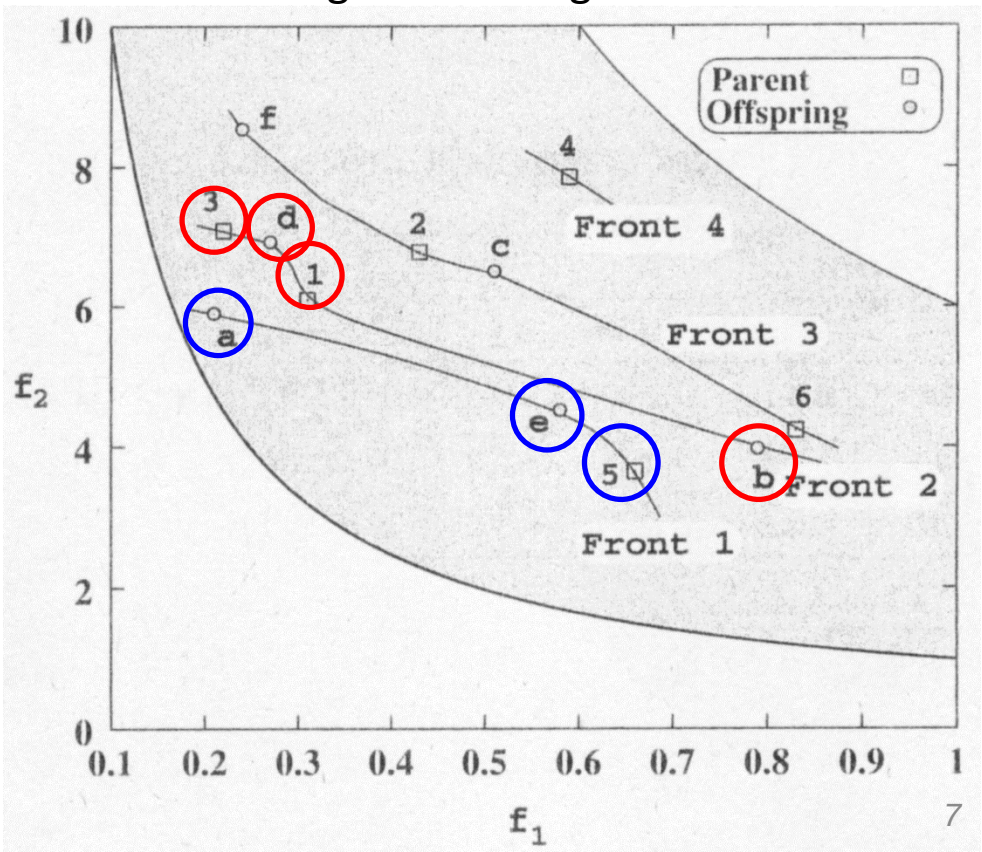
- Front 1の個体(a,e,5)を次世代の親集団に入れる

  Front 1 solutions (a,e,5) are placed into the next generation

- Front2をすべていれると6を超えてしまうため，Front 2のなかからCrowding distanceの値に基づいて3つの個体を選択する

  Number of Front 1 and Front 2 solutions (a,e,5,b,1,d,3) is over 6. Therefore, three solutions among the Front 2 solutions (b,1,d,3) are selected according to crowding distance.



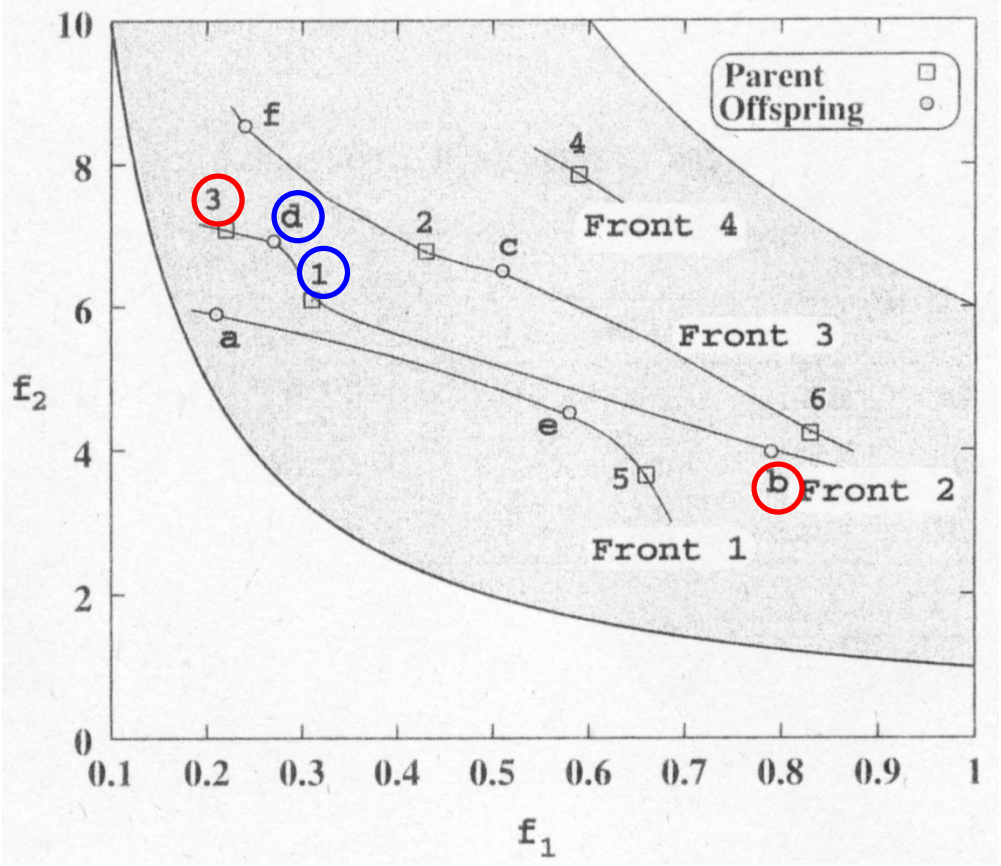


解答（混合距離ランキング）
Answer (Crowding distance sort)

1. 個体3とbは無限大のcrowding distanceを与える

   give crowding distance of infinity to solutions b and 3

$$d_3 = \infty \qquad d_b = \infty$$

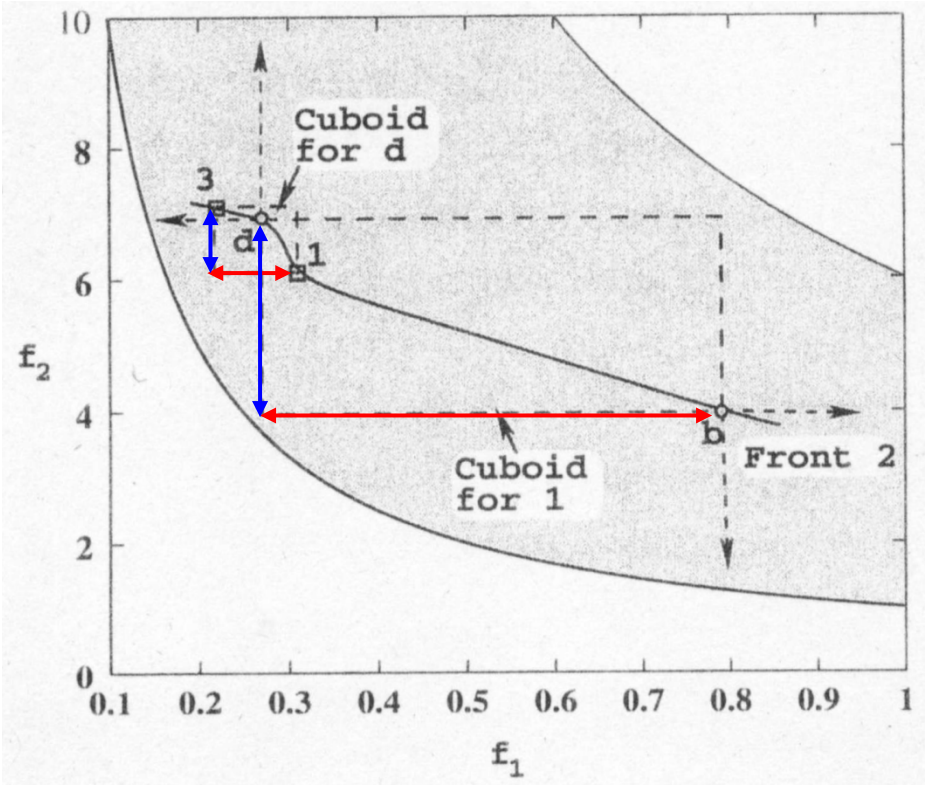


解答（混合距離ランキング）
Answer (Crowding distance sort)

2. 残りの個体(1,d)についてcrowding distanceを計算 Calculate Crowding distance

$$d_d = \frac{f_1^{(1)} - f_1^{(3)}}{f_1^{\max} - f_1^{\min}} + \frac{f_2^{(1)} - f_2^{(3)}}{f_2^{\max} - f_2^{\min}} = 0.10$$

$$d_1 = \frac{f_1^{(b)} - f_1^{(d)}}{f_1^{\max} - f_1^{\min}} + \frac{f_2^{(b)} - f_2^{(d)}}{f_2^{\max} - f_2^{\min}} = 0.58$$



解答（混合距離ランキング）

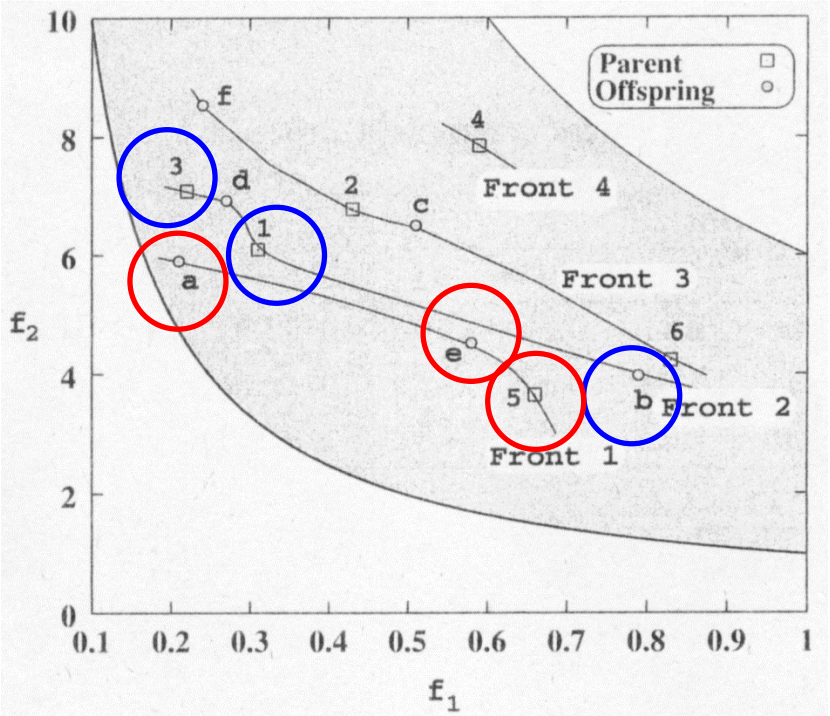
3. Crowding distanceの大きい上位3つの個体(3,b,1)を次世代の親集団P_{t+1}に組み込む

   Three solutions with higher crowding distance are placed into the next generation

$$d_3 = \infty \quad d_1 = 0.63 \quad d_d = 0.12 \quad d_b = \infty$$

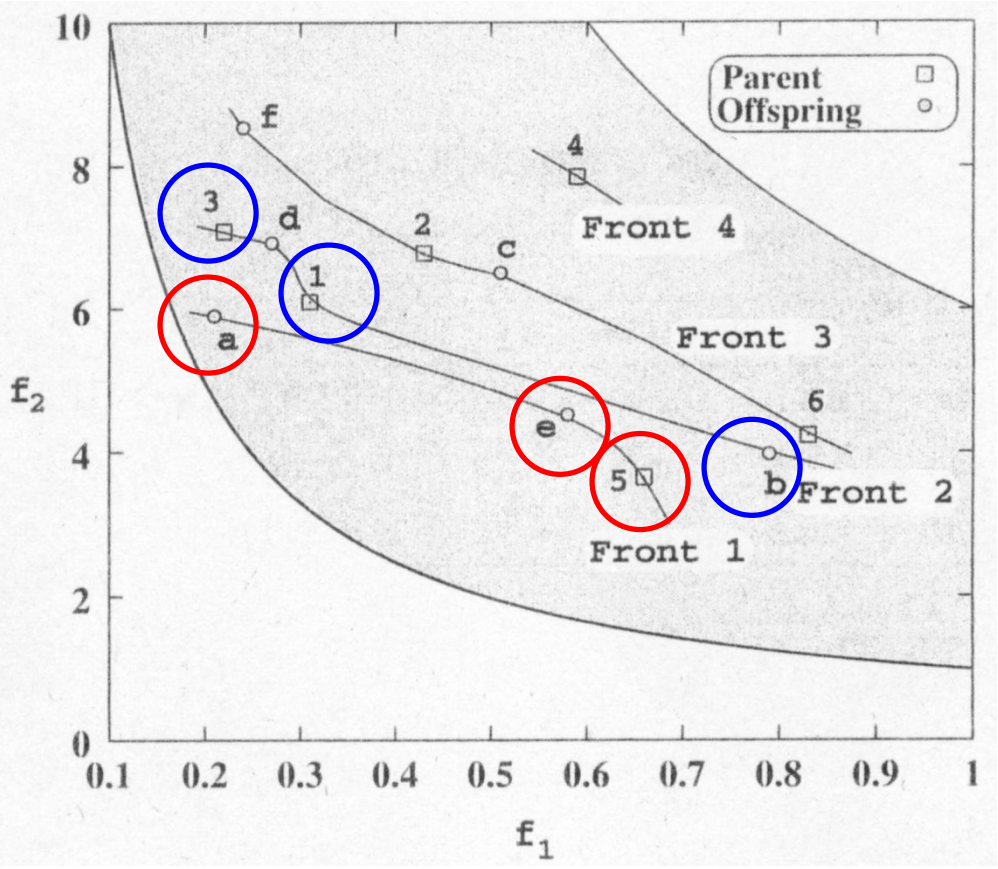
→次世代の親集団P_{t+1}は(a,e,5,3,b,1)

The next generation P_{t+1} is (a,e,5,3,b,1)



子集団の生成　Generation of new population

1. 次世代の親集団P_{t+1}から交配する個体をTournament selectionにより選択
   (1) 乱数によりN個のペアを選択 (5,e), (a,3) (1,b), (a,1), (e,b), (3,5)
   (2) それぞれのペアのうち，優れた方を選択
   (5,e)→5, (a,3)→a, (1,b)→b, (a,1) →a, (e,b) →e, (3,5) →5
2. 選ばれた個体間で交叉を行う
   (5,a) →2つの子供個体, (b,a) →2つの子供個体, (e,5)→2つの子供個体



多目的進化アルゴリズムの生い立ち History of MOEA

- 最初の進化アルゴリズム　The first EC algorithms (1970's)
- 最初の多目的進化アルゴリズムThe first MOEC (Schaffer, 1984)

  Vector-Evaluated Genetic Algorithm
- 多目的進化計算への非劣解・劣解の概念の導入(Goldberg, 1989)

  Concept of non-dominated/dominated solutions
- 実用的な多目的進化アルゴリズム

  Multi-Objective Genetic Algorithm (Fonseca and Fleming, 1993)
  - 非劣解・劣解の概念の導入　　Concept of domination
  - 適応度シェアリングの概念の導入　　Concept of sharing
- SPEA2 (Zitzler et al, 2001), NSGA-II(Deb et al., 2002), …
- MOEA/D(2007), IBEA(2004), NSGA-III (Deb, 2013), etc…

支配関係に基づく多目的進化アルゴリズムの問題点
Difficulty of dominance-based MOEAs

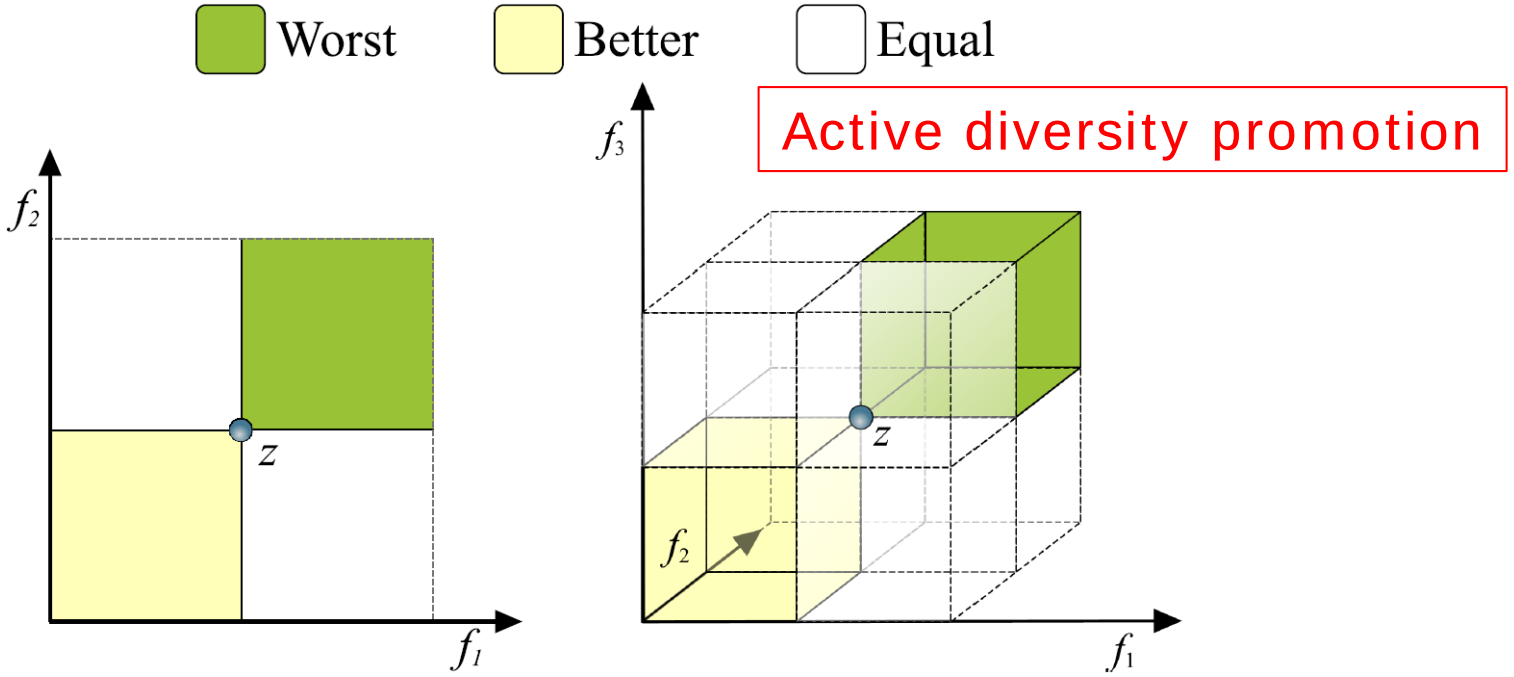
- 目的関数の数が4つ以上になった場合にパレート最適解を見つけることが難しくなる

  It is difficult to get Pareto-optimal solutions when the number of design objectives is greater than 3.

多数目的最適化問題　many-objective optimization problem

（原因）パレート支配に基づくランキングでは多くの個体が非劣解（ランクが1位）になってしまい優劣がつかない

(reason) most solutions becomes non-dominated solutions because rank of most solutions becomes 1.



$$d(2) = 2/4 = 0.5 \qquad d(3) = 6/8 = 0.75$$

$$d(4)=14/16=0.88 \qquad d(5) = 30/32 \approx 0.94$$

多目的進化アルゴリズムの生い立ち History of MOEA

- 最初の進化アルゴリズム　The first EC algorithms (1970's)
- 最初の多目的進化アルゴリズムThe first MOEC (Schaffer, 1984)

  Vector-Evaluated Genetic Algorithm
- 多目的進化計算への非劣解・劣解の概念の導入(Goldberg, 1989)

  Concept of non-dominated/dominated solutions
- 実用的な多目的進化アルゴリズム

  Multi-Objective Genetic Algorithm (Fonseca and Fleming, 1993)
  - 非劣解・劣解の概念の導入　　Concept of domination
  - 適応度シェアリングの概念の導入　　Concept of sharing
- SPEA2 (Zitzler et al, 2001), NSGA-II(Deb et al., 2002), …
- MOEA/D(2007), IBEA(2004), NSGA-III (Deb, 2013), etc…

MOEA/D

- One of the decomposition-based MOEA
- It decomposes a given MOP with M objectives into N single-objective sub-problems

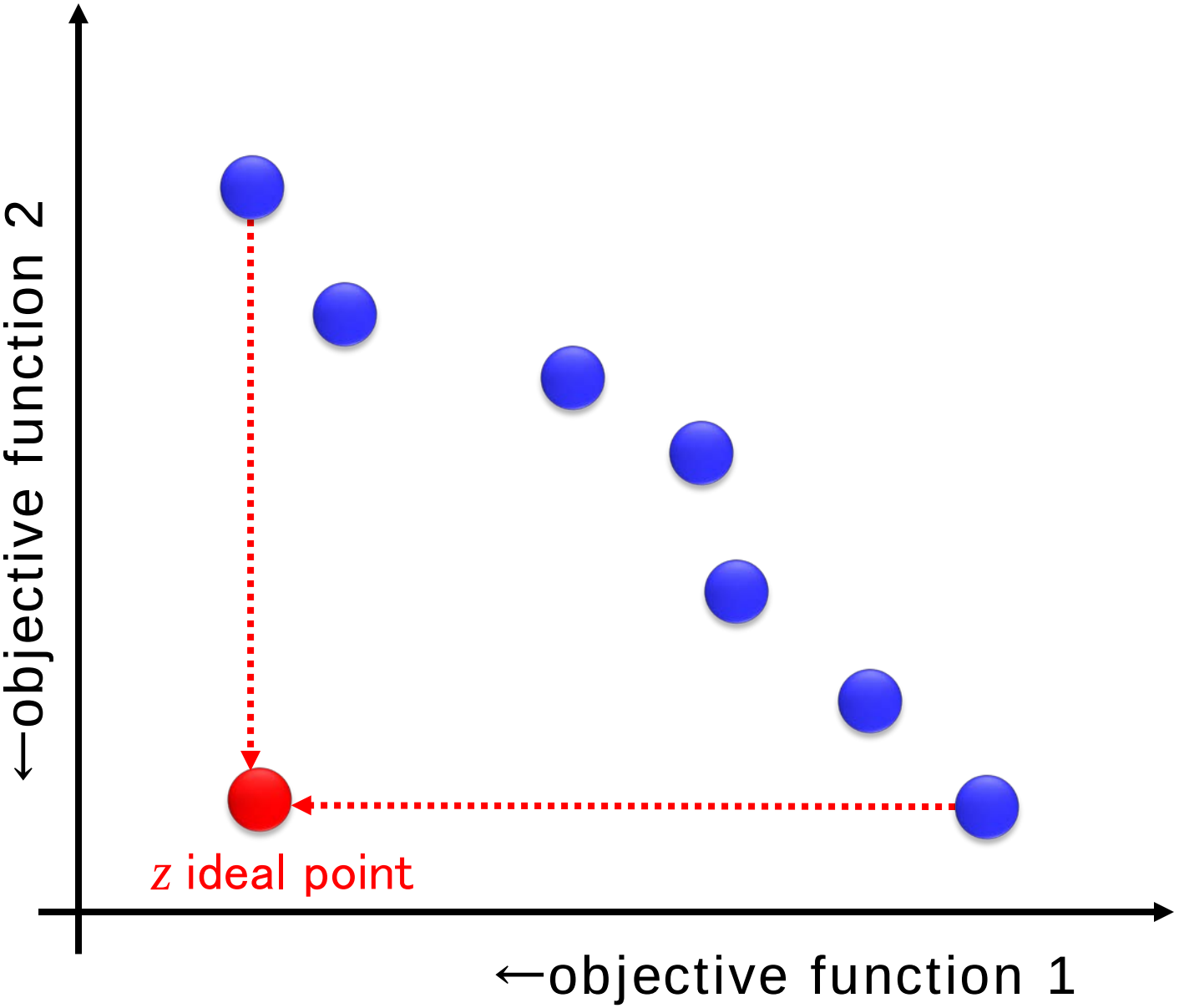
  $g(\boldsymbol{x}|\boldsymbol{w}^1), \ldots, g(\boldsymbol{x}|\boldsymbol{w}^N)$

  Using a set of weight vectors $W=\{\boldsymbol{w}^1, \ldots, \boldsymbol{w}^N\}$ and a scalarizing function $g(\boldsymbol{x}|\boldsymbol{w})$, where $\boldsymbol{w}^i=(w^i_1, \ldots, w^i_M)^{\mathrm{T}}$, $\sum_{j=1}^{M} w^i_j = 1$

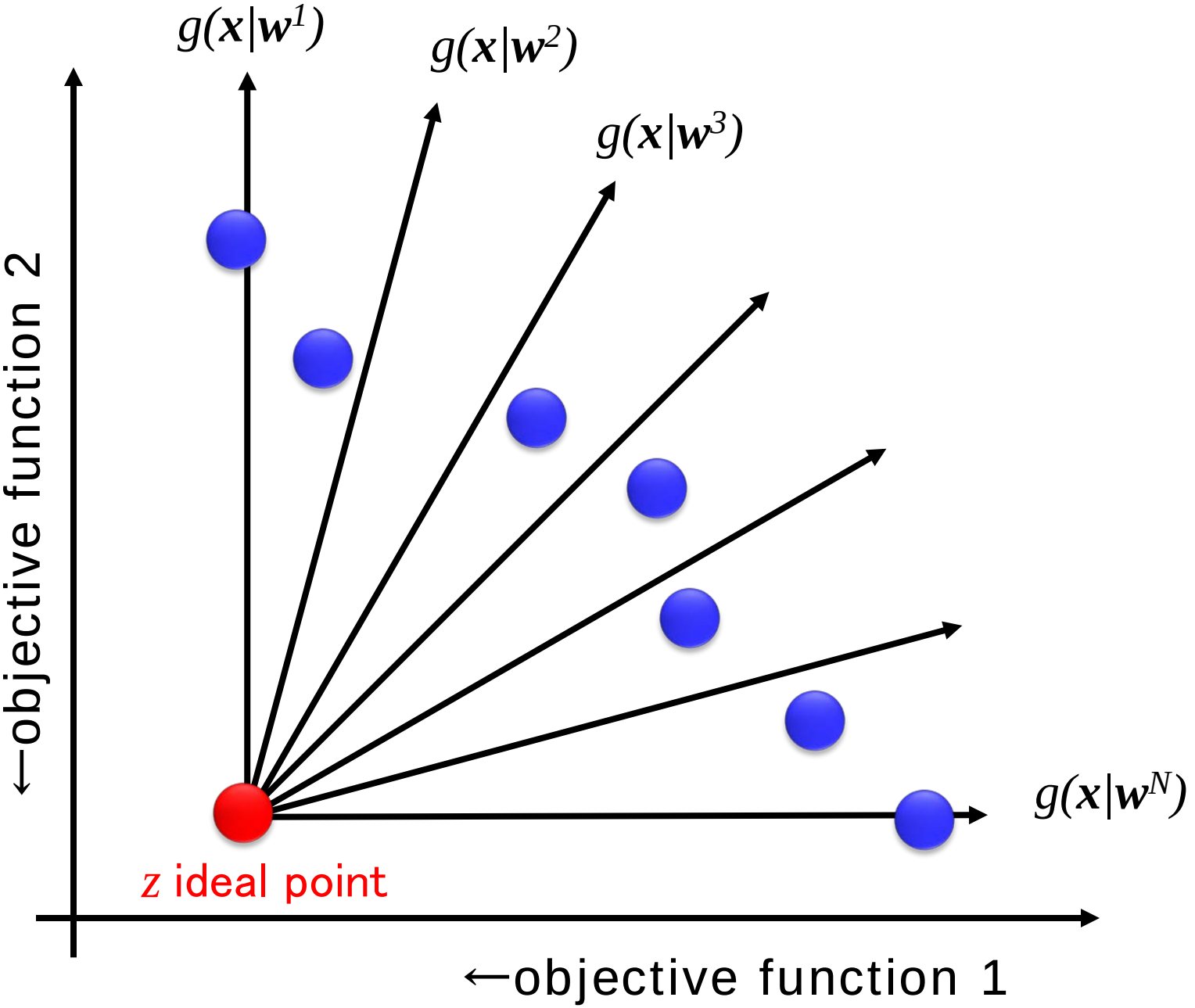
- Parent individuals are selected from a set of neighborhood individuals of each weight vector $\boldsymbol{w}^i$

Q. Zhang and H. Li, "MOEA/D: A multiobjective evolutionary algorithm based on decomposition," IEEE Trans. Evol. Comput., vol. 11, no. 6, pp. 712731, Dec. 2007.

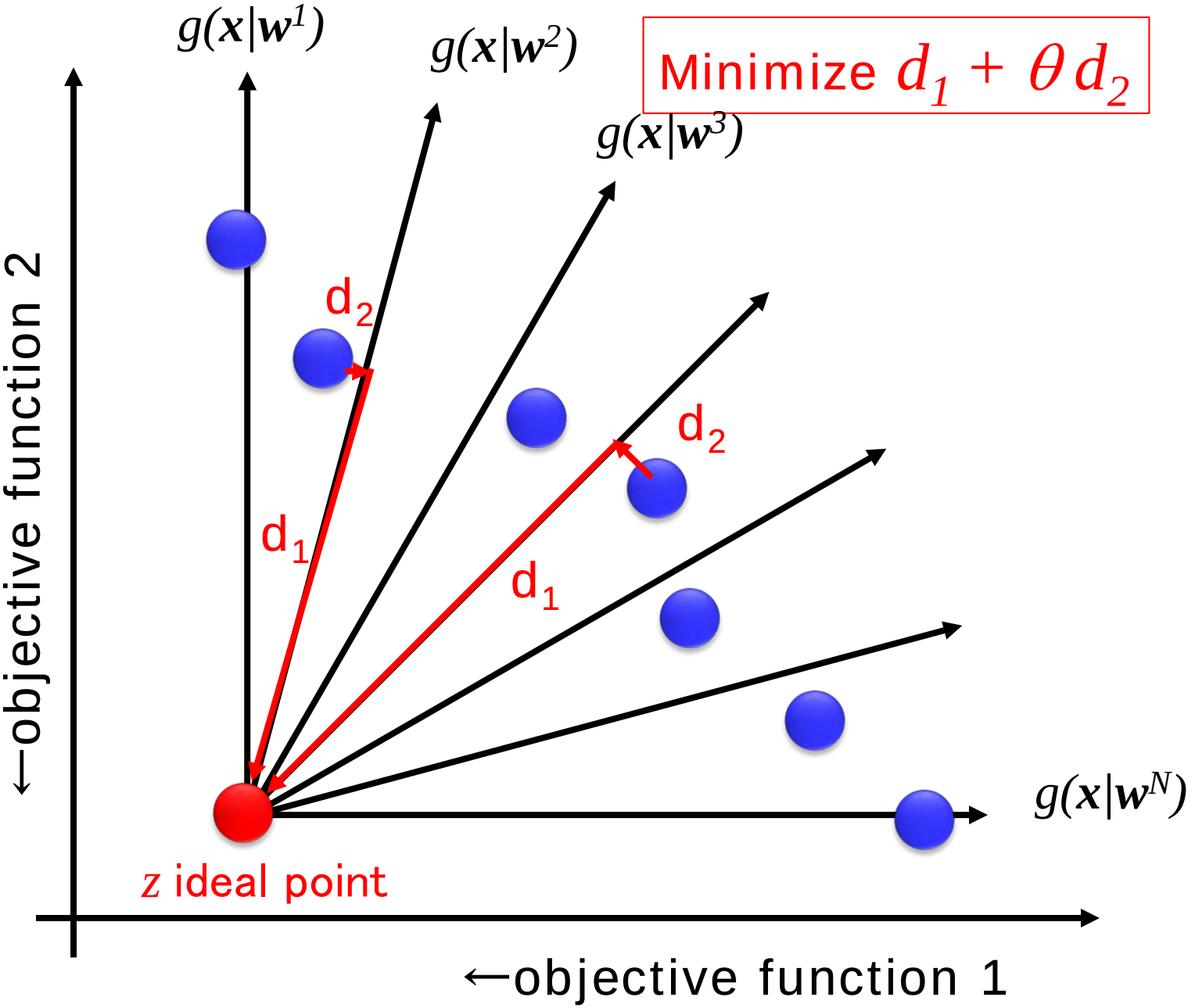
MOEA/D



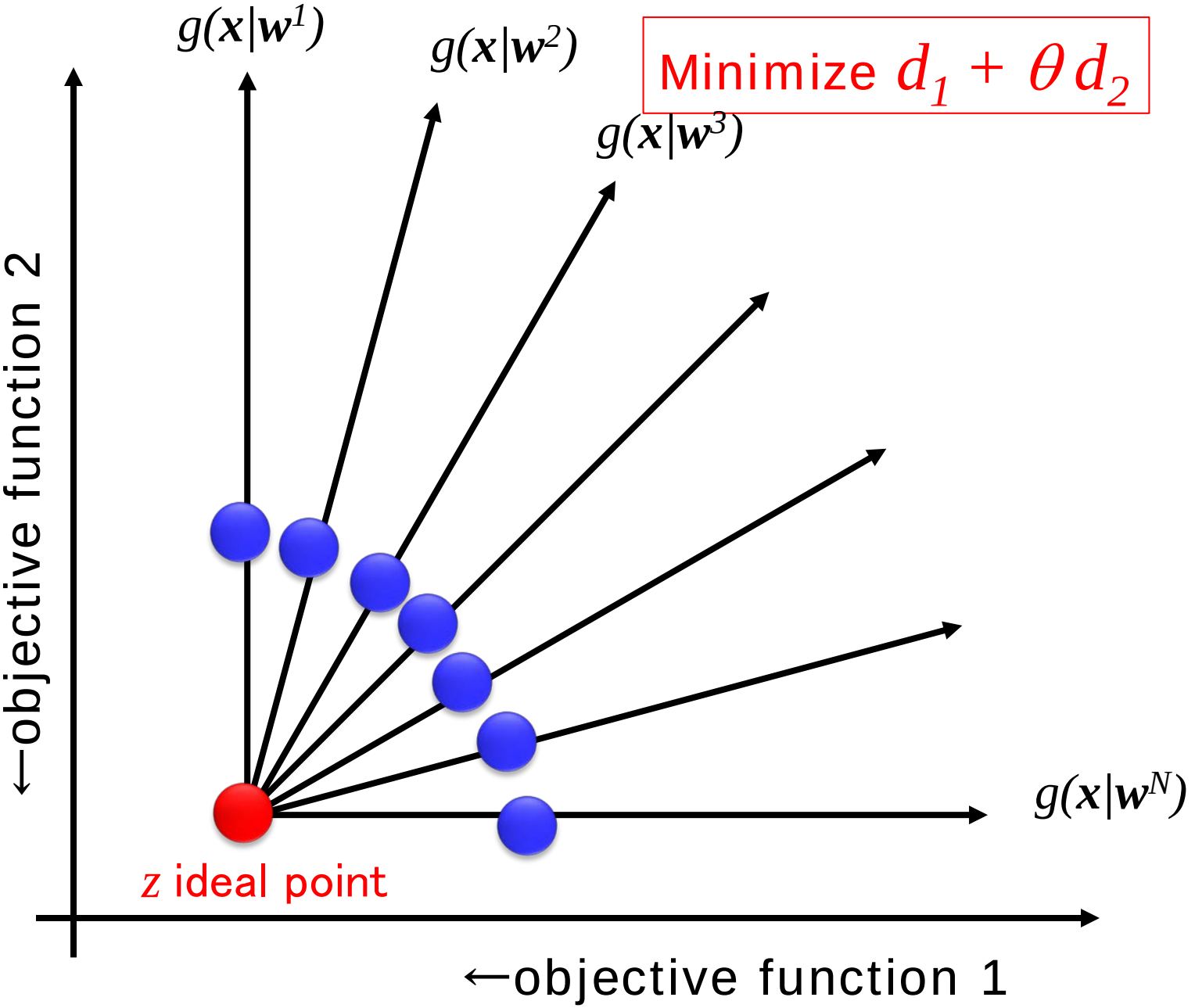
MOEA/D


$g(x|w^1)$
$g(x|w^2)$
$g(x|w^3)$
$g(x|w^N)$
←objective function 2
←objective function 1
z ideal point

MOEA/D


$g(x|w^1)$
$g(x|w^2)$
$g(x|w^3)$
$g(x|w^N)$
Minimize $d_1 + \theta d_2$
d_2
d_2
d_1
d_1
z ideal point
←objective function 2
←objective function 1

MOEA/D


$g(x|w^1)$
$g(x|w^2)$
$g(x|w^3)$
Minimize $d_1 + \theta d_2$
$g(x|w^N)$
z ideal point
←objective function 2
←objective function 1

MOEA/D

Multiobjective optimization is decomposed of a set of single objective optimization problems



Good performance even in many-objective optimization problems

(It was not intension of Prof. Zhang though…*)*

Indicator-Based Evolutionary Algorithm (IBEA)

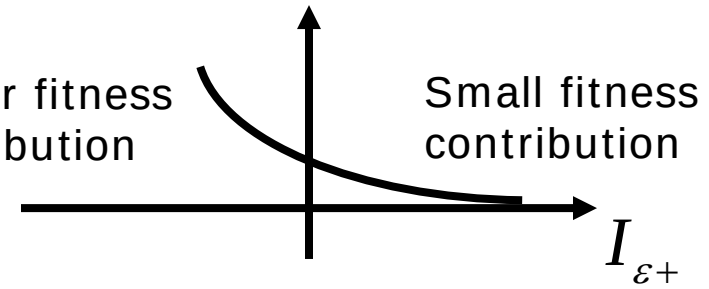
- One of the indicator-based MOEAs
- In the environmental selection, fitness is assigned by a paired comparison based on **quality indicator values**.
- Fitness is used for the parent selection as well.
- $IBEA_{HD}$ bases on **hyper volume difference indicator** and $IBEA_{\varepsilon+}$ bases on **binary additive ε-indicator**

E. Zitzler and S. Künzli, "Indicator-based selection in multiobjective search," in Proc. Int. Conf. PPSN, 2004, pp. 832-842.

Fitness of $IBEA_{\varepsilon+}$

$$F(x^1) = \sum_{x^2 \in P \setminus \{x^1\}} e^{-I_{\varepsilon+}(x^1, x^2)/\kappa}$$

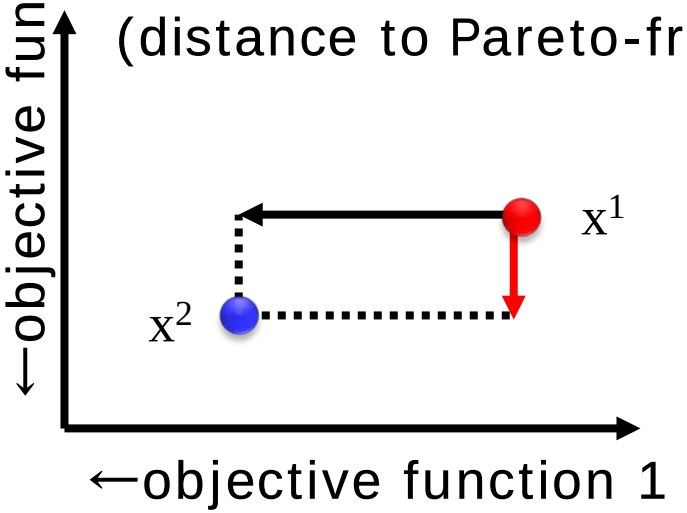
$$F = e^{-I_{\varepsilon+}}$$



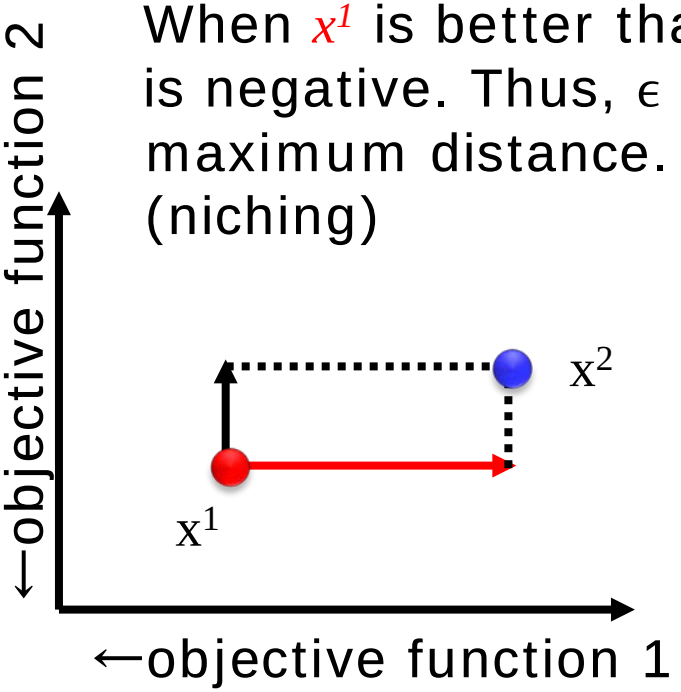
$$I_{\varepsilon+}(x^1, x^2) = min_{\epsilon}\{ f_i(x^1) - \epsilon \leq f_i(x^2) \; for \; i \in \{1, \ldots, n\}\}$$

➡ $I_{\varepsilon+}(x^1, x^2)$ indicates how much improvement is required for x^1 to be nondominated.

When x^1 is worse than x^2, ϵ is positive. Thus, ϵ is minimum distance. (distance to Pareto-front).



When x^1 is better than x^2, ϵ is negative. Thus, ϵ is maximum distance. (niching)



Indicator-Based Evolutionary Algorithm (IBEA)

Fitness is determined based on distance to the (current) Pareto-front and neighbors



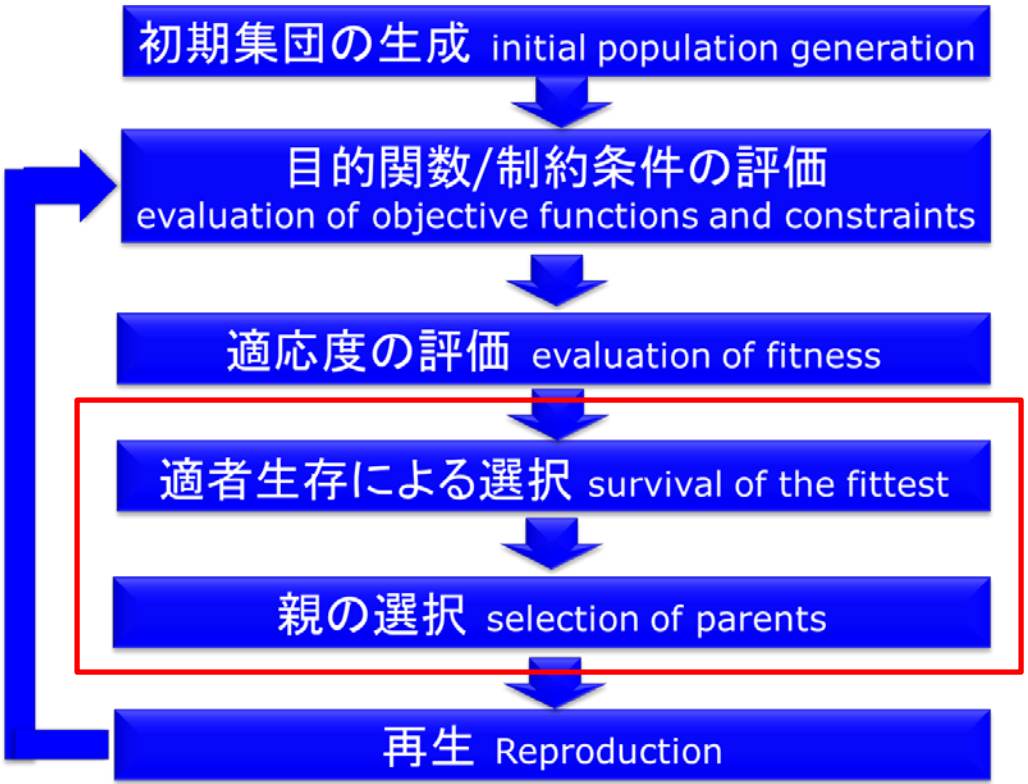
Good performance even in many-objective optimization problems because rank of each individual is appropriately given.

(It was not intension of Prof. Zitzler though…*)*

NSGA-III

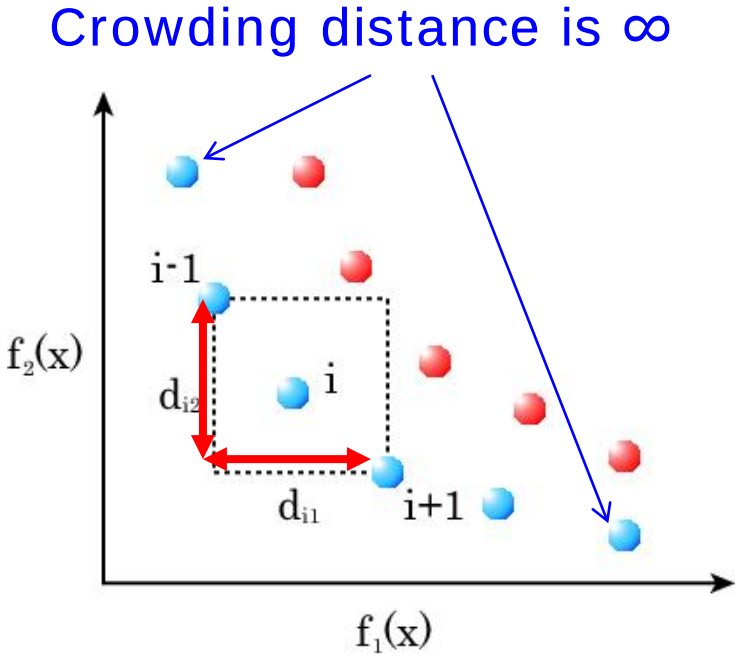
- An improved version of the dominance-based MOEAs
- Basic framework remains similar to NSGA-II except for selection mechanism

K. Deb and H. Jain, "An evolutionary many-objective optimization algorithm using reference-point-based nondominated sorting approach, part I: Solving problems with box constraints," IEEE Trans. Evol. Comput., vol. 18, no. 4, pp. 577601, Aug. 2014.



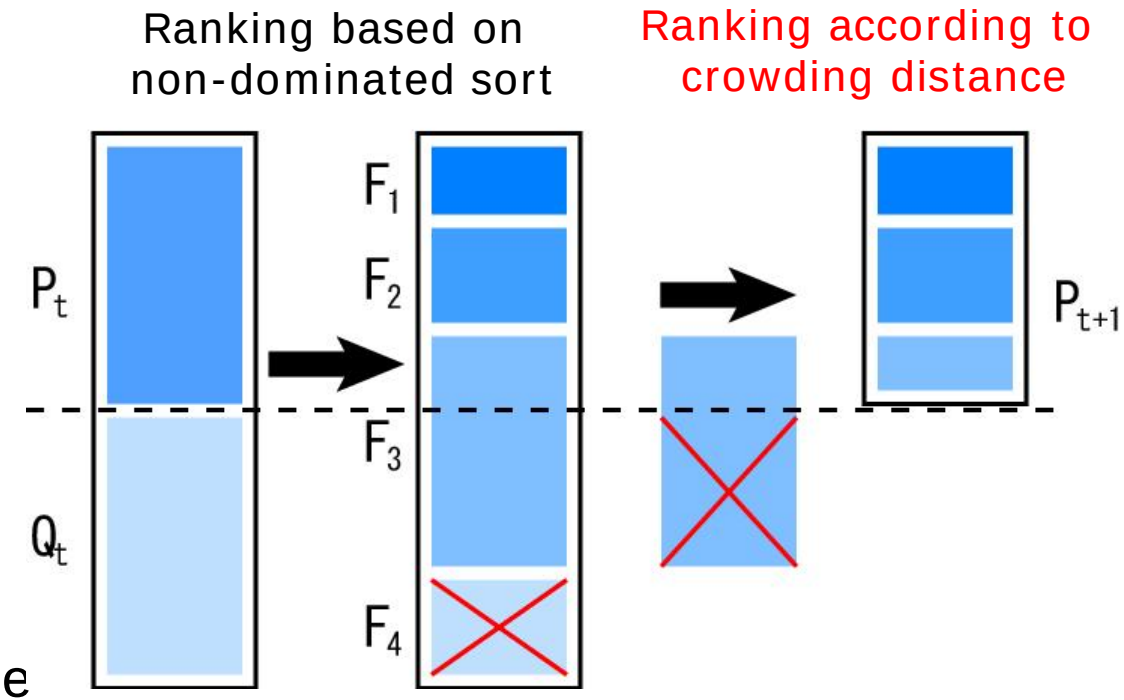
survival of the fittest in NSGA-III

In NSGA-II, the last solutions are selected based on crowding distance



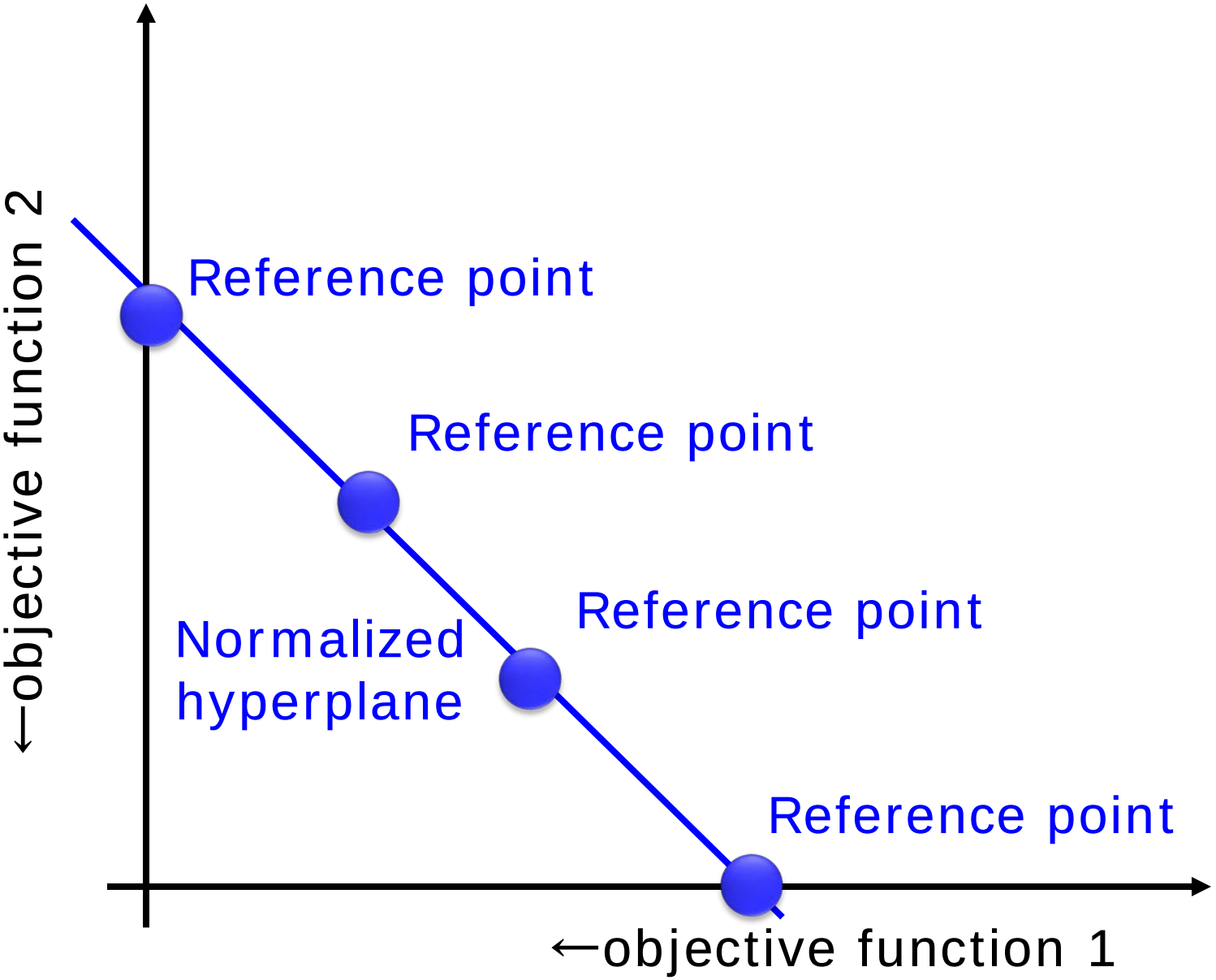
Concept of crowding distance

Crowding distance = d_{i1} + d_{i2}

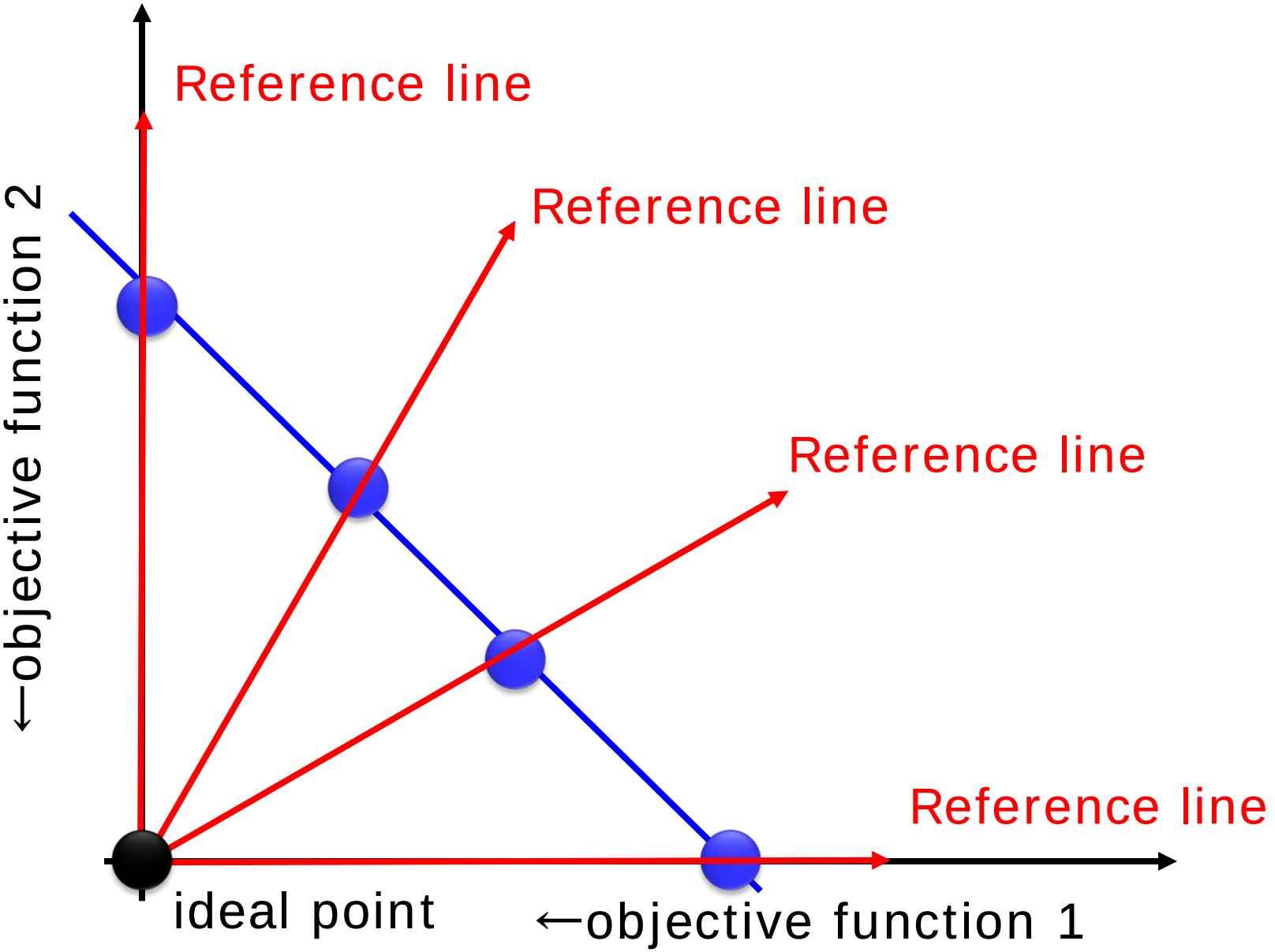


In NSGA-III, the last solutions are selected using reference lines

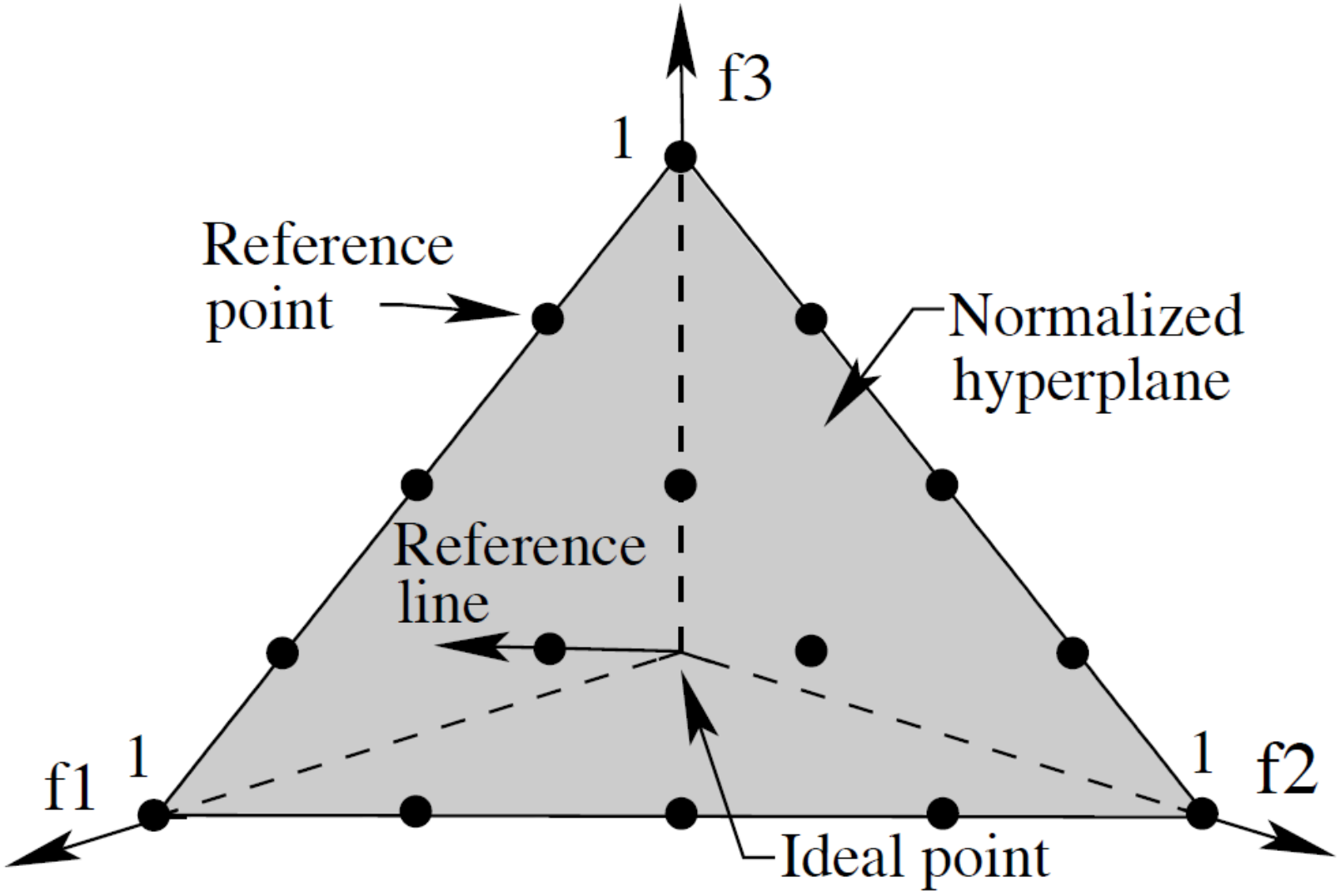
Reference points and reference lines in NSGA-III



Reference points and reference lines in NSGA-III



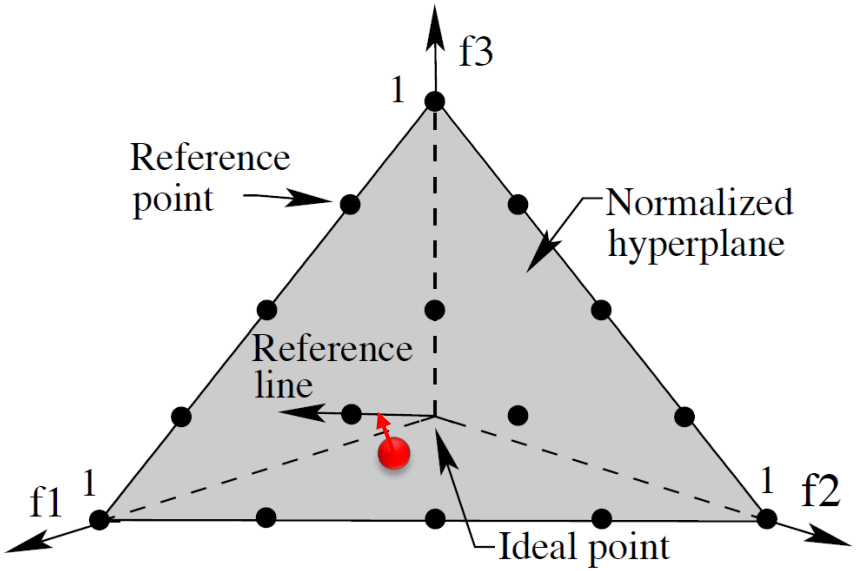
Reference points and reference lines in NSGA-III

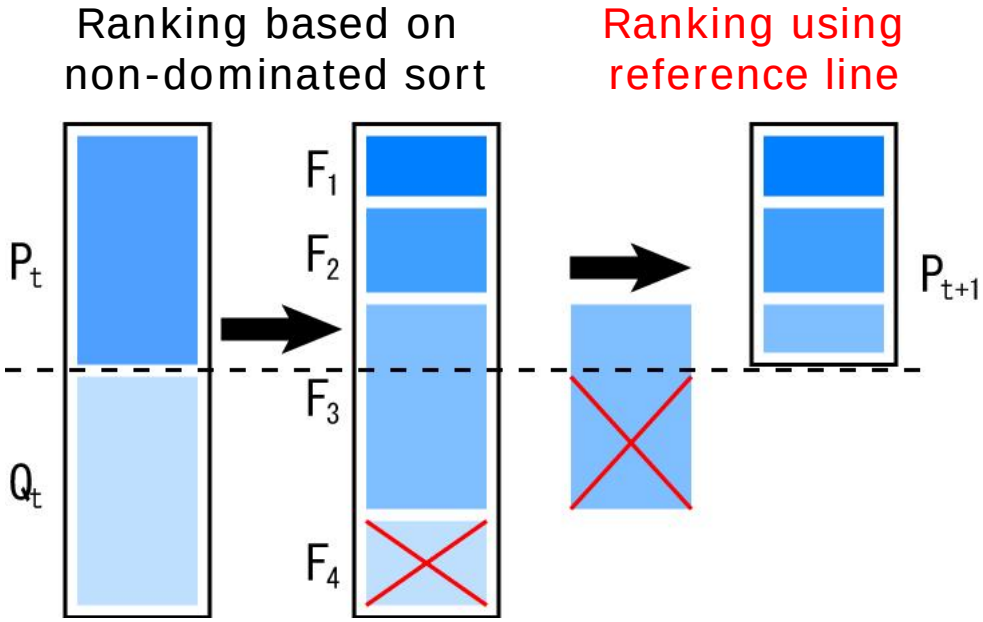


In case of minimization problems

Selection in NSGA-III

1. Each solution is assigned to the closest reference line
2. The reference line with minimum number of assigned solutions is selected and the closest solution to that line is selected.
3. Remove the selected line and the solution
4. Repeat





NSGA-III

Some of the solutions are selected based on distance to reference lines



Good diversity even in many-objective optimization problems

多目的進化アルゴリズムの比較

comparison of multiobjective evolutionary algorithms

IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received July 27, 2017, accepted August 31, 2017, date of publication September 11, 2017, date of current version October 12, 2017.

Digital Object Identifier 10.1109/ACCESS.2017.2751071

Benchmarking Multi- and Many-Objective Evolutionary Algorithms Under Two Optimization Scenarios

RYOJI TANABE[1], (Member, IEEE), HISAO ISHIBUCHI[1], (Fellow, IEEE), AND AKIRA OYAMA[2], (Member, IEEE)

[1]Department of Computer Science and Engineering, Southern University of Science and Technology, Shenzhen 518055, China
[2]Institute of Space and Astronautical Science, Japan Aerospace Exploration Agency, Sagamihara 252-5210, Japan

Corresponding author: Hisao Ishibuchi (hisao@sustc.edu.cn)

This work was supported by MEXT—Development of Innovative Design and Production Processes that Lead the Way for the Manufacturing Industry in the Near Future through Priority Issue on Post-K computer.

多目的進化アルゴリズム

multiobjective evolutionary algorithms

MOEAs	C1	C2	C3	C4	C5	Year	$FEvals^{max}$
NSGA-II [6]	✓					2002	2.5×10^4
NSGA-III [10]	✓				✓	2014	5.5×10^5
SPEA2 [7]	✓					2001	1.0×10^6
SPEA2+SDE [11]	✓					2014	1.0×10^5
GrEA [12]		✓				2013	1.0×10^5
VaEA [13]	✓				✓	2016	5.5×10^5
θ-DEA [14]	✓		✓		✓	2016	5.5×10^5
MOEA/D-07 [15]			✓			2007	7.5×10^4
MOEA/D-09 [16]			✓			2009	3.0×10^5
MOEA/D-DRA [17]			✓			2009	3.0×10^5
MOEA/D-STM [18]			✓			2014	3.0×10^5
MOEA/DD [19]	✓		✓			2015	5.5×10^5
MOEA/D-DU [20]			✓			2016	2.6×10^5
I-DBEA [21]	✓		✓			2015	1.4×10^6
EFR-RR [20]			✓			2016	2.6×10^5
RVEA [22]			✓		✓	2016	2.8×10^5
$IBEA_{\epsilon+}$ [23]				✓		2004	2.0×10^4
$IBEA_{HD}$ [23]				✓		2004	2.0×10^4
HypE [24]	✓			✓		2011	1.0×10^4
BiGE [25]	✓			✓		2015	1.0×10^5
MOMBI-II [26]				✓		2015	4.4×10^5

- 使われたベンチーマーク問題が異なる
  Different benchmark problems are used
- 目的関数評価回数（集団サイズx世代数）が異なる
  Different number of objective function evaluations (population size x number of generations) is used.

どれがよいかわからない．．．
No one tell what is the best...

（自分のアルゴリズムにとって一番都合のよい条件を選んでいる？）

多目的進化アルゴリズム

multiobjective evolutionary algorithms

MOEAs	C1	C2	C3	C4	C5	Year	$FEvals^{max}$
NSGA-II [6]	✓					2002	2.5×10^4
NSGA-III [10]	✓				✓	2014	5.5×10^5
SPEA2 [7]	✓					2001	1.0×10^6
SPEA2+SDE [11]	✓					2014	1.0×10^5
GrEA [12]		✓				2013	1.0×10^5
VaEA [13]	✓				✓	2016	5.5×10^5
θ-DEA [14]	✓		✓		✓	2016	5.5×10^5
MOEA/D-07 [15]			✓			2007	7.5×10^4
MOEA/D-09 [16]			✓			2009	3.0×10^5
MOEA/D-DRA [17]			✓			2009	3.0×10^5
MOEA/D-STM [18]			✓			2014	3.0×10^5
MOEA/DD [19]	✓		✓			2015	5.5×10^5
MOEA/D-DU [20]			✓			2016	2.6×10^5
I-DBEA [21]	✓		✓			2015	1.4×10^6
EFR-RR [20]			✓			2016	2.6×10^5
RVEA [22]			✓		✓	2016	2.8×10^5
$IBEA_{\epsilon+}$ [23]				✓		2004	2.0×10^4
$IBEA_{HD}$ [23]				✓		2004	2.0×10^4
HypE [24]	✓			✓		2011	1.0×10^4
BiGE [25]	✓			✓		2015	1.0×10^5
MOMBI-II [26]				✓		2015	4.4×10^5

- 最終世代集団のパレート解の比較 Final population scenario
  進化アルゴリズムの比較では一般的
  Often used for algorithm comparison

- 最終世代までに得られたすべてのパレート最適解の比較 Unbounded external archive scenario
  工学的にはこちらが重要
  This performance is important in engineering view

多目的進化アルゴリズム

multiobjective evolutionary algorithms

MOEAs	C1	C2	C3	C4	C5	Year	$FEvals^{max}$
NSGA-II [6]	✓					2002	2.5×10^4
NSGA-III [10]	✓				✓	2014	5.5×10^5
SPEA2 [7]	✓					2001	1.0×10^6
SPEA2+SDE [11]	✓					2014	1.0×10^5
GrEA [12]		✓				2013	1.0×10^5
VaEA [13]	✓				✓	2016	5.5×10^5
θ-DEA [14]	✓		✓		✓	2016	5.5×10^5
MOEA/D-07 [15]			✓			2007	7.5×10^4
MOEA/D-09 [16]			✓			2009	3.0×10^5
MOEA/D-DRA [17]			✓			2009	3.0×10^5
MOEA/D-STM [18]			✓			2014	3.0×10^5
MOEA/DD [19]	✓		✓			2015	5.5×10^5
MOEA/D-DU [20]			✓			2016	2.6×10^5
I-DBEA [21]	✓		✓			2015	1.4×10^6
EFR-RR [20]			✓			2016	2.6×10^5
RVEA [22]			✓		✓	2016	2.8×10^5
$IBEA_{\epsilon+}$ [23]				✓		2004	2.0×10^4
$IBEA_{HD}$ [23]				✓		2004	2.0×10^4
HypE [24]	✓			✓		2011	1.0×10^4
BiGE [25]	✓			✓		2015	1.0×10^5
MOMBI-II [26]				✓		2015	4.4×10^5

- 同一のテスト問題を使う
  same benchmark problems are used
- 各目的関数評価回数での性能を評価する
  Anytime performance is evaluated
- 最終世代までに得られたすべてのパレート最適解を比較する
  Unbounded external archive scenario is used.

Websites that we downloaded the source codes

MOEA	Download site
SPEA2, NSGA-II, IBEA$_{HD}$, MOEA/D-09, and MOEA/D-DRA	jMetal 4.5[5]
RVEA and I-DBEA	MOEA Framework 2.11[6]
MOEA/D-STM and MOEA/DD	Ke Li's website [7]
NSGA-III, θ-DEA, MOEA/D-DU, and EFR-RR	Yao's website[8]
HypE	PISA[9]
GrEA, SPEA2+SDE, and BiGE	Miqing Li's website [10]
MOMBI-II	Gómez's website [11]
VaEA	Yi's website[12]

[5] http://jmetal.sourceforge.net/
[6] http://moeaframework.org/index.html
[7] http://www.cs.bham.ac.uk/~likw/publications.html
[8] http://www.cs.bham.ac.uk/~xin/journal_papers.html
[9] http://www.tik.ee.ethz.ch/sop/download/supplementary/hype/
[10] http://www.cs.bham.ac.uk/~limx/
[11] https://www.cs.cinvestav.mx/~EVOCINV/software/MOMBI-II/MOMBI-II.html
[12] https://www.researchgate.net/profile/Xiang_Yi9/publications

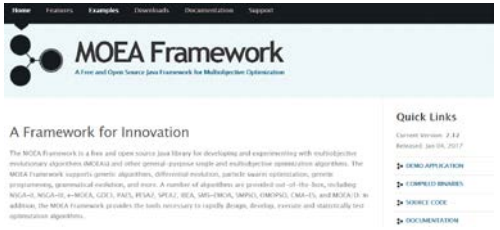
websites on evolutionary algorithms

- K. Deb's group http://www.iitk.ac.in/kangal/

  NSGA-II in C (Real and Binary-Coded + constraint Handling)

- **MOEA Framework (last update: Jan 4, 2017)**
  http://moeaframework.org/

  MOEA Framework is free and open source Java library for developing and experimenting with multiobjective evolutionary algorithms



- **Jmetal 5 (last update: Dec 13, 2017)**
  http://jmetal.github.io/jMetal/

  jMetal stands for Metaheuristic Algorithms in Java, and it is an object-oriented Java-based framework for multi-objective optimization with metaheuristics.



- **PlatEMO (last update: 2017)**
  http://bimk.ahu.edu.cn/index.php?s=/Index/Software/index.html

  Evolutionary multi-objective optimization platform on MATLAB

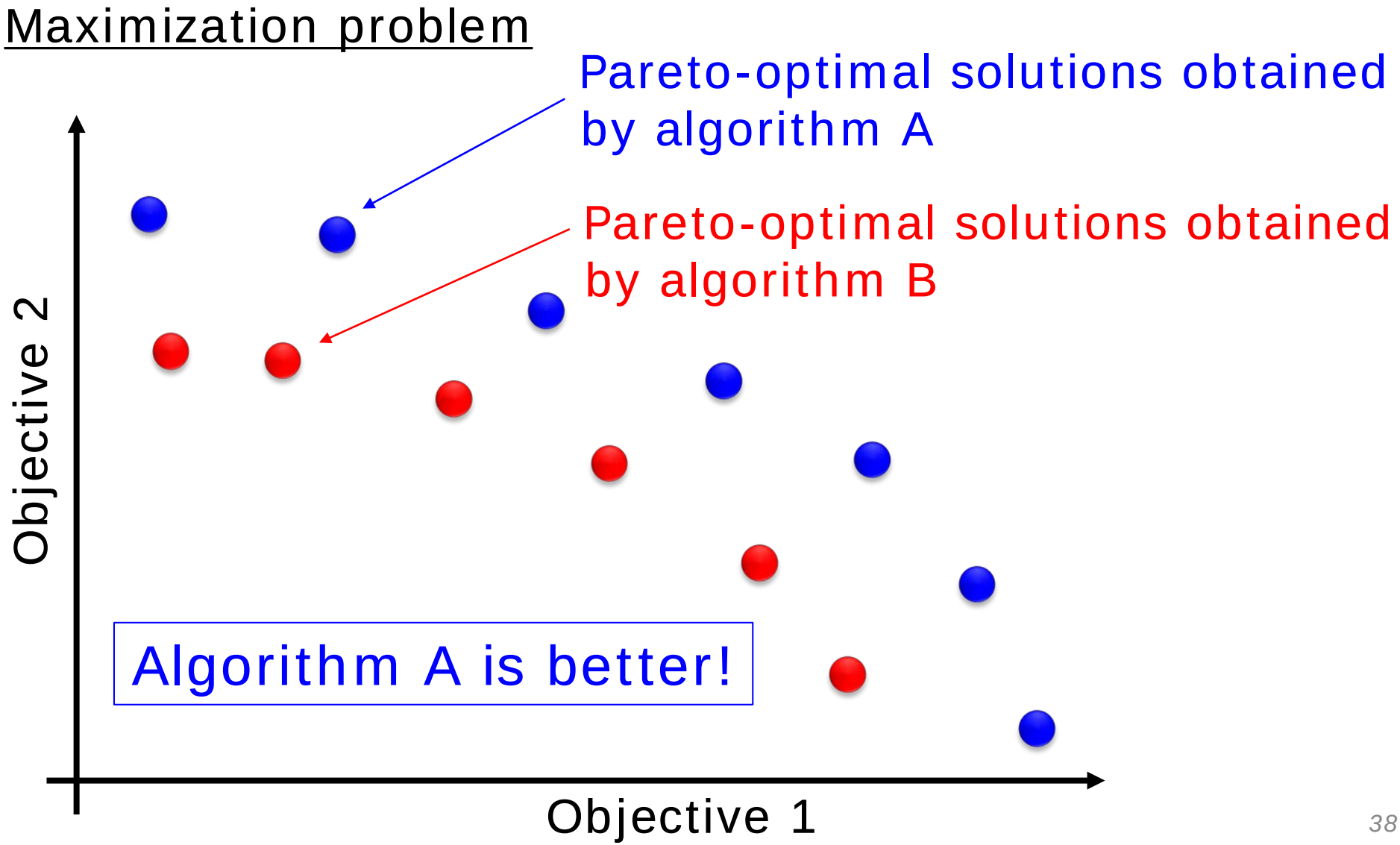
  50+ open source evolutionary algorithms

  100+ open source multi-objective test problems



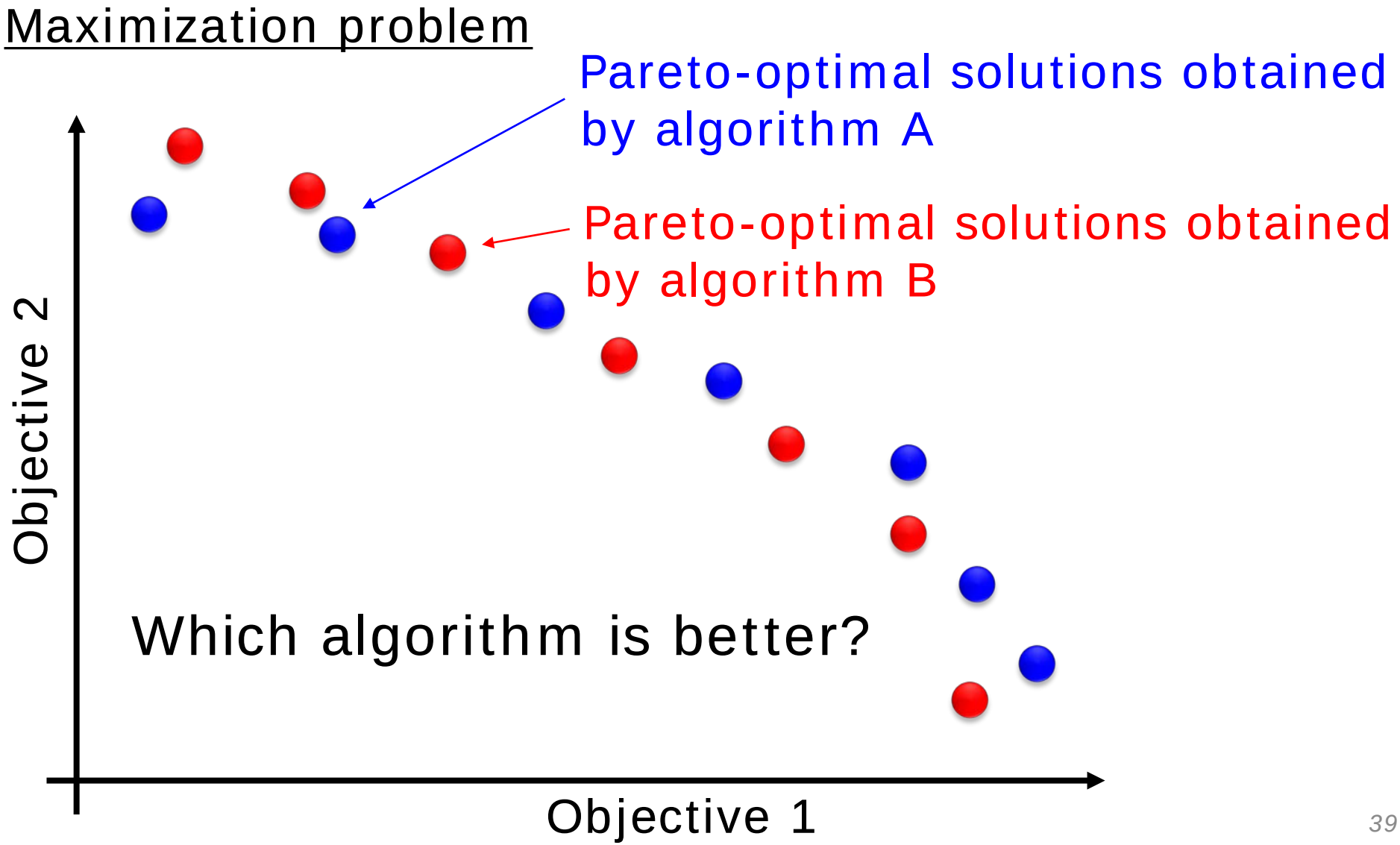
どうやって異なる多目的最適化アルゴリズムの性能を評価するか？

How to compare performance of different multiobjective algorithms?

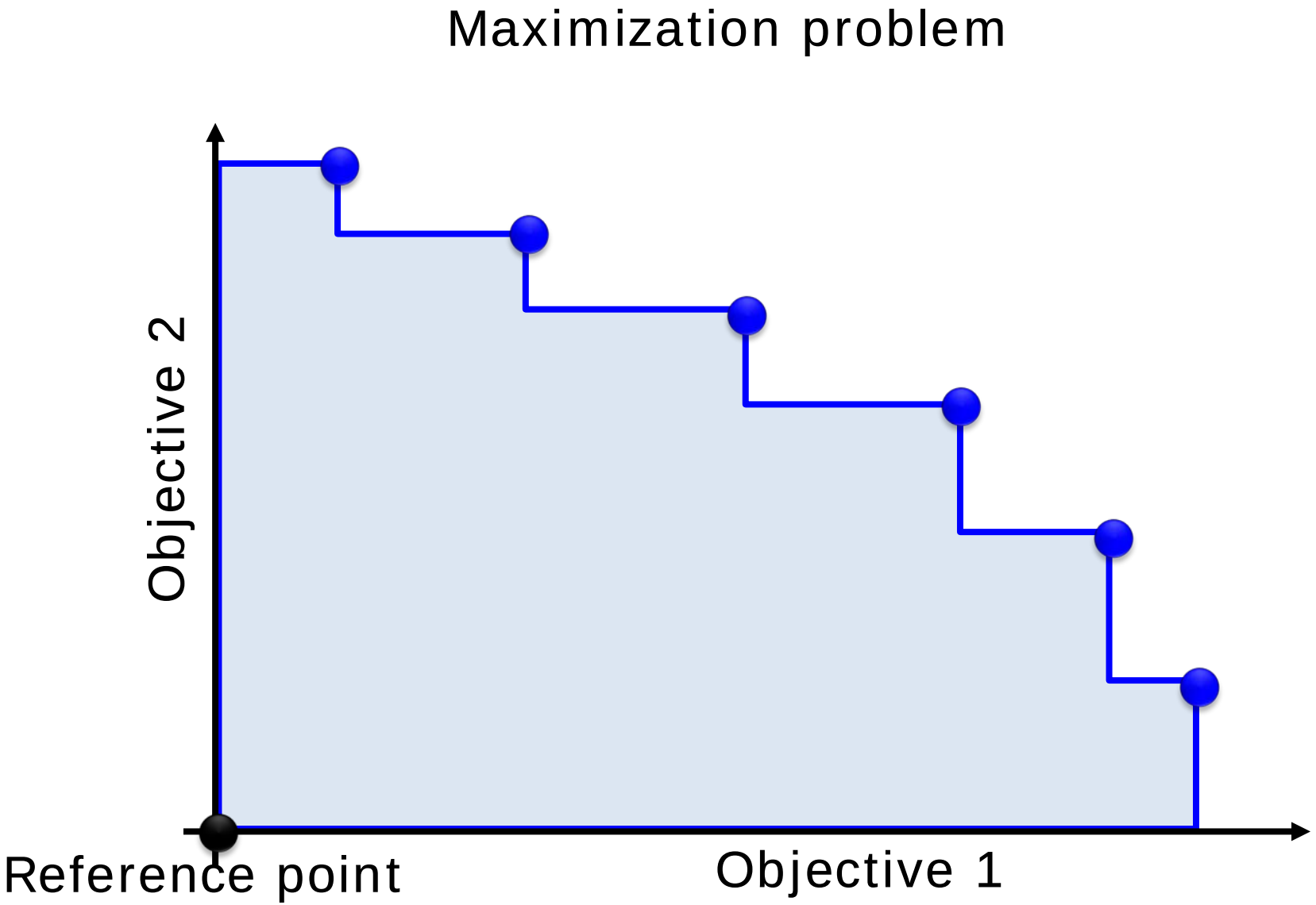


どうやって異なる多目的最適化アルゴリズムの性能を評価するか？

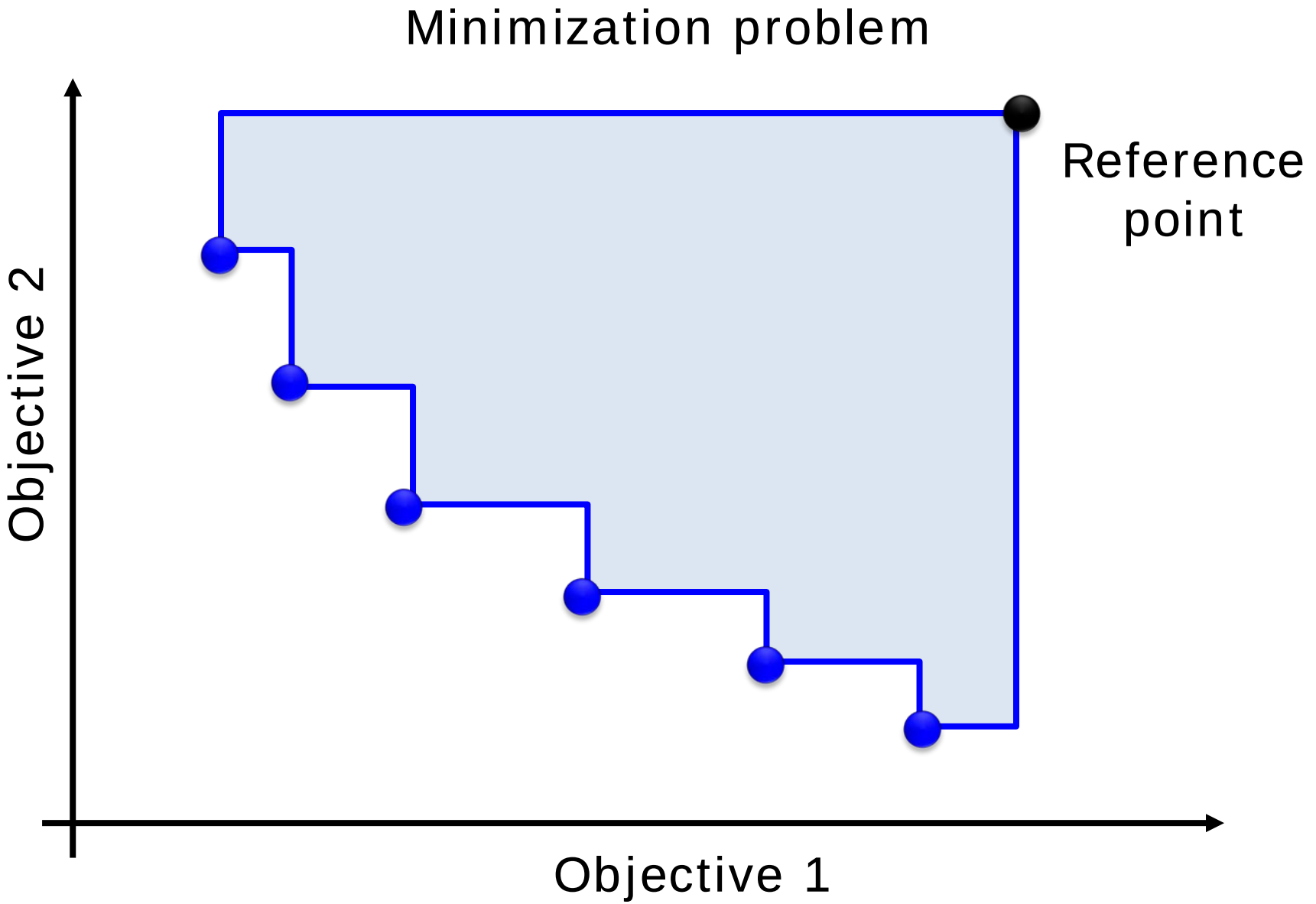
How to compare performance of different multiobjective algorithms?



Hypervolume indicator

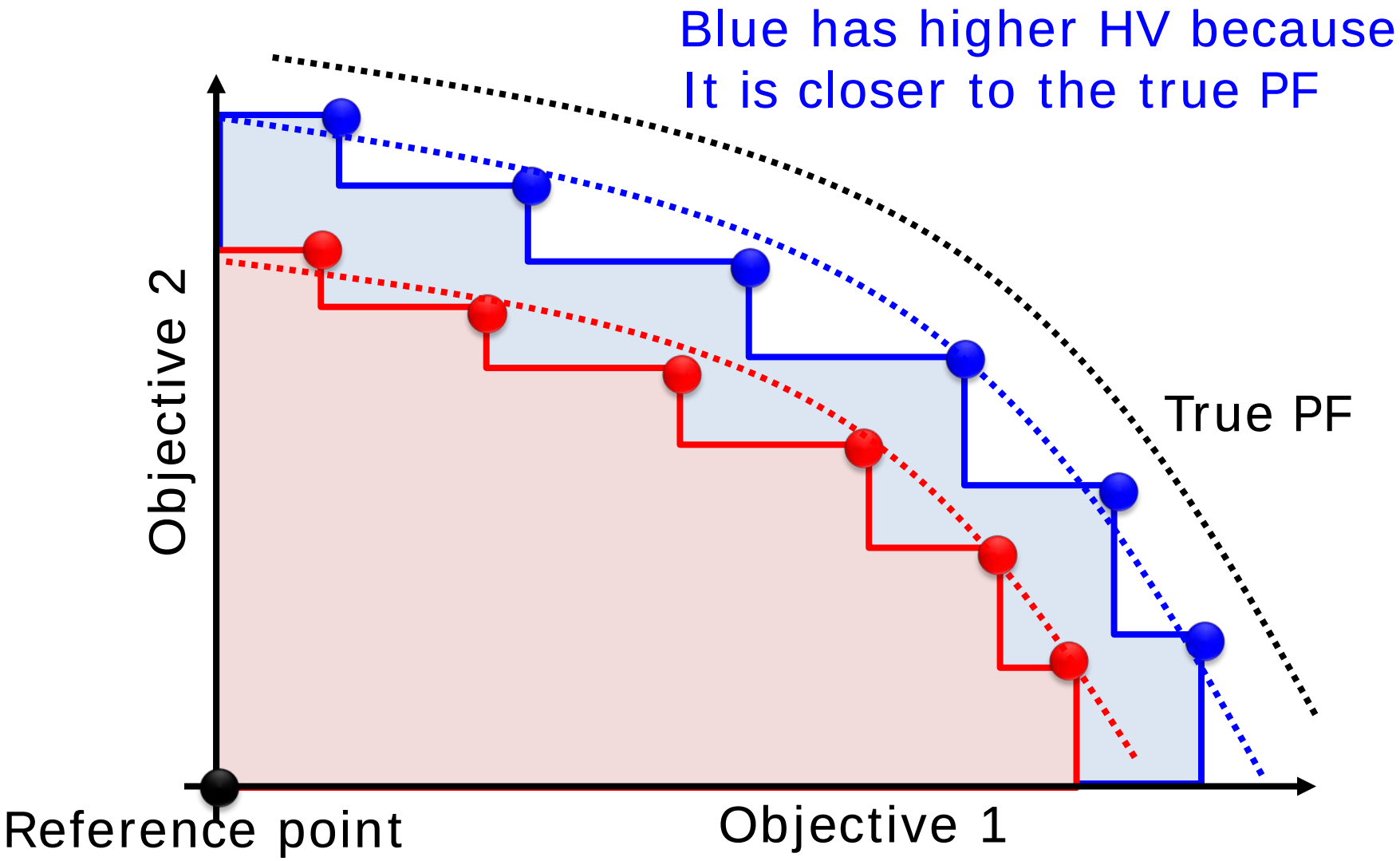


Hypervolume indicator



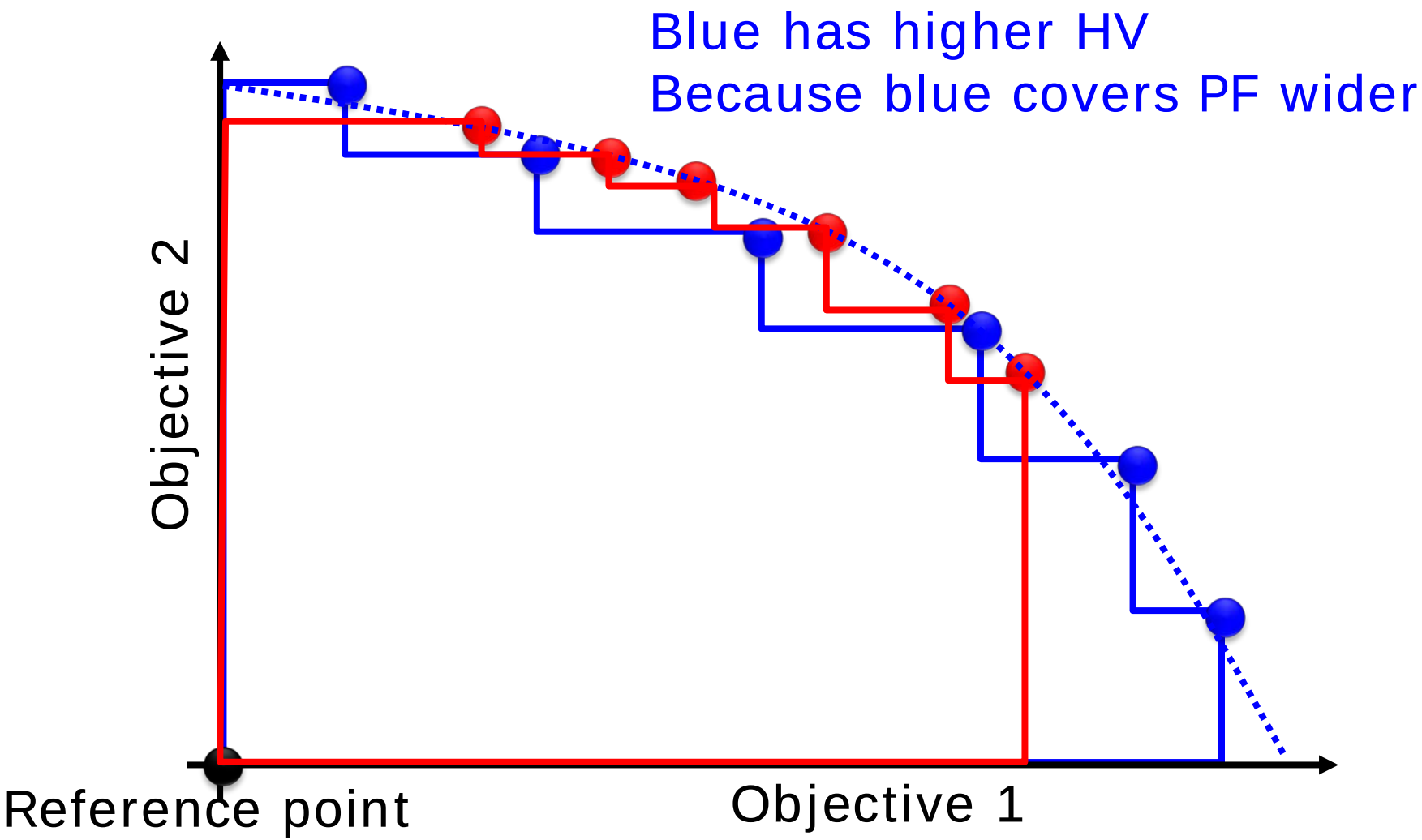
Hypervolume (HV) indicator

Maximization problem



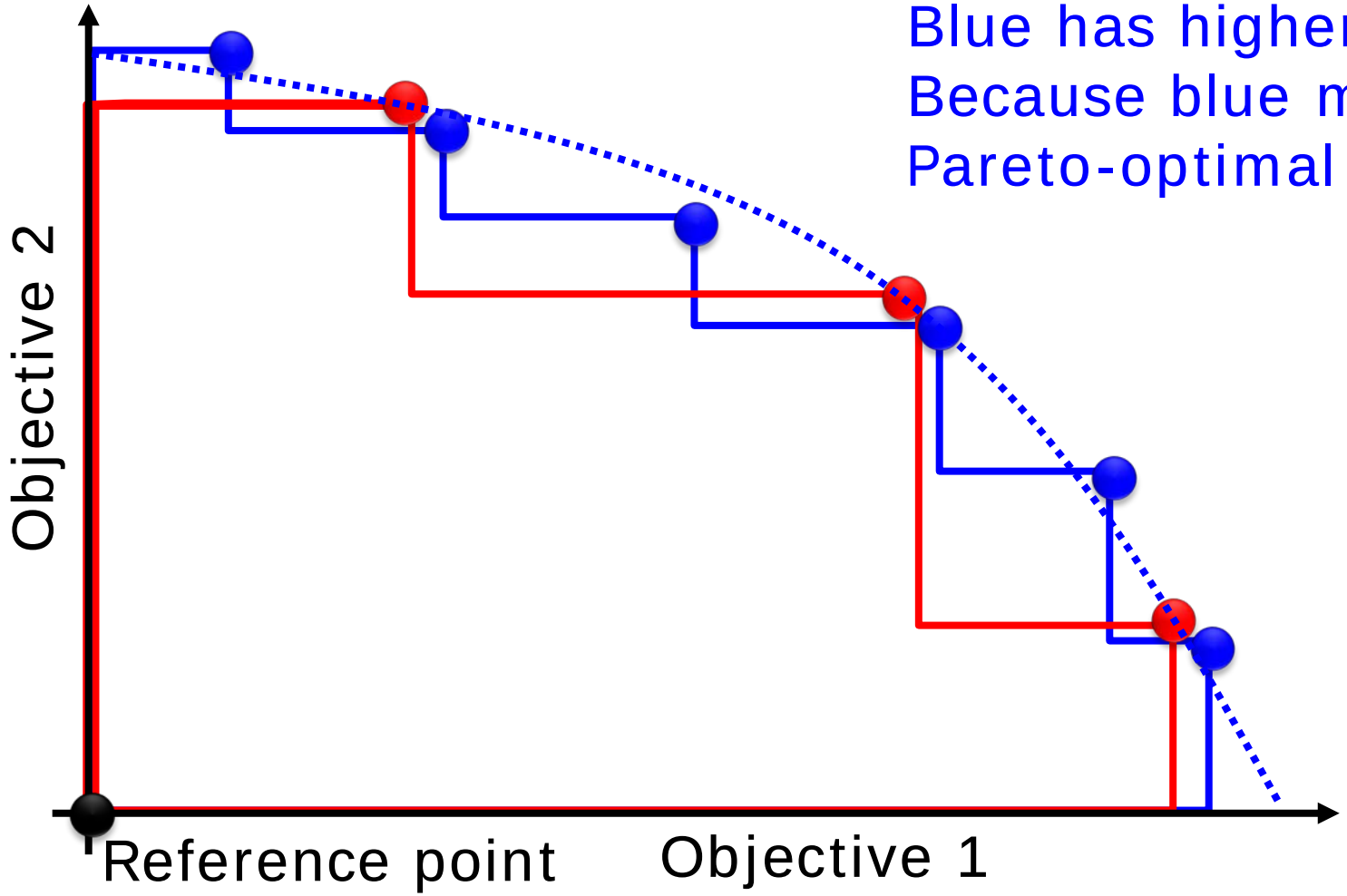
Hypervolume (HV) indicator

Maximization problem



Hypervolume (HV) indicator

Maximization problem



Hypervolume indicator

- HV indicator of a set of Pareto-optimal solutions becomes higher when
  - It is closer to the true Pareto-front
  - It convers Pareto-front wider
  - It contains larger number of Pareto-optimal solutions
- Advantage: Computation of HV indicator does not require to know true Pareto-front
- Drawback: Contributions of convergence, diversity, and number of solutions can not be separated.

Benchmark problems?

ベンチマーク問題 benchmark problems

- さまざまな最適化手法が提案されているし，今後もいろいろな手法が提案されていく．どの手法が優れているのか？

  Many optimization algorithms have been proposed and will be proposed. Which one is the best?

- 最適化手法を比較するためのベンチマーク問題が提案されている．

  Benchmark problems have been proposed for comparison of optimization algorithms

Benchmark problems

- DTLZ test problems
- WFG test problems
- Real-world-like optimization problems
  - Two-bar truss design [19]
  - Speed reducer design [19][26]
  - Disk brake design [19],[26]
  - Welded beam design [28],[26]
  - Car side impact design [7]
  - Ship design [29],[27]
  - Water design [20], [7]

DTLZ test problems

- It is created by Deb et al. in 2001.
- Nine test problems are included (DTLZ8 and DTLZ9 is side-constrained)
- It is **scalable** to any number of objectives
- DTLZ1-DTLZ6 are scalable with respect to the number of distance parameters, but have a fixed number of M-1 position parameters.
- No deceptive test problem.

[23] K. Deb, L. Thiele, M. Laumanns, and E. Zitzler, *Scalable Test Problems for Evolutionary Multi-Objective Optimization*. Kanpur, India: Kanpur Genetic Algorithms Lab. (KanGAL), Indian Inst. Technol., 2001. KanGAL Report 2 001 001.
[33] K. Deb, L. Thiele, M. Laumanns, and E. Zitzler, "Scalable multi-objective optimization test problems," in Proc. Congr. Evol. Comput., May 2002, vol. 1, pp. 825–830.

[3]In fact, the number of DTLZ problems depends on which paper is being referred to. In this paper, we are using the nine DTLZ problems from the original technical report [23] A more recent conference paper version of the technical report also exists [33], in which only seven of the original nine problems were reproduced (DTLZ5 and DTLZ9 from the original paper were dropped). As a result, DTLZ6 from the original technical report is DTLZ5 in the conference paper, DTLZ7 is DTLZ6 in the conference paper, and DTLZ8 is DTLZ7 in the conference paper.

DTLZ test problems

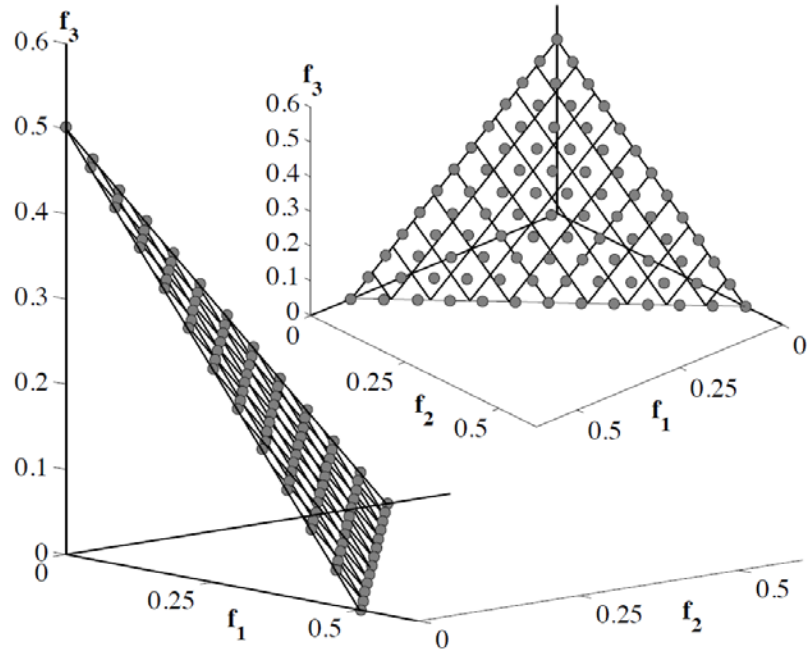
Name	Problem	Parameter Domains
DTLZ1	$f_1 = (1+g)0.5\prod_{i=1}^{M-1} y_i$ $f_{m=2:M-1} = (1+g)0.5\left(\prod_{i=1}^{M-m} y_i\right)(1-y_{M-m+1})$ $f_M = (1+g)0.5(1-y_1)$ $g = 100\left[k+\sum_{i=1}^{k}\left((z_i-0.5)^2-\cos(20\pi(z_i-0.5))\right)\right]$	$[0,1]$
DTLZ2	$f_1 = (1+g)\prod_{i=1}^{M-1}\cos(y_i\pi/2)$ $f_{m=2:M-1} = (1+g)\left(\prod_{i=1}^{M-m}\cos(y_i\pi/2)\right)\sin(y_{M-m+1}\pi/2)$ $f_M = (1+g)\sin(y_1\pi/2)$ $g = \sum_{i=1}^{k}(z_i-0.5)^2$	$[0,1]$
DTLZ3	As DTLZ2, except the equation for g is replaced by the one from DTLZ1.	$[0,1]$
DTLZ4	As DTLZ2, except all $y_i \in y$ are replaced by y_i^α, where $\alpha > 0$.	$[0,1]$
DTLZ5	As DTLZ2, except all $y_2,\ldots,y_{M-1} \in \mathbf{y}$ are replaced by $\frac{1+2gy_i}{2(1+g)}$.	$[0,1]$
DTLZ6	As DTLZ5, except the equation for g is replaced by $g = \sum_{i=1}^{k} z_i^{0.1}$.	$[0,1]$
DTLZ7	$f_{m=1:M-1} = y_m$ $f_M = (1+g)\left(M-\sum_{i=1}^{M-1}\left[\frac{f_i}{1+g}(1+\sin(3\pi f_i))\right]\right)$ $g = 1+9\sum_{i=1}^{k} z_i/k$	$[0,1]$

DTLZ test problems

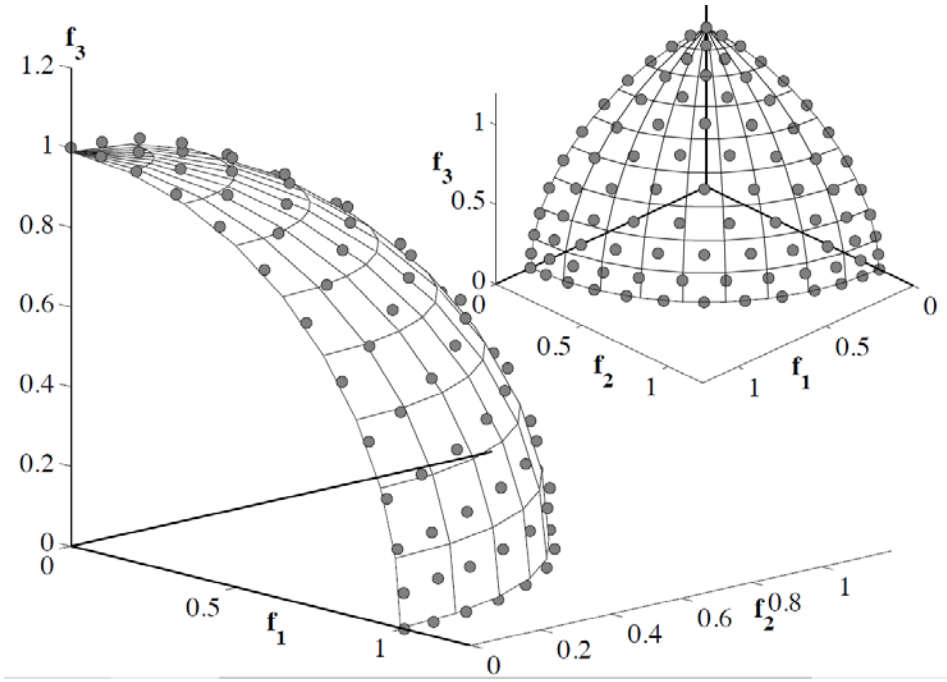
Name	Objective	R3: # Parameters	F2: Separability	F5: Modality	R1: No Extremal	R2: No Medial	R5: Diss. Domains	R6: Diss. Ranges	R7: Optima Known	F1: Geometry	F3: Bias	F4a: Pareto Many-to-one	F4b: Flat Regions
DTLZ1	$f_{1:M}$	✓*	S*	M	✓	×	×	×	✓	linear	−	+	−
DTLZ2*	$f_{1:M}$	✓*	S*	U	✓	×	×	×	✓	concave	−	+	−
DTLZ3	$f_{1:M}$	✓*	S*	M	✓	×	×	×	✓	concave	−	+	−
DTLZ4	$f_{1:M}$	✓*	S*	U	✓	×	×	×	✓	concave	+	+	−
DTLZ5*	$f_{1:M}$	✓*	?	U	?	?	×	?	×	?	−	+	−
DTLZ6*	$f_{1:M}$	✓*	?	U	?	?	×	?	×	?	+	+	−
DTLZ7	$f_{1:M-1}$	1	NA	U	×	✓	×	×*	✓	disconnected*	−	−	−
	f_M	✓	S	M									

S: separable
NS: nonseparable
U: unimodal
M: multimodal
D: deceptiveness

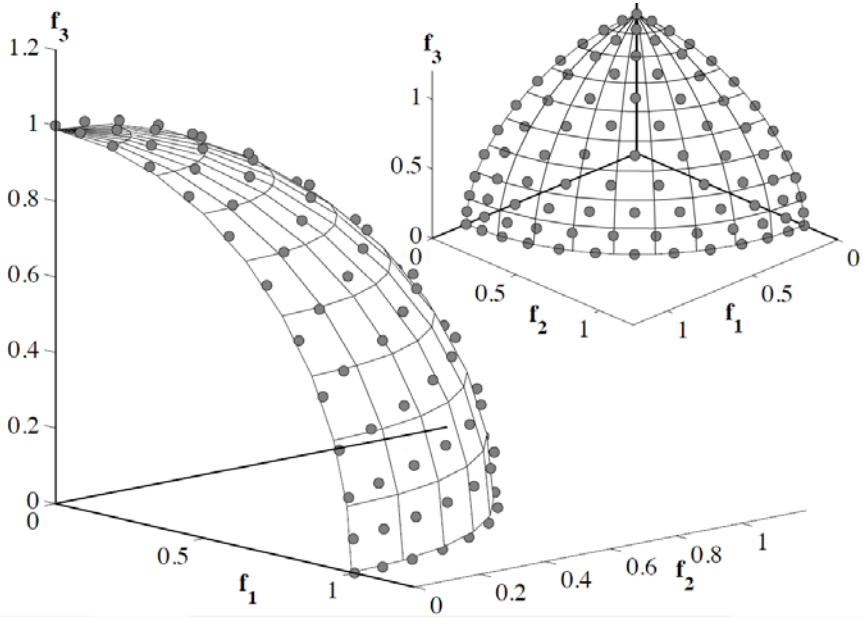
Pareto-front of DTLZ problems (minimization problems)



DTLZ1 (linear)



DTLZ2 (concave)



DTLZ4 (concave)

Benchmark problems: WFG test problems

- It is introduced by Huband et al in 2005.
- It consists of 9 test problems with $M \in \{2,3,4,5,6\}$ (Number of objective functions)

S. Huband, L. Barone, L. While, and P. Hingston, “A scalable multiobjective test problem toolkit,” Proc. Evolutionary Multi-Criterion Optimization: 3rd Int. Conf., 2005, pp. 280–294

Benchmark problems: WFG test problems

All problems created by the WFG Toolkit conform to the following format:

$$\text{Given } \mathbf{z} = \{z_1, \ldots, z_k, z_{k+1}, \ldots, z_n\}$$

$$\text{Minimize } f_{m=1:M}(\mathbf{x}) = Dx_M + S_m h_m(x_1, \ldots, x_{M-1})$$

Objective functions

$$\text{where } \mathbf{x} = \{x_1, \ldots, x_M\}$$

$$= \{\max(t^p_M, A_1)(t^p_1 - 0.5) + 0.5, \ldots,$$

$$\max(t^p_M, A_{M-1})(t^p_{M-1} - 0.5) + 0.5, t^p_M\}$$

$$\mathbf{t}^p = \{t^p_1, \ldots, t^p_M\} \longleftarrow |\mathbf{t}^{p-1} \longleftarrow | \ldots \longleftarrow |\mathbf{t}^1 \longleftarrow |\mathbf{z}_{[0,1]}$$

$$\mathbf{z}_{[0,1]} = \{z_{1,[0,1]}, \ldots, z_{n,[0,1]}\}$$

$$= \{z_1/z_{1,\max}, \ldots, z_n/z_{n,\max}\}$$

Benchmark problems: WFG test problems

Problem	Type	Setting
All	Constants	$S_{m=1:M} = 2m$ $D = 1$ $A_1 = 1$ $A_{2:M-1} = \begin{cases} 0, & \text{for WFG3} \\ 1, & \text{otherwise} \end{cases}$ The settings for $S_{1:M}$ ensures the Pareto optimal fronts have dissimilar tradeoff magnitudes, and the settings for $A_{1:M-1}$ ensures the Pareto optimal fronts are not degenerate, except in the case of WFG3, which has a one dimensional Pareto optimal front.
All	Domains	$z_{i=1:n,\max} = 2i$ The working parameters have domains of dissimilar magnitude.
WFG1	Shape	$h_{m=1:M-1} = \text{convex}_m$ $h_M = \text{mixed}_M$ (with $\alpha = 1$ and $A = 5$)
	$\mathbf{t}^1$	$t^1_{i=1:k} = y_i$ $t^1_{i=k+1:n} = \text{s_linear}(y_i, 0.35)$
	$\mathbf{t}^2$	$t^2_{i=1:k} = y_i$ $t^2_{i=k+1:n} = \text{b_flat}(y_i, 0.8, 0.75, 0.85)$
	$\mathbf{t}^3$	$t^3_{i=1:n} = \text{b_poly}(y_i, 0.02)$
	$\mathbf{t}^4$	$t^4_{i=1:M-1} = \text{r_sum}(\{y_{(i-1)k/(M-1)+1}, \ldots, y_{ik/(M-1)}\}, \{2((i-1)k/(M-1)+1), \ldots, 2ik/(M-1)\})$ $t^4_M = \text{r_sum}(\{y_{k+1}, \ldots, y_n\}, \{2(k+1), \ldots, 2n\})$
WFG2	Shape	$h_{m=1:M-1} = \text{convex}_m$ $h_M = \text{disc}_M$ (with $\alpha = \beta = 1$ and $A = 5$)
	$\mathbf{t}^1$	As $\mathbf{t}^1$ from WFG1. (Linear shift.)
	$\mathbf{t}^2$	$t^2_{i=1:k} = y_i$ $t^2_{i=k+1:k+l/2} = \text{r_nonsep}(\{y_{k+2(i-k)-1}, y_{k+2(i-k)}\}, 2)$
	$\mathbf{t}^3$	$t^3_{i=1:M-1} = \text{r_sum}(\{y_{(i-1)k/(M-1)+1}, \ldots, y_{ik/(M-1)}\}, \{1, \ldots, 1\})$ $t^3_M = \text{r_sum}(\{y_{k+1}, \ldots, y_{k+l/2}\}, \{1, \ldots, 1\})$

Benchmark problems: WFG test problems

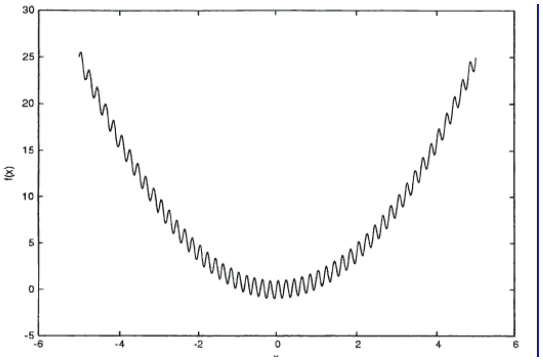
WFG3	Shape	$h_{m=1:M} = \text{linear}_m$ (degenerate)
	$\mathbf{t}^{1:3}$	As $\mathbf{t}^{1:3}$ from WFG2. (Linear shift, non-separable reduction, and weighted sum reduction.)
WFG4	Shape	$h_{m=1:M} = \text{concave}_m$
	$\mathbf{t}^1$	$t^1_{i=1:n} = \text{s_multi}(y_i, 30, 10, 0.35)$
	$\mathbf{t}^2$	$t^2_{i=1:M-1} = \text{r_sum}(\{y_{(i-1)k/(M-1)+1}, \ldots, y_{ik/(M-1)}\}, \{1, \ldots, 1\})$
		$t^2_M = \text{r_sum}(\{y_{k+1}, \ldots, y_n\}, \{1, \ldots, 1\})$
WFG5	Shape	$h_{m=1:M} = \text{concave}_m$
	$\mathbf{t}^1$	$t^1_{i=1:n} = \text{s_decept}(y_i, 0.35, 0.001, 0.05)$
	$\mathbf{t}^2$	As $\mathbf{t}^2$ from WFG4. (Weighted sum reduction.)
WFG6	Shape	$h_{m=1:M} = \text{concave}_m$
	$\mathbf{t}^1$	As $\mathbf{t}^1$ from WFG1. (Linear shift.)
	$\mathbf{t}^2$	$t^2_{i=1:M-1} = \text{r_nonsep}(\{y_{(i-1)k/(M-1)+1}, \ldots, y_{ik/(M-1)}\}, k/(M-1))$
		$t^2_M = \text{r_nonsep}(\{y_{k+1}, \ldots, y_n\}, l)$
WFG7	Shape	$h_{m=1:M} = \text{concave}_m$
	$\mathbf{t}^1$	$t^1_{i=1:k} = \text{b_param}(y_i, \text{r_sum}(\{y_{i+1}, \ldots, y_n\}, \{1, \ldots, 1\}), \frac{0.98}{49.98}, 0.02, 50)$
		$t^1_{i=k+1:n} = y_i$
	$\mathbf{t}^2$	As $\mathbf{t}^1$ from WFG1. (Linear shift.)
	$\mathbf{t}^3$	As $\mathbf{t}^2$ from WFG4. (Weighted sum reduction.)
WFG8	Shape	$h_{m=1:M} = \text{concave}_m$
	$\mathbf{t}^1$	$t^1_{i=1:k} = y_i$
		$t^1_{i=k+1:n} = \text{b_param}(y_i, \text{r_sum}(\{y_1, \ldots, y_{i-1}\}, \{1, \ldots, 1\}), \frac{0.98}{49.98}, 0.02, 50)$
	$\mathbf{t}^2$	As $\mathbf{t}^1$ from WFG1. (Linear shift.)
	$\mathbf{t}^3$	As $\mathbf{t}^2$ from WFG4. (Weighted sum reduction.)
WFG9	Shape	$h_{m=1:M} = \text{concave}_m$
	$\mathbf{t}^1$	$t^1_{i=1:n-1} = \text{b_param}(y_i, \text{r_sum}(\{y_{i+1}, \ldots, y_n\}, \{1, \ldots, 1\}), \frac{0.98}{49.98}, 0.02, 50)$
		$t^1_n = y_n$
	$\mathbf{t}^2$	$t^2_{i=1:k} = \text{s_decept}(y_i, 0.35, 0.001, 0.05)$
		$t^2_{i=k+1:n} = \text{s_multi}(y_i, 30, 95, 0.35)$
	$\mathbf{t}^3$	As $\mathbf{t}^2$ from WFG6. (Non-separable reduction.)

Benchmark problems: WFG test problems

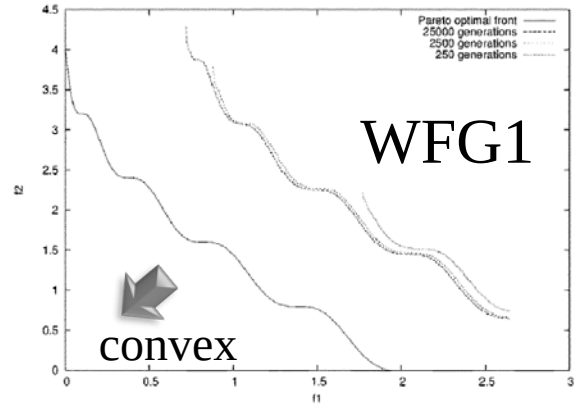
Problem	Obj.	Separability	Modality	Bias	Geometry
WFG1	$f_{1:M}$	S	U	Polynomial, flat region	convex, mixed
WFG2	$f_{1:M-1}$	NS	U	–	convex, disconnected
	f_M	NS	M		
WFG3	$f_{1:M}$	NS	U	–	linear, degenerate
WFG4	$f_{1:M}$	S	M	–	concave
WFG5	$f_{1:M}$	S	D	–	concave
WFG6	$f_{1:M}$	NS	U	–	concave
WFG7	$f_{1:M}$	S	U	parameter dependent	concave
WFG8	$f_{1:M}$	NS	U	parameter dependent	concave
WFG9	$f_{1:M}$	NS	M,D	parameter dependent	concave

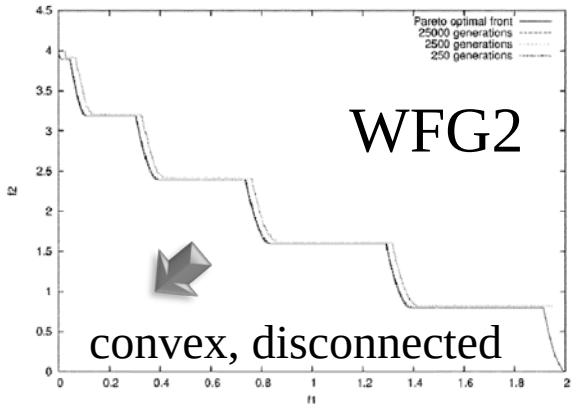
S: separable
NS: nonseparable

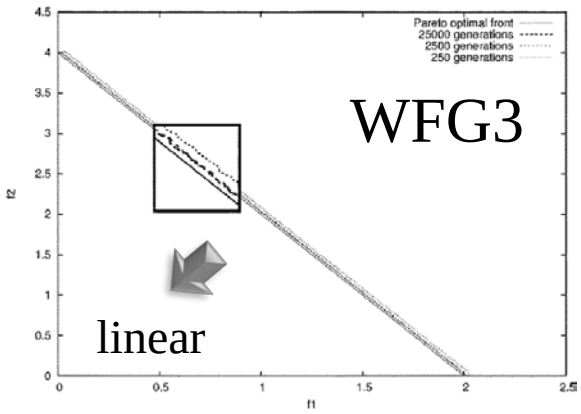
U: unimodal
M: multimodal
D: deceptiveness

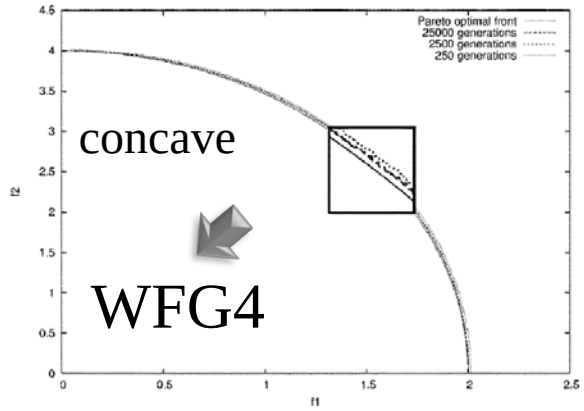


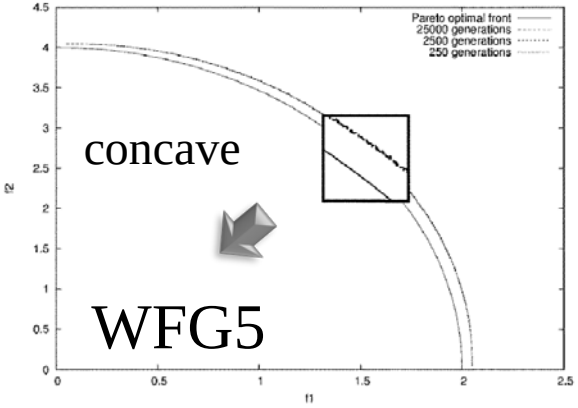
Pareto front of WFG test problems and result with NSGA-II

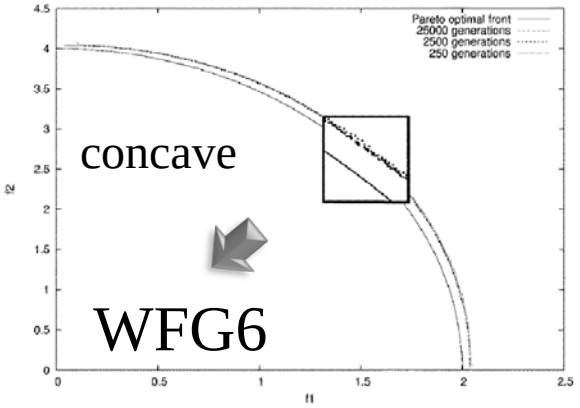


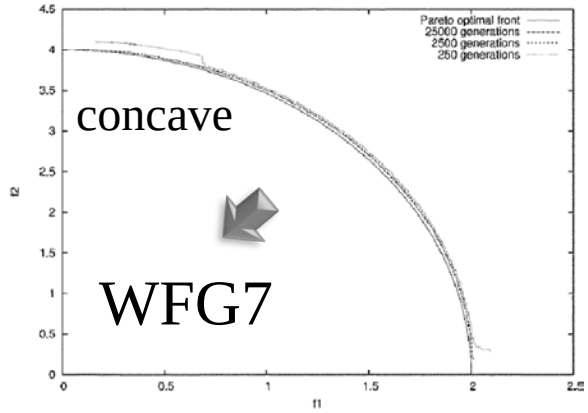




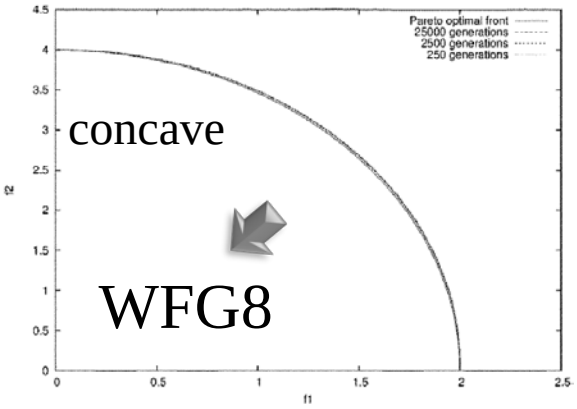




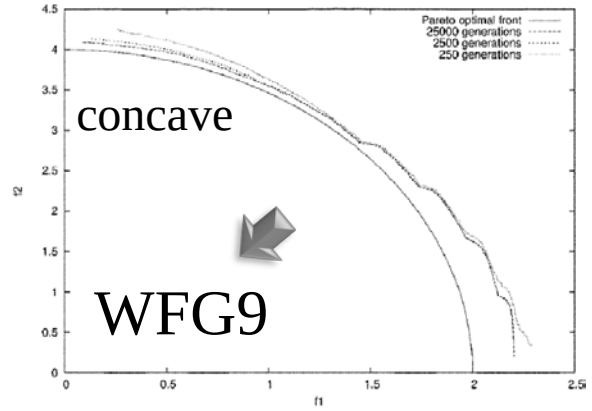




(i)



(g)



(h)

Benchmark problems

- DTLZ test problems
- WFG test problems
- Real-world-like optimization problems
  - Two-bar truss design [19]
  - Speed reducer design [19][26]
  - Disk brake design [19],[26]
  - **Welded beam design [28],[26]**
  - **Car side impact design [7]**
  - Ship design [29],[27]
  - Water design [20], [7]

[7] H. Jain and K. Deb, “An Evolutionary Many-Objective Optimization Algorithm Using Reference-Point Based Nondominated Sorting Approach, Part II: Handling Constraints and Extending to an Adaptive Approach,” IEEE TEVC, vol. 18, no. 4, pp. 602–622, 2014.

[19] C. A. C. Coello and G. T. Pulido, “Multiobjective structural optimization using a microgenetic algorithm,” Stru. and Multi. Opt., vol. 30, no. 5, pp. 388–403, 2005.

[20] T. Ray, K. Tai, and C. Seow, “An evolutionary algorithm for multiobjective optimization,” Eng. opt., vol. 33, no. 3, pp. 399–424, 2001.

[26] W. Gong, Z. Cai, and Z. Li, “An efficient multiobjective differential evolution algorithm for engineering design,” Stru. and Multi. Opt., vol. 38, no. 2, pp. 137–157, 2009.

[27] H. K. Singh, A. Isaacs, T. Ray, and W. Smith, “Infeasibility Driven Evolutionary Algorithm (IDEA) for Engineering Design Optimization,” in AI2008, 2008, pp. 104–115.

[28] T. Ray and K. M. Liew, “A swarm metaphor for multiobjective design optimization,” Eng. opt., vol. 34, no. 2, pp. 141–153, 2002.

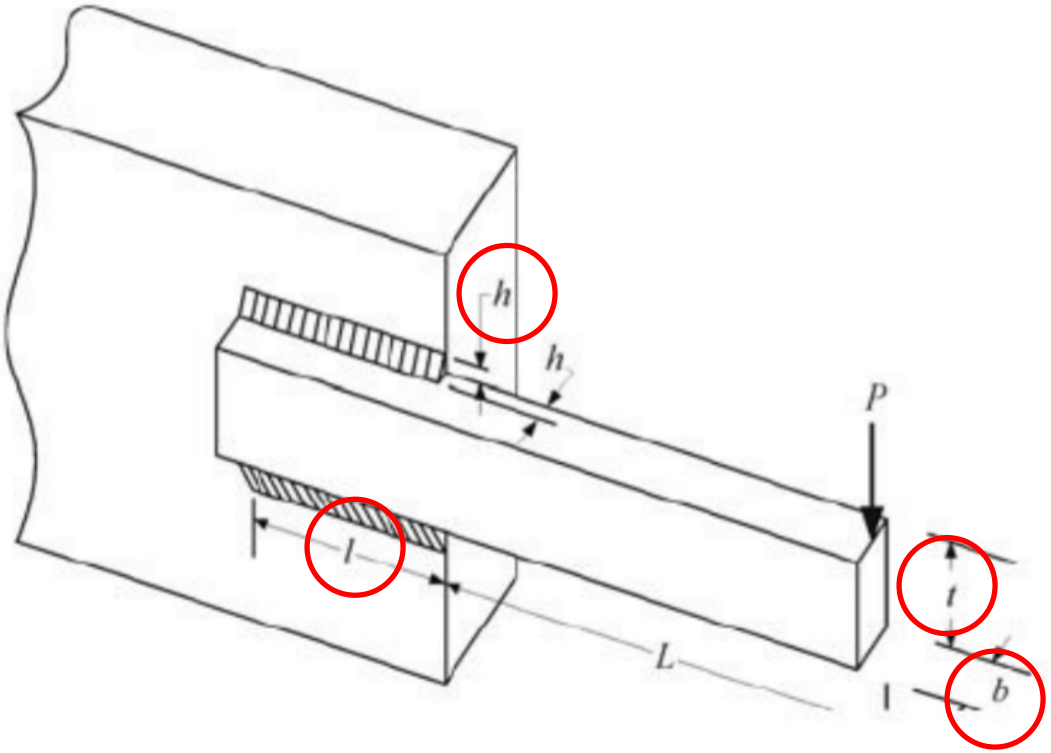
[29] M. G. Parsons and R. L. Scott, “Formulation of Multi-criterion Design Optimization Problems for Solution With Scalar Numerical Optimization Methods,” J. Ship Research, vol. 48, no. 1, pp. 61–76, 2004.

Welded beam design problem

Objective functions : 1. cost to be minimized and 2. end deflection

Constraints : 1. shear stress, 2. bending stress, and 3. buckling load.

Design variables : h, l, t, and b



Wenyin Gong · Zhihua Cai · Li Zhu, “An Efficient Multiobjective Differential Evolution Algorithm for Engineering Design”
https://pdfs.semanticscholar.org/0841/044f92a047ae4b6b6f5dce1b9049b0006505.pdf

Welded beam design

minimize

$$\begin{cases} f_1(\boldsymbol{x}) = 1.10471h^2l + 0.04811tb(14.0+t) \\ f_2(\boldsymbol{x}) = \delta(\boldsymbol{x}) = 2.1952/(t^3b) \end{cases}$$

subject to

$$\begin{cases} e_1(\boldsymbol{x}) = 13,600 - \tau(\boldsymbol{x}) \geq 0 \\ e_2(\boldsymbol{x}) = 30,000 - \sigma(\boldsymbol{x}) \geq 0 \\ e_3(\boldsymbol{x}) = b - h \geq 0 \\ e_4(\boldsymbol{x}) = \mathrm{Pc}(\boldsymbol{x}) - 6,000 \geq 0 \end{cases}$$

where

$$\begin{cases} \tau(\boldsymbol{x}) = \sqrt{(\tau')^2 + (\tau'')^2 + \frac{l\tau'\tau''}{\sqrt{0.25(l^2+(h+t)^2)}}} \\ \tau' = 6,000/(\sqrt{2}hl) \\ \tau'' = \frac{6,000(14.0+0.5l)\sqrt{0.25(l^2+(h+t)^2)}}{2[0.707hl(l^2/12+0.25(l^2+(h+t)^2))]} \\ \sigma(\boldsymbol{x}) = 504,000/(t^2b) \\ \mathrm{Pc}(\boldsymbol{x}) = 64,764.022(1-0.0282346t)tb^3 \\ 0.125 \leq h, b \leq 5.0, \mathrm{and} 0.1 \leq l, t \leq 10.0 \end{cases}$$

Welded beam design

Minimize $f_1(\boldsymbol{x}) = 1.10471 x_1^2 x_2 + 0.04811 x_3 x_4 (14.0 + x_2)$

Minimize $f_2(\boldsymbol{x}) = \delta(x)$

Subject to

$\tau(\mathbf{x}) - \tau_{\max} \le 0$

$\sigma(\mathbf{x}) - \sigma_{\max} \le 0$

$x_1 - x_4 \le 0$

$0.125 - x_1 \le 0$

$P - P_C(x) \le 0$

where $\tau(\boldsymbol{x}) = \sqrt{(\tau')^2 + \dfrac{2\tau'\tau'' x_2}{2R} + (\tau'')^2}$

$\tau' = \dfrac{P}{\sqrt{2} x_1 x_2}$

$\tau'' = \dfrac{MR}{J}$

$M = P(L + \dfrac{x_2}{2})$

$R = \sqrt{\dfrac{x_2^2}{4} + \left(\dfrac{x_1 + x_3}{2}\right)^2}$

$J = 2\left\{\sqrt{2} x_1 x_2 \left[\dfrac{x_2^2}{12} + \left(\dfrac{x_1 + x_3}{2}\right)^2\right]\right\}$

$\sigma(\boldsymbol{x}) = \dfrac{6PL}{x_4 x_3^2}$

$\delta(\boldsymbol{x}) = \dfrac{4PL^3}{E x_4 x_3^3}$

$P_C(\boldsymbol{x}) = \dfrac{4.013E\sqrt{\dfrac{x_3^2 x_4^6}{36}}}{L^2}\left(1 - \dfrac{x_3}{2L}\sqrt{\dfrac{E}{4G}}\right)$

P = 6000 lb, L = 14 in, $\delta_{\max}$ = 0.25 in, E = 30 × 10^6 psi, G = 12 × 10^6 psi, $\tau_{\max}$ = 13,600 psi, $\sigma_{\max}$ = 30,000 psi, $0.125 \le x_1 \le 5.0$, $0.1 \le x_2 \le 10.0$, $0.1 \le x_3 \le 10$ and $0.125 \le x_4 \le 5.0$

Car side impact problem

Objective functions : 1. minimization of weight of car, 2. minimization of pubic force experienced by a passenger, and 3. average velocity of the V-Pillar

Constraints : 10 constraints limiting values of abdomen load, pubic force, velocity of V-Pillar, rib deflection, etc.

Design variables : 11 design variables describing thickness of B-Pillars, floor, cross-members, door beam, roof rail, etc.

Himanshu Jain and Kalyanmoy Deb, “An Evolutionary Many-Objective Optimization Algorithm Using Reference-point Based Non-dominated Sorting Approach, Part II: Handling Constraints and Extending to an Adaptive Approach”
https://www.egr.msu.edu/~kdeb/papers/k2012010.pdf

Car side impact problem

$$
\begin{aligned}
f_1(\mathbf{x}) =& 1.98 + 4.9x_1 + 6.67x_2 + 6.98x_3 + 4.01x_4 + 1.78x_5 \\
&+ 0.00001x_6 + 2.73x_7, \\
f_2(\mathbf{x}) =& F \\
f_3(\mathbf{x}) =& 0.5(V_{MBP} + V_{FD}) \\
g_1(\mathbf{x}) =& 1.16 - 0.3717x_2x_4 - 0.0092928x_3 \leq 1 \\
g_2(\mathbf{x}) =& 0.261 - 0.0159x_1x_2 - 0.06486x_1 - 0.019x_2x_7 + 0.0144x_3x_5 \\
&+ 0.0154464x_6 \leq 0.32 \\
g_3(\mathbf{x}) =& 0.214 + 0.00817x_5 - 0.045195x_1 - 0.0135168x_1 \\
&+ 0.03099x_2x_6 - 0.018x_2x_7 + 0.007176x_3 \\
&+ 0.023232x_3 - 0.00364x_5x_6 - 0.018x_2^2 \leq 0.32 \\
g_4(\mathbf{x}) =& 0.74 - 0.61x_2 - 0.031296x_3 - 0.031872x_7 + 0.227x_2^2 \leq 0.32 \\
g_5(\mathbf{x}) =& 28.98 + 3.818x_3 - 4.2x_1x_2 + 1.27296x_6 - 2.68065x_7 \leq 32 \\
g_6(\mathbf{x}) =& 33.86 + 2.95x_3 - 5.057x_1x_2 - 3.795x_2 - 3.4431x_7 \\
&+ 1.45728 \leq 32 \\
g_7(\mathbf{x}) =& 46.36 - 9.9x_2 - 4.4505x_1 \leq 32 \\
g_8(\mathbf{x}) \equiv& F = 4.72 - 0.5x_4 - 0.19x_2x_3 \leq 4 \\
g_9(\mathbf{x}) \equiv& V_{MBP} = 10.58 - 0.674x_1x_2 - 0.67275x_2 \leq 9.9 \\
g_{10}(\mathbf{x}) \equiv& V_{FD} = 16.45 - 0.489x_3x_7 - 0.843x_5x_6 \leq 15.7
\end{aligned}
$$

Car side impact problem

Variable bounds are given given as follows:

$$0.5 \leq x_1 \leq 1.5, \quad 0.45 \leq x_2 \leq 1.35, \quad 0.5 \leq x_3 \leq 1.5,$$
$$0.5 \leq x_4 \leq 1.5, \quad 0.875 \leq x_5 \leq 2.625, \quad 0.4 \leq x_6 \leq 1.2,$$
$$0.4 \leq x_7 \leq 1.2$$

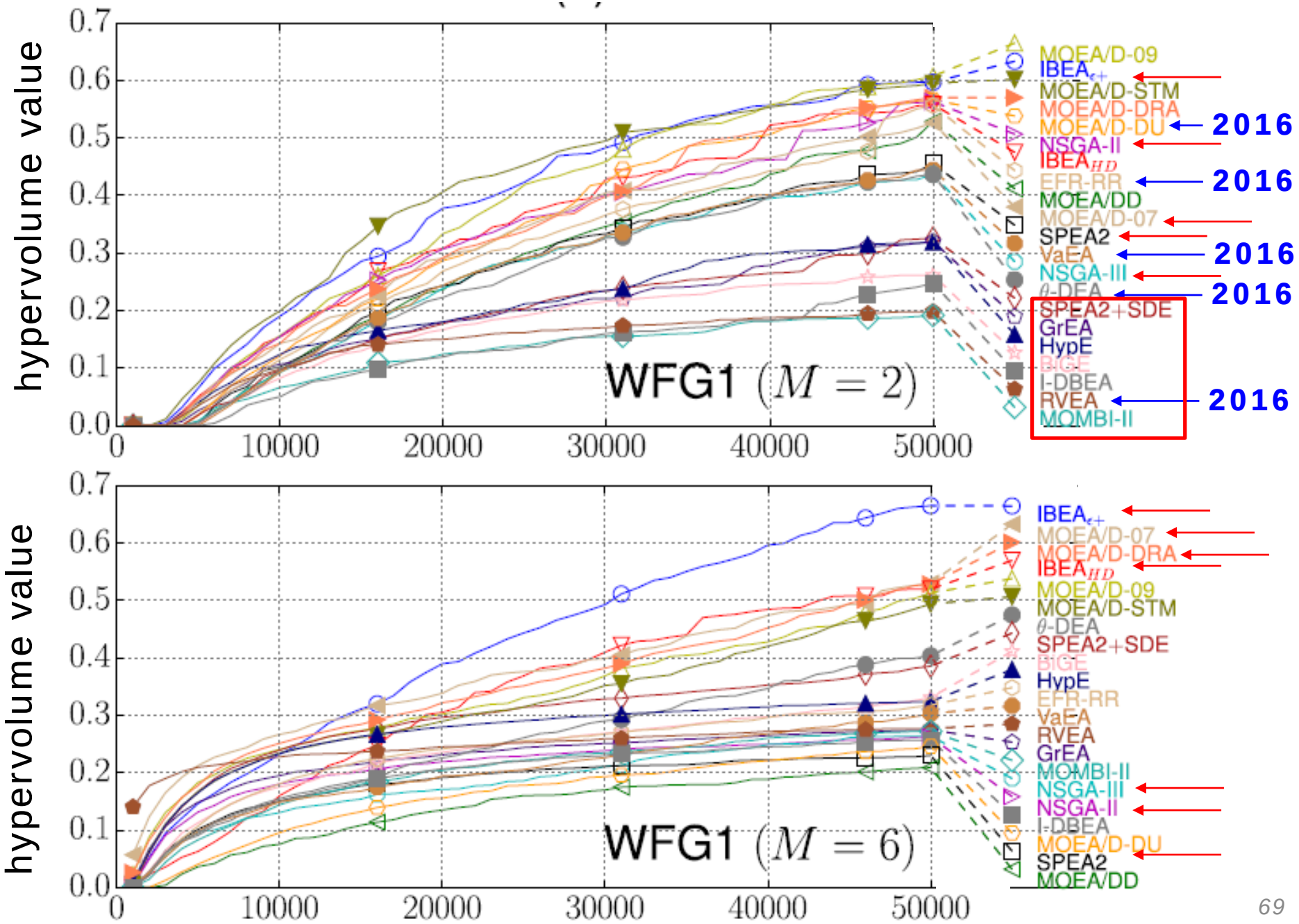
Results of comparison of evolutionary algorithms

1. Comparison using WFG benchmark problems

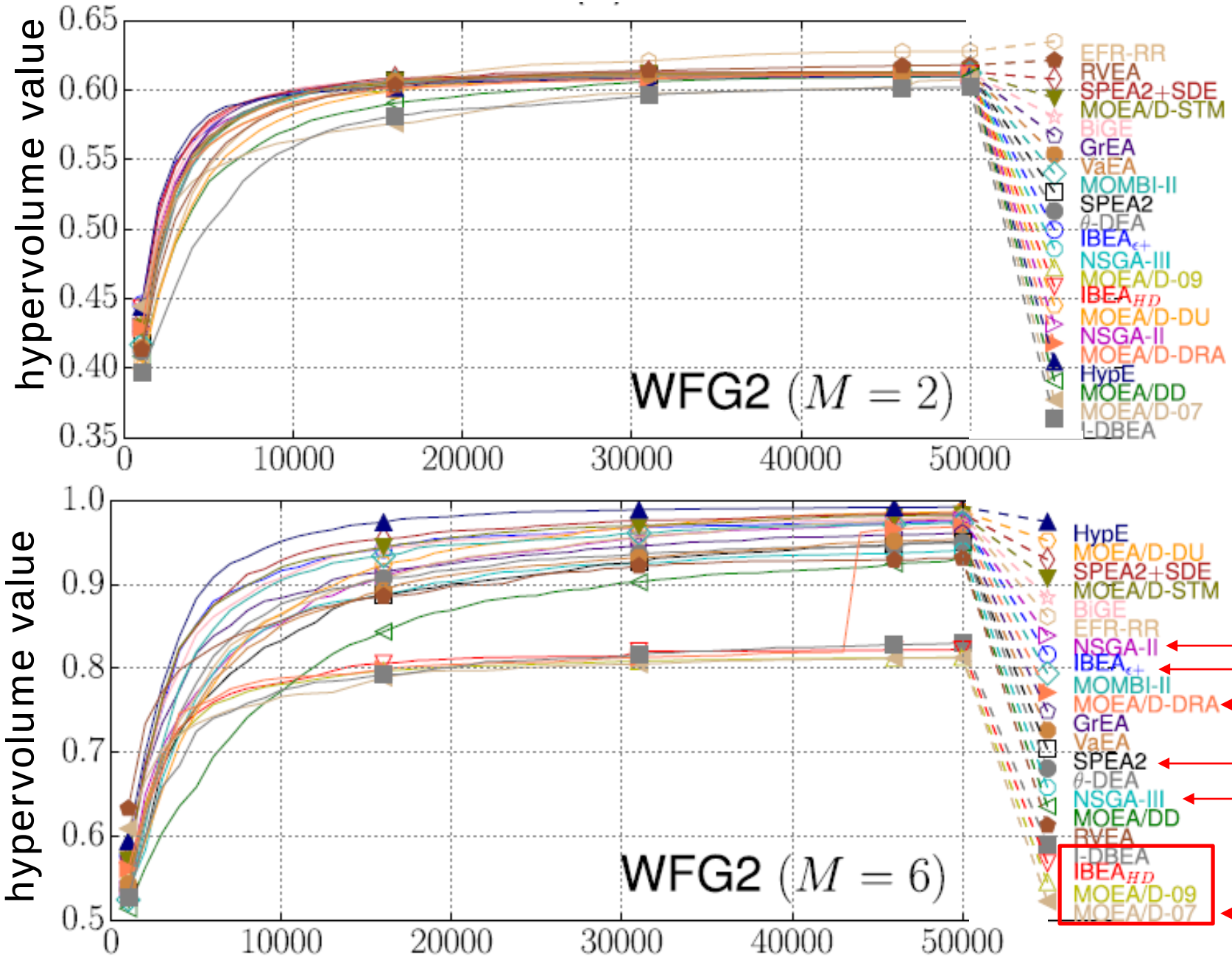
2. Comparison based on anytime performance

3. Comparison of Pareto-optimal solutions obtained in the entire optimization process

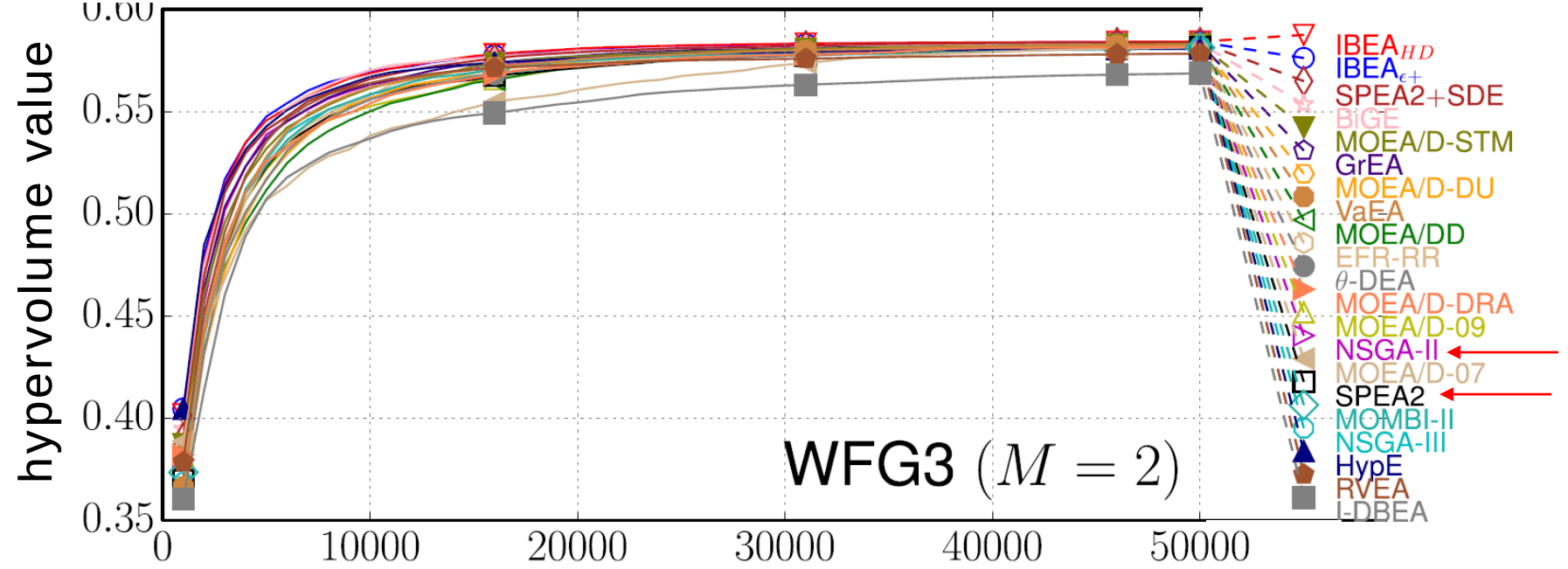
Performance of the MOEAs on the WFG1 problem (M=2 and 6)

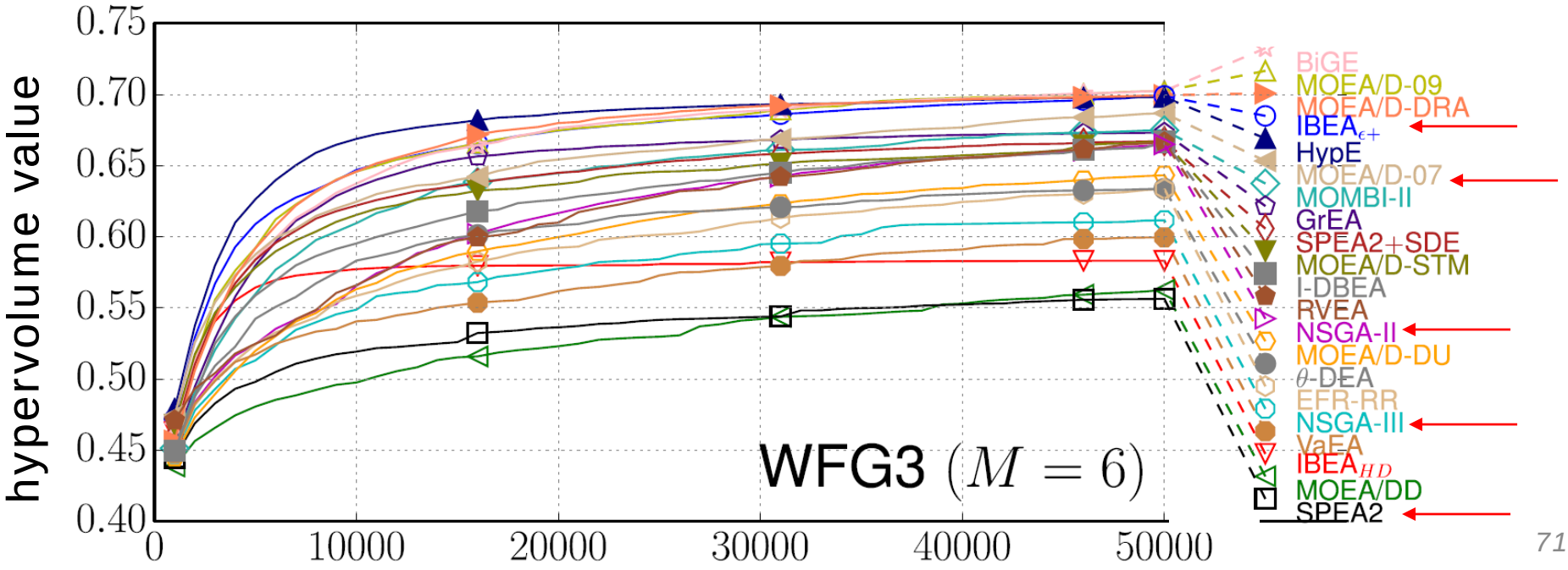


Performance of the MOEAs on the WFG2 problem (M=2 and 6)

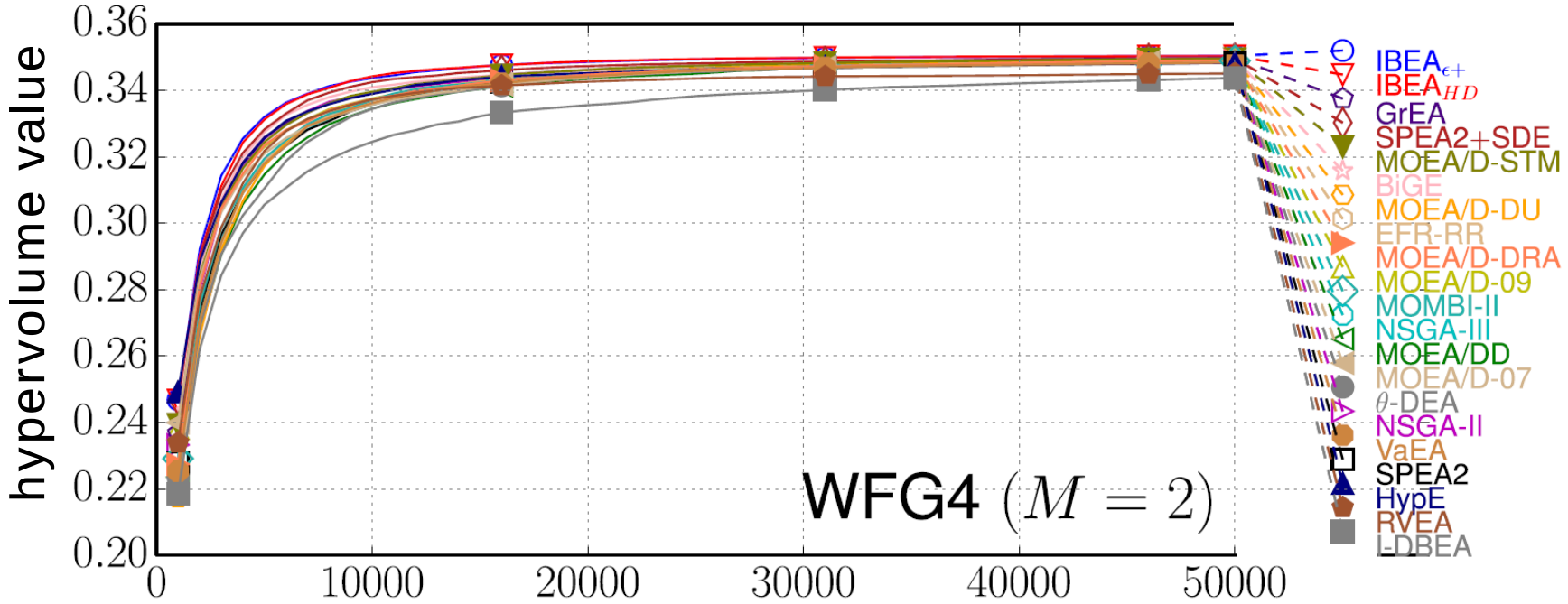


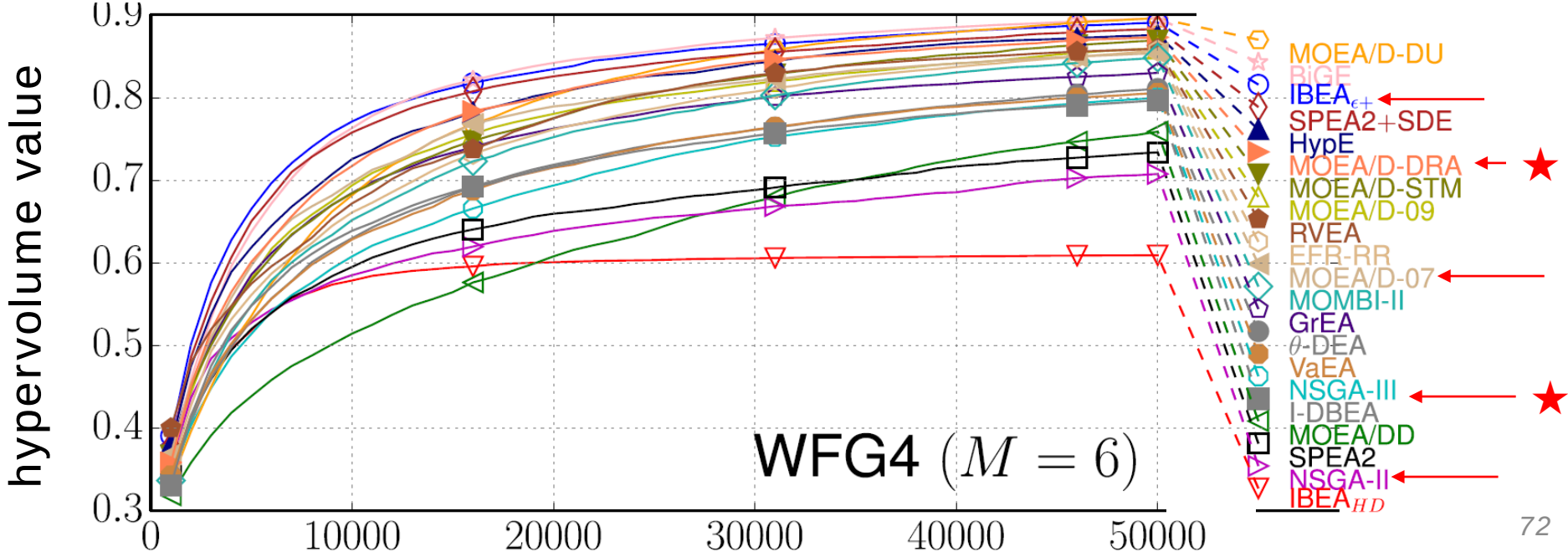
Performance of the MOEAs on the WFG3 problem (M=2 and 6)



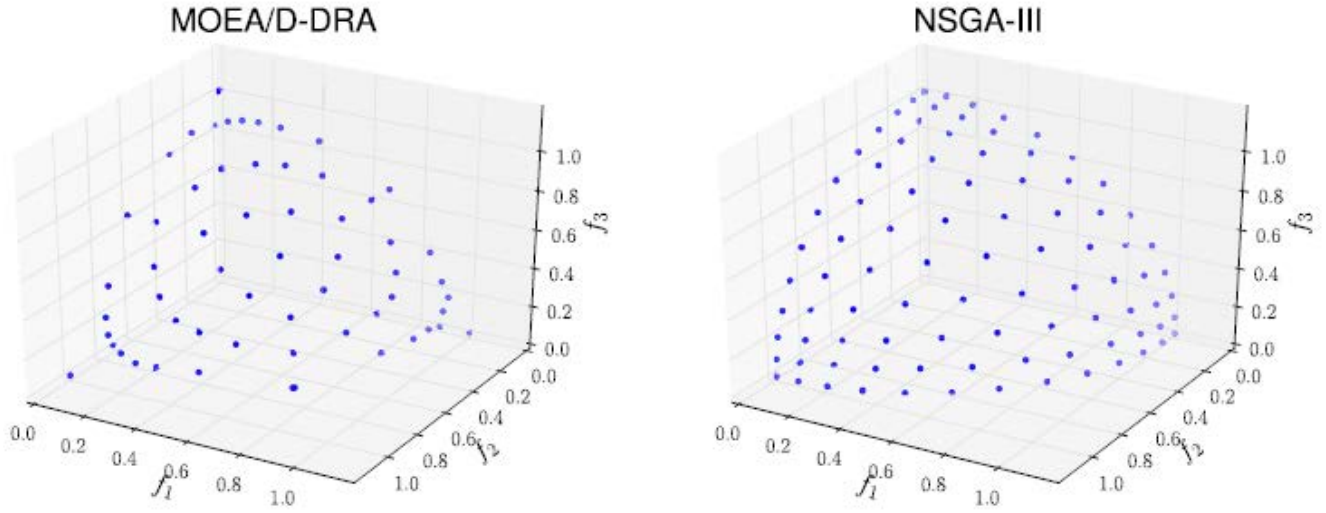


Performance of the MOEAs on the WFG4 problem (M=2 and 6)

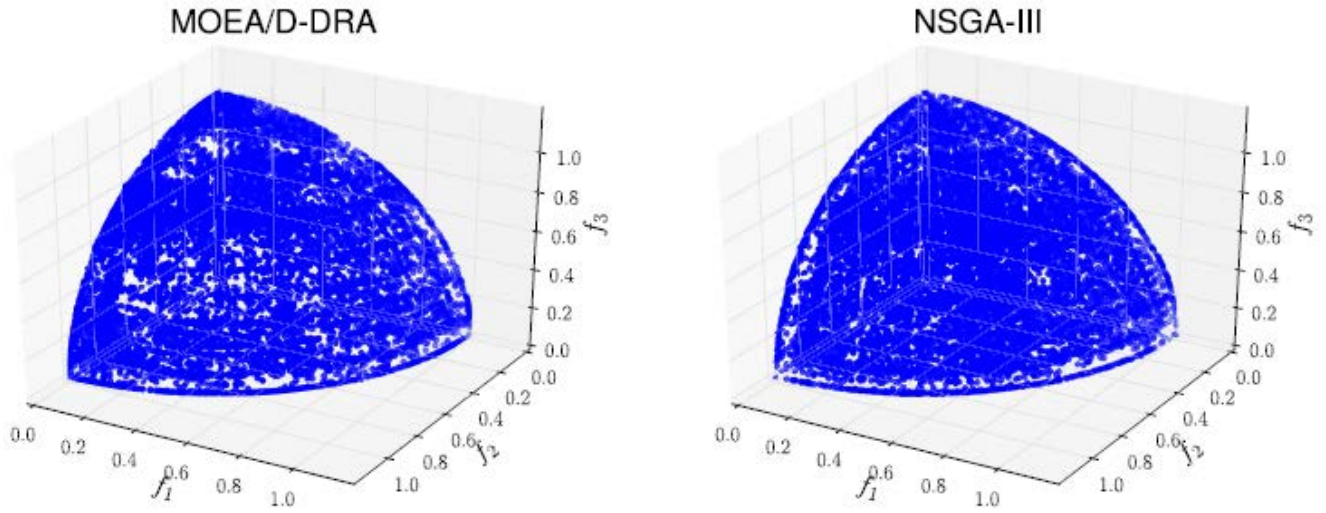




Three-objective WFG4 problem with 50,000 of fitness evaluations

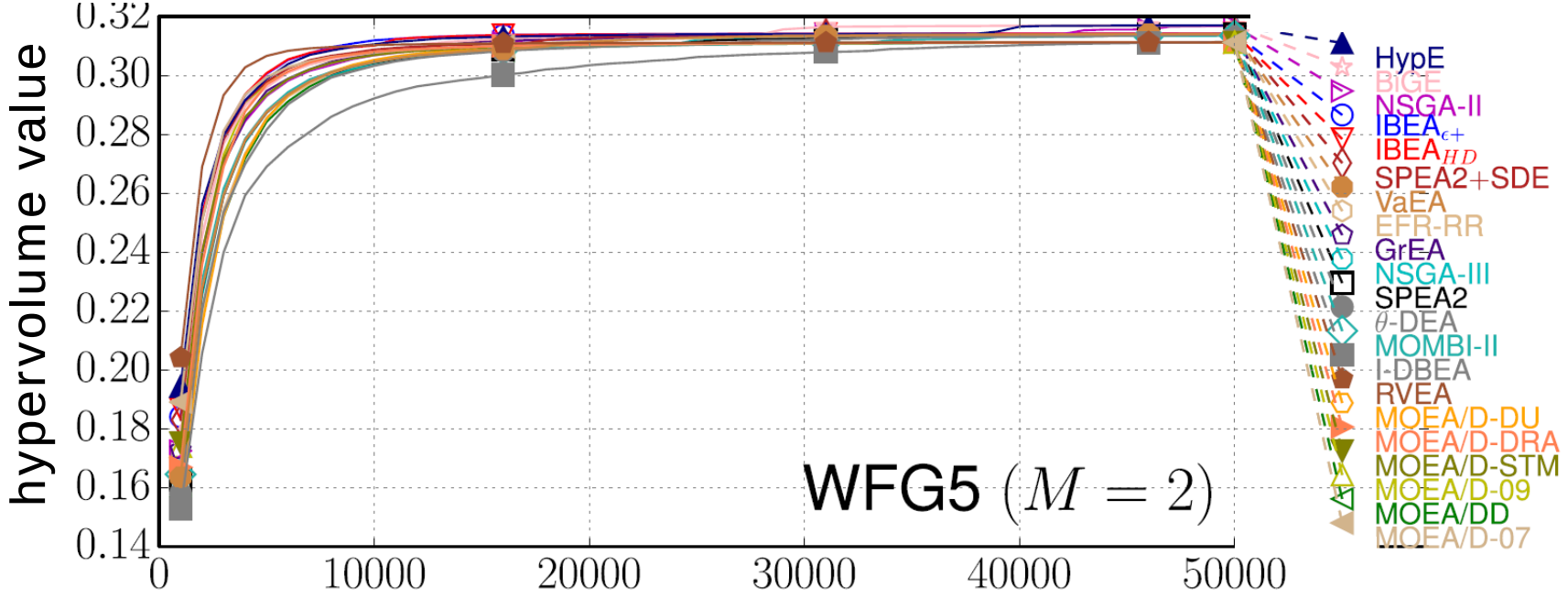


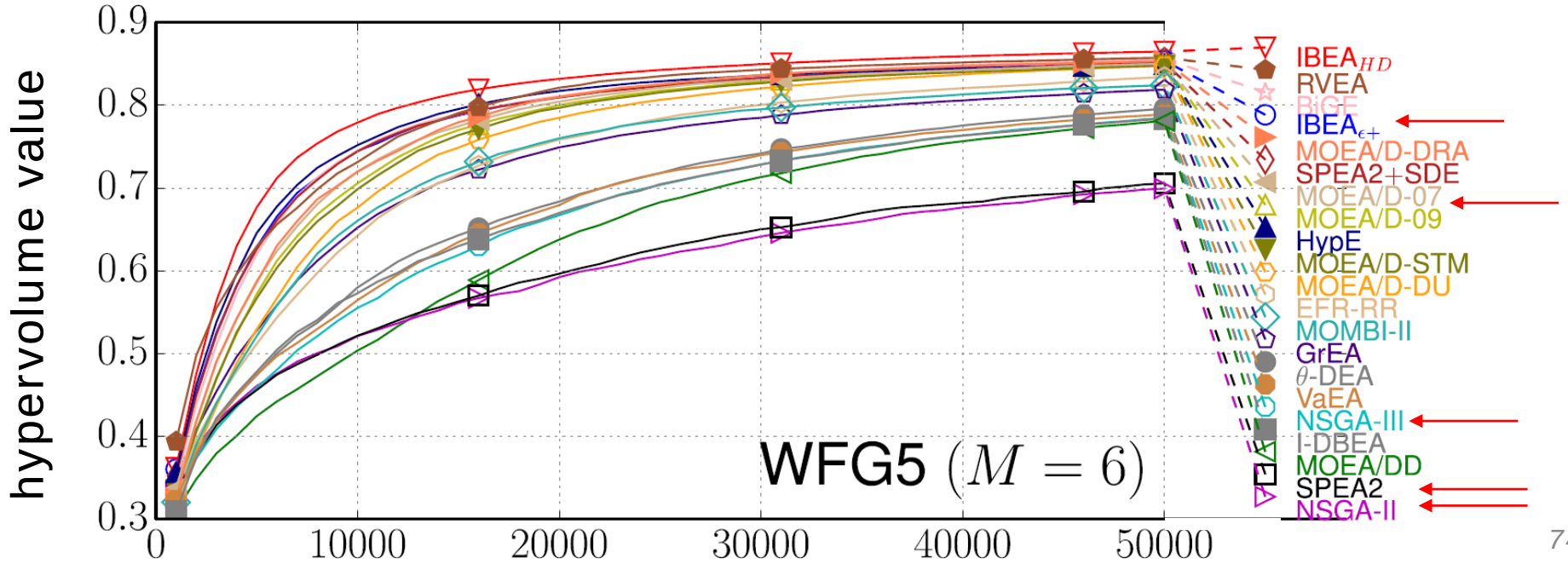
(a) Comparison of Pareto-optimal solutions at final generation



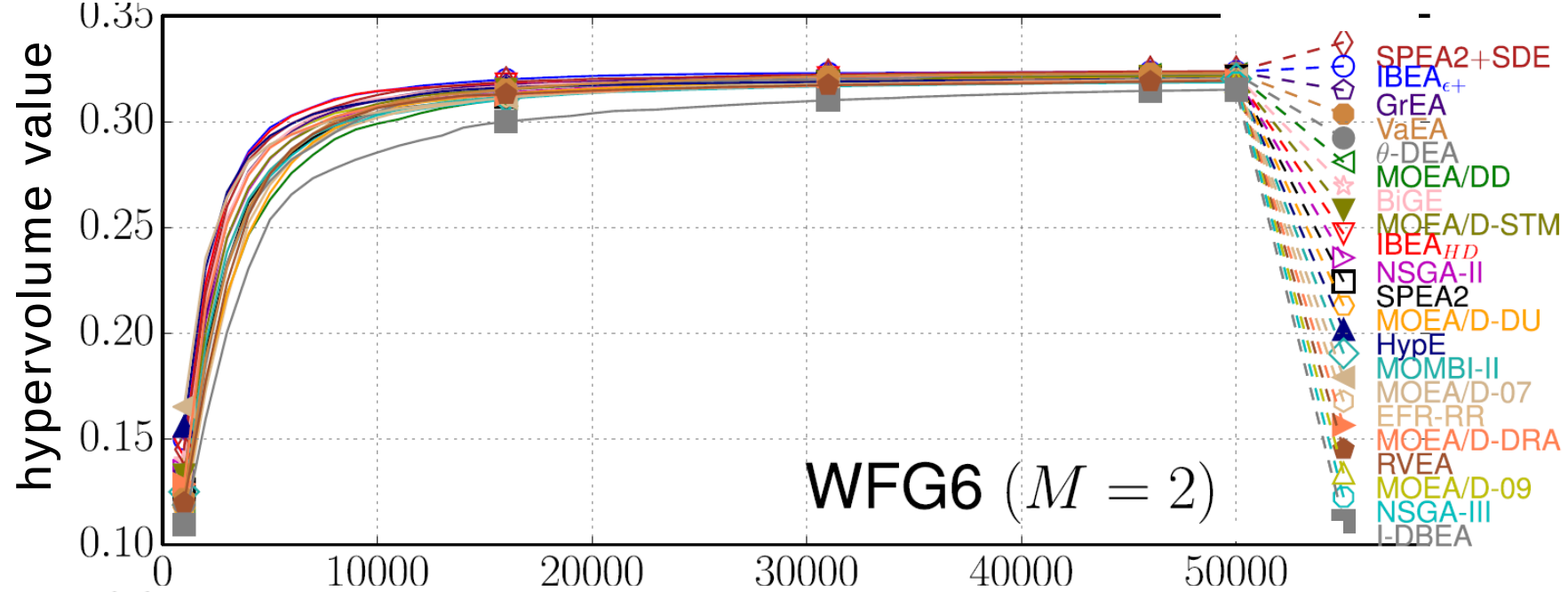
(b) Comparison of Pareto-optimal solutions of unbounded external archives

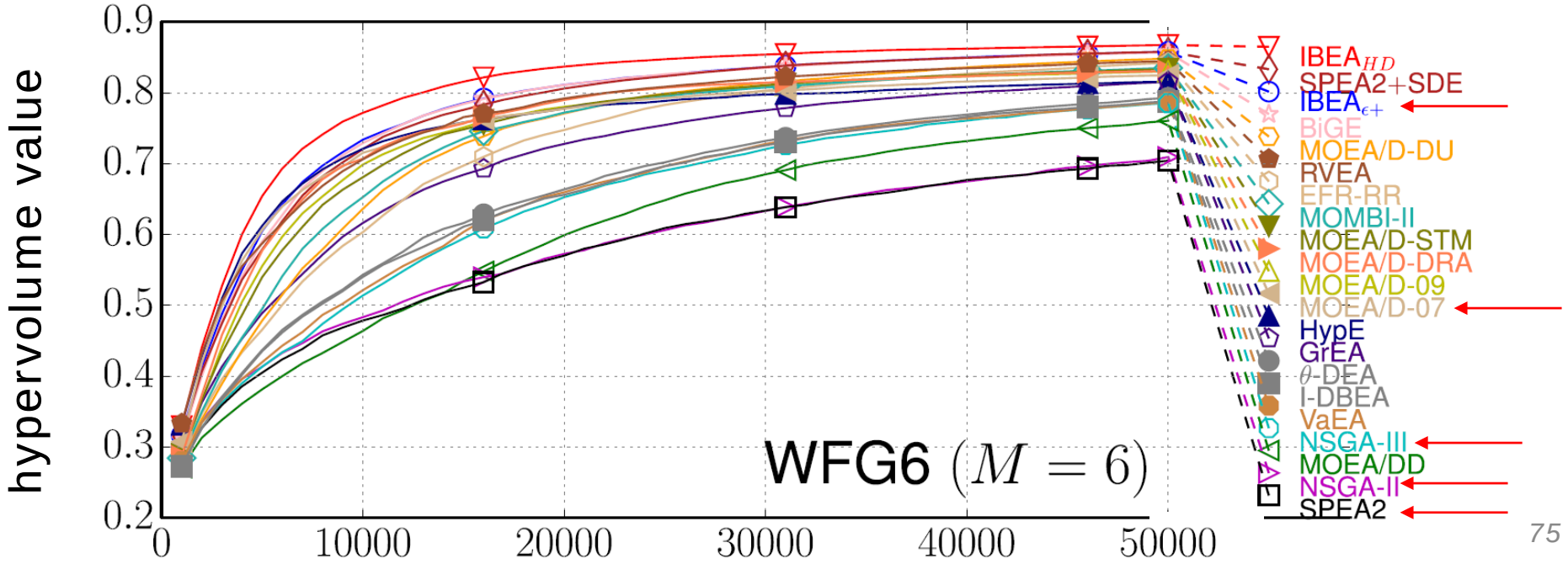
Performance of the MOEAs on the WFG5 problem (M=2 and 6)



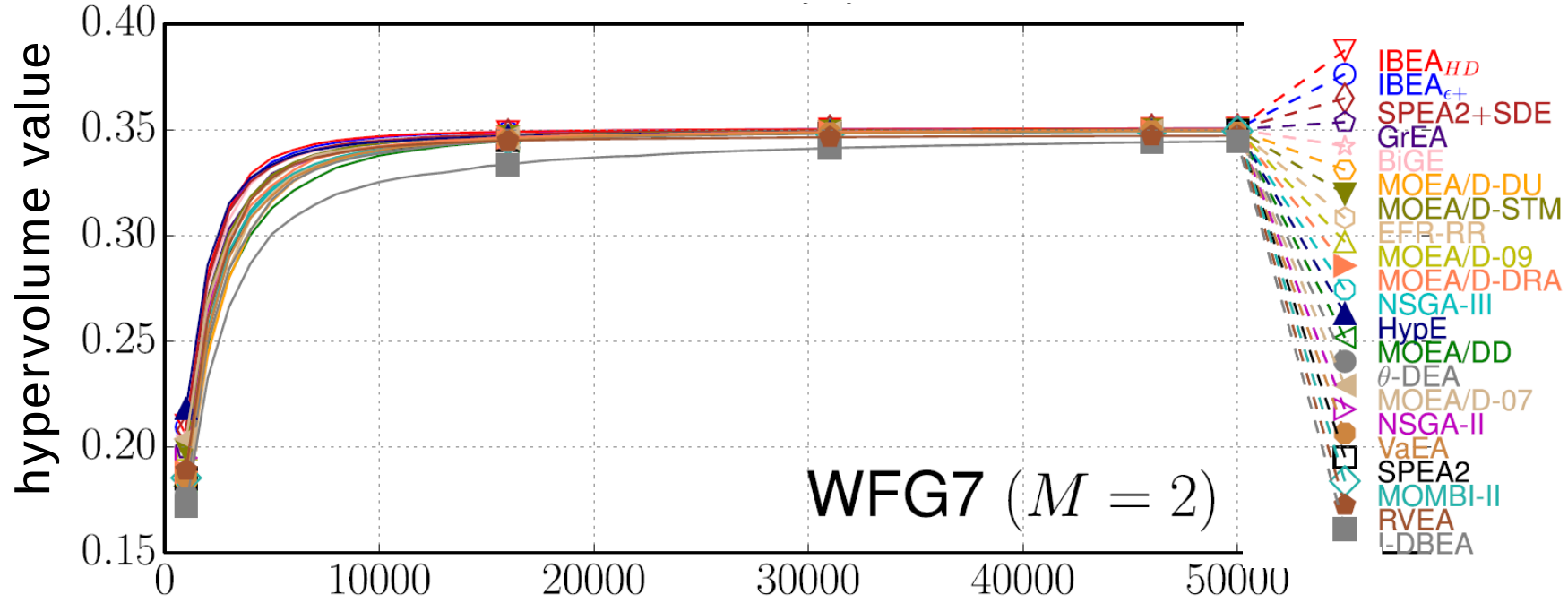


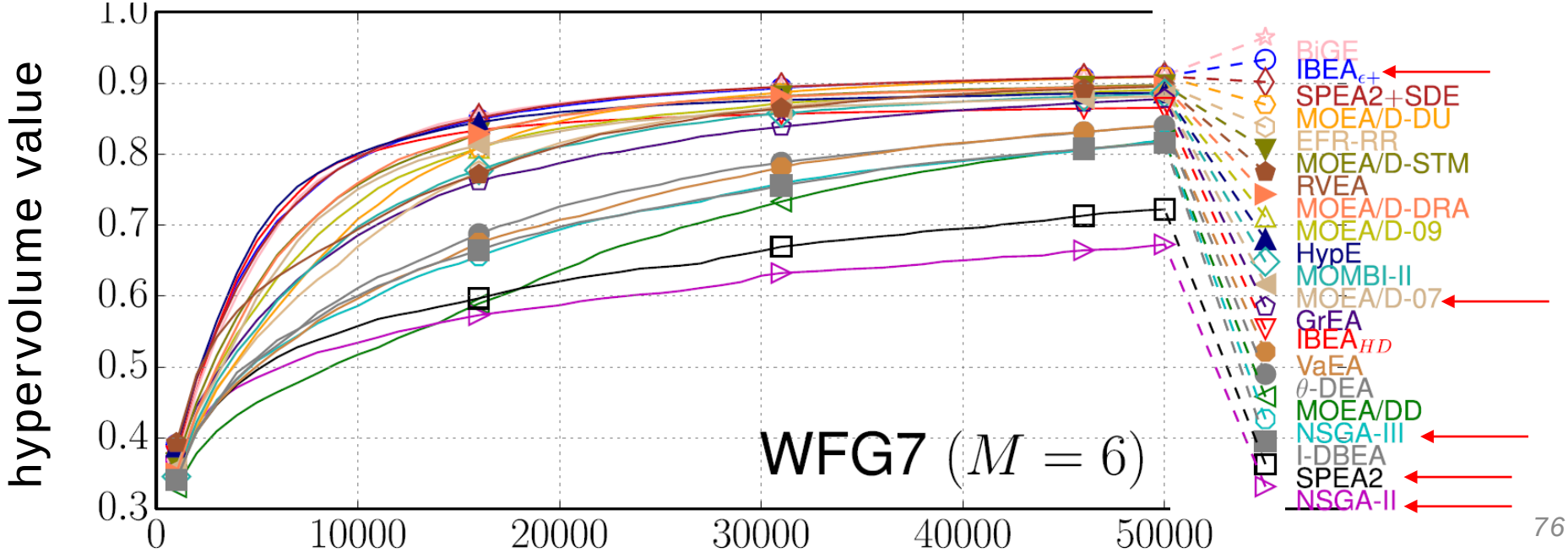
Performance of the MOEAs on the WFG6 problem (M=2 and 6)



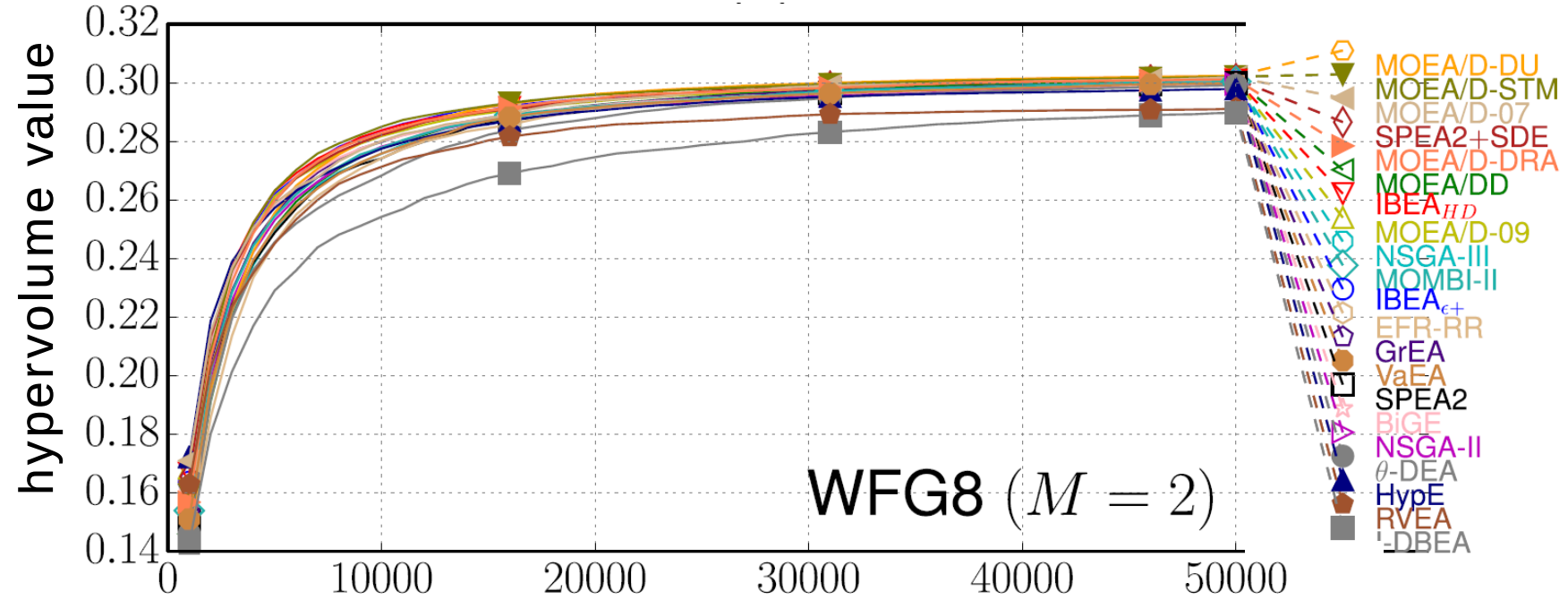


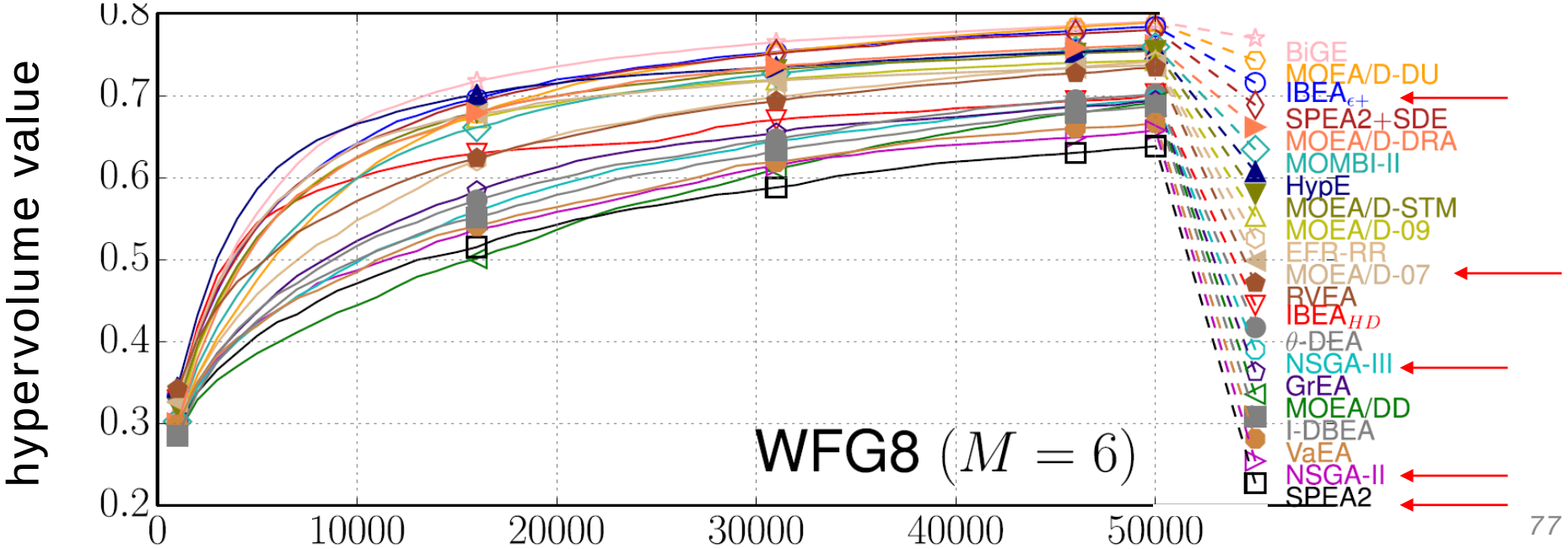
Performance of the MOEAs on the WFG7 problem (M=2 and 6)



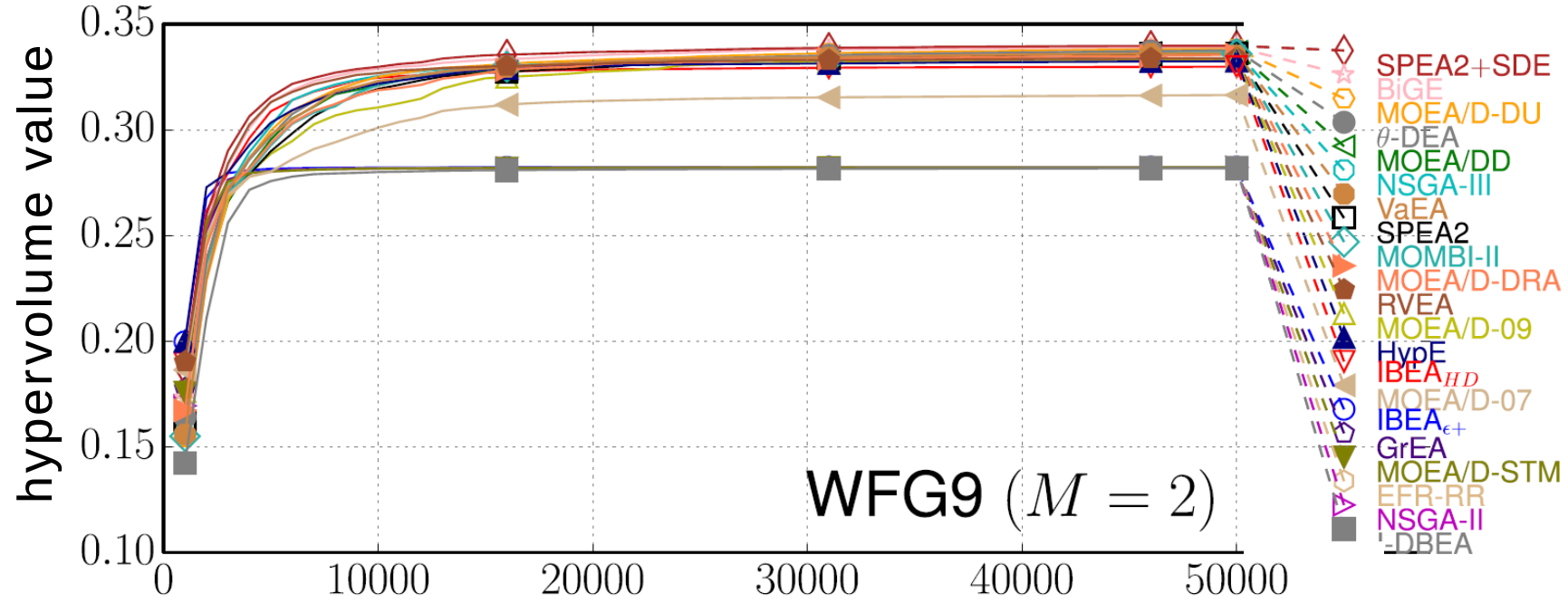


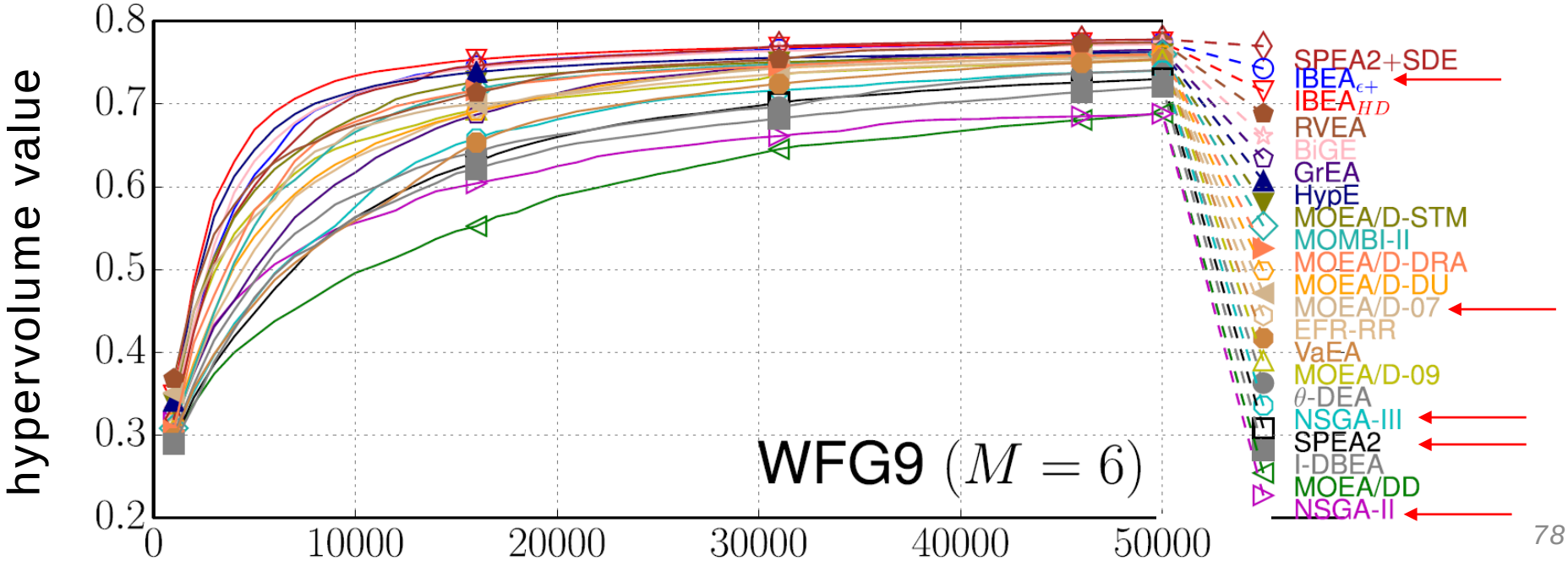
Performance of the MOEAs on the WFG8 problem (M=2 and 6)





Performance of the MOEAs on the WFG9 problem (M=2 and 6)





Conclusions of the comparison

- Nondominance-based algorithms (NSGA-II, SPEA2, ...) works poorly when number of objective functions is greater than 3
- Performance of recently proposed MOEAs (VaEA, θ-DEA, EFR-RR, and RVEA) works poorly when compared with Pareto-optimal solutions obtained in the entire optimization process.
- Evaluation using anytime performance is important

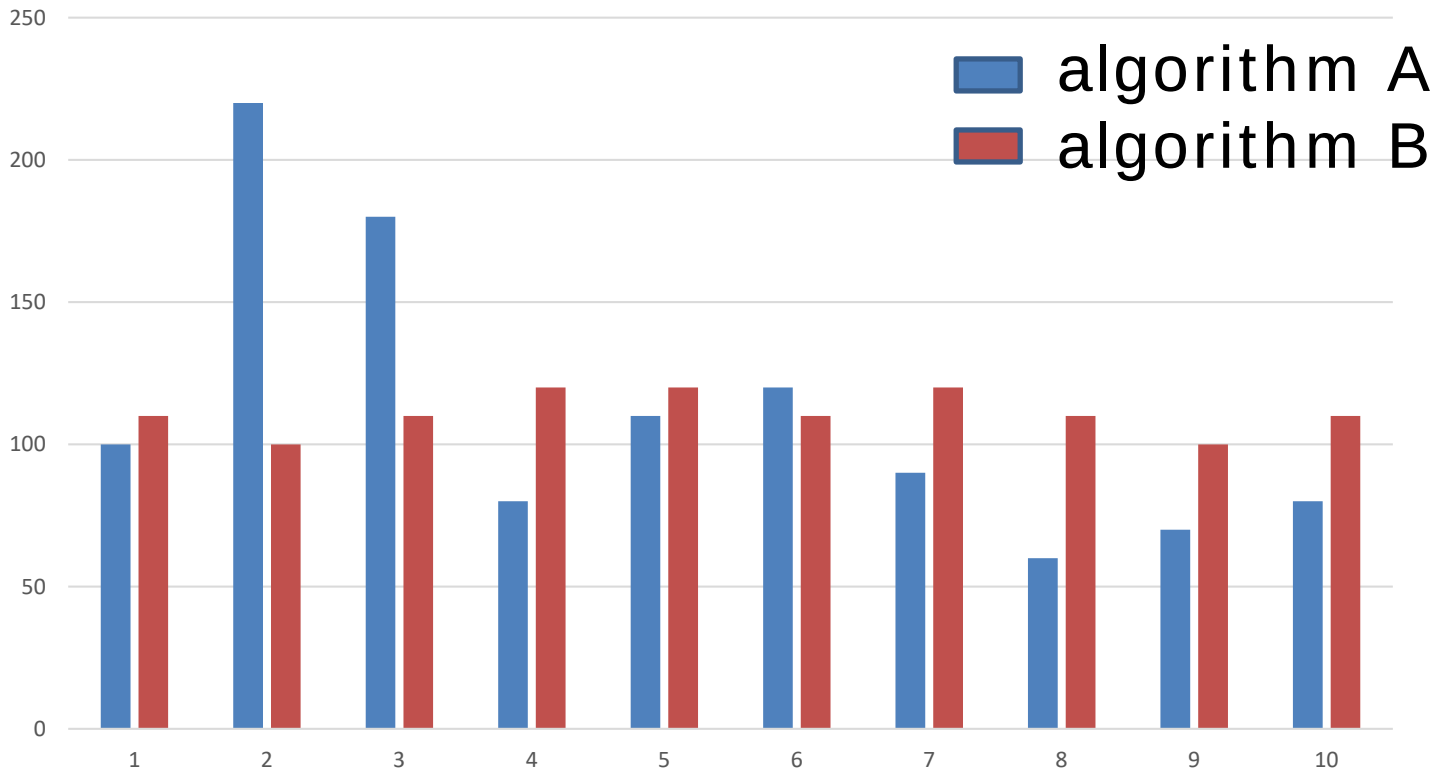
Benchmark problems are valuable in developing MOEAs

ノーフリーランチ定理

No free lunch theorem (NFLT)

コスト関数の極値を探索するあらゆるアルゴリズムは、全ての可能なコスト関数に適用した結果を平均すると同じ性能となる

the computational cost of finding an optimum solution, averaged over all problems in the class, is the same for any solution method.



多項式や三角関数の問題を効率よくアルゴリズムを開発すればいいの？
Is it enough to develop algorithm that efficiently solve problems consisted by polynomials and trigonometric functions?

設計変数の数が多い問題での性能は気にしなくていいの？
Is it enough to develop algorithm that efficiently solve problems with small number of design parameters?

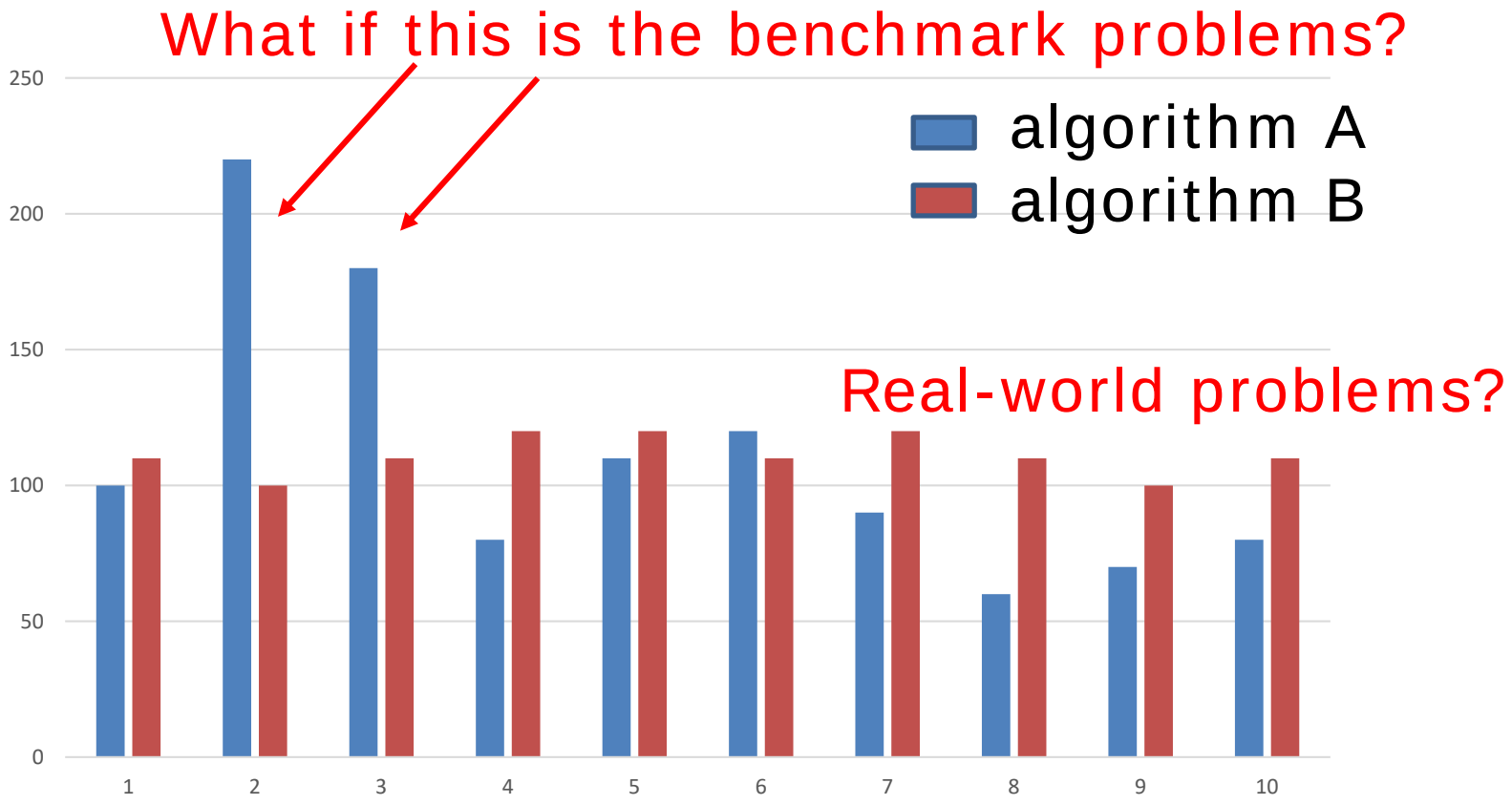
設計制約条件がある問題での性能は気にしなくていいの？
Is it enough to develop algorithm that efficiently solve problems without any design constraints?

ノーフリーランチ定理

No free lunch theorem (NFLT)

コスト関数の極値を探索するあらゆるアルゴリズムは、全ての可能なコスト関数に適用した結果を平均すると同じ性能となる

the computational cost of finding an optimum solution, averaged over all problems in the class, is the same for any solution method.



進化計算学会論文誌 Vol.8 No.1

論　文

応答曲面法を用いた複数車種の同時最適化ベンチマーク問題の提案

Proposal of Simultaneous Design Optimization Benchmark Problem of Multiple Car Structures Using Response Surface Method

小平　剛央
Takehisa Kohira
マツダ株式会社　技術研究所
Technical Research Center, Mazda Motor Corporation
kohira.t@mazda.co.jp

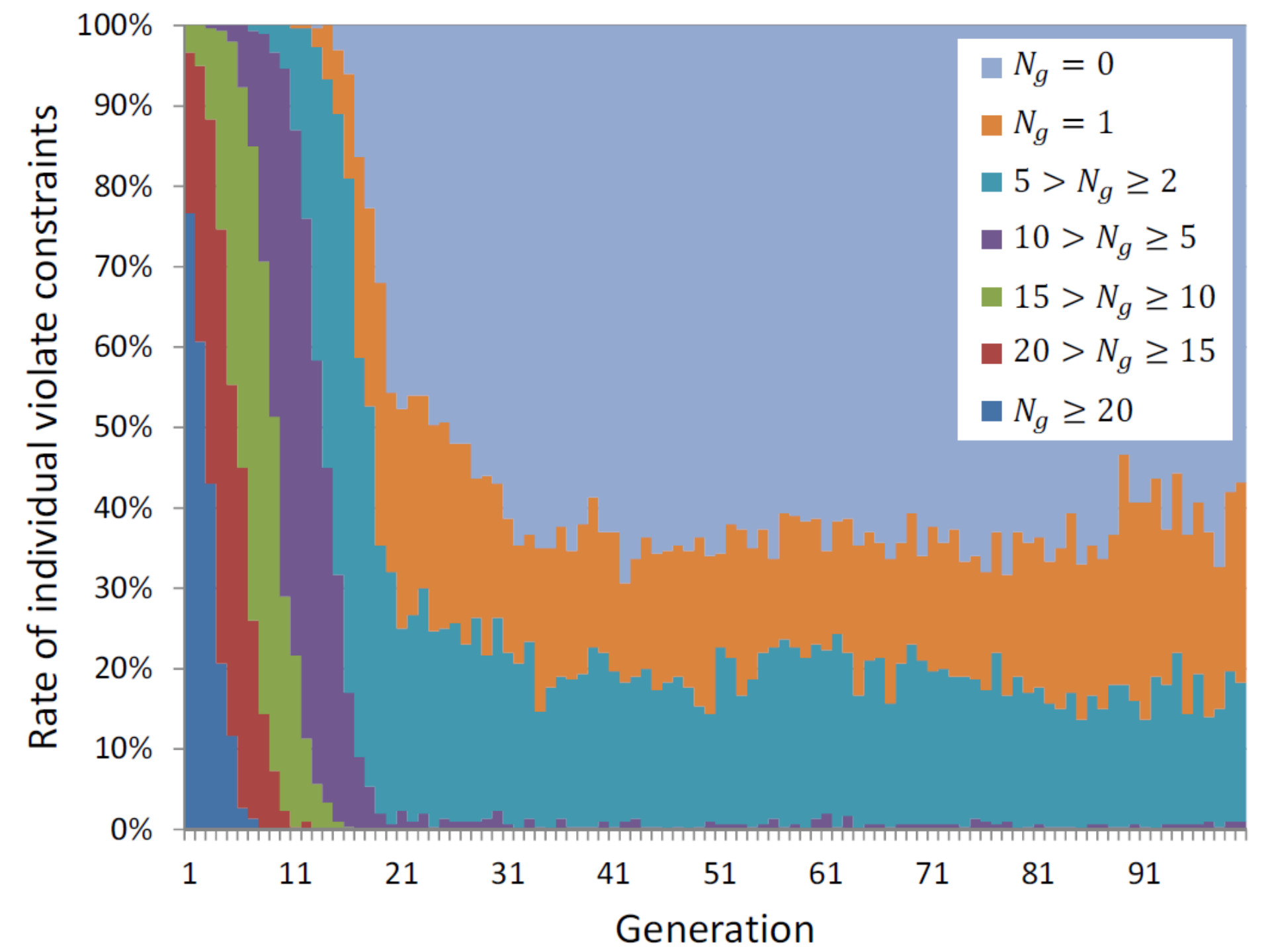
釼持　寛正
Hiromasa Kemmotsu
（同　　上）
kemmotsu.h@mazda.co.jp

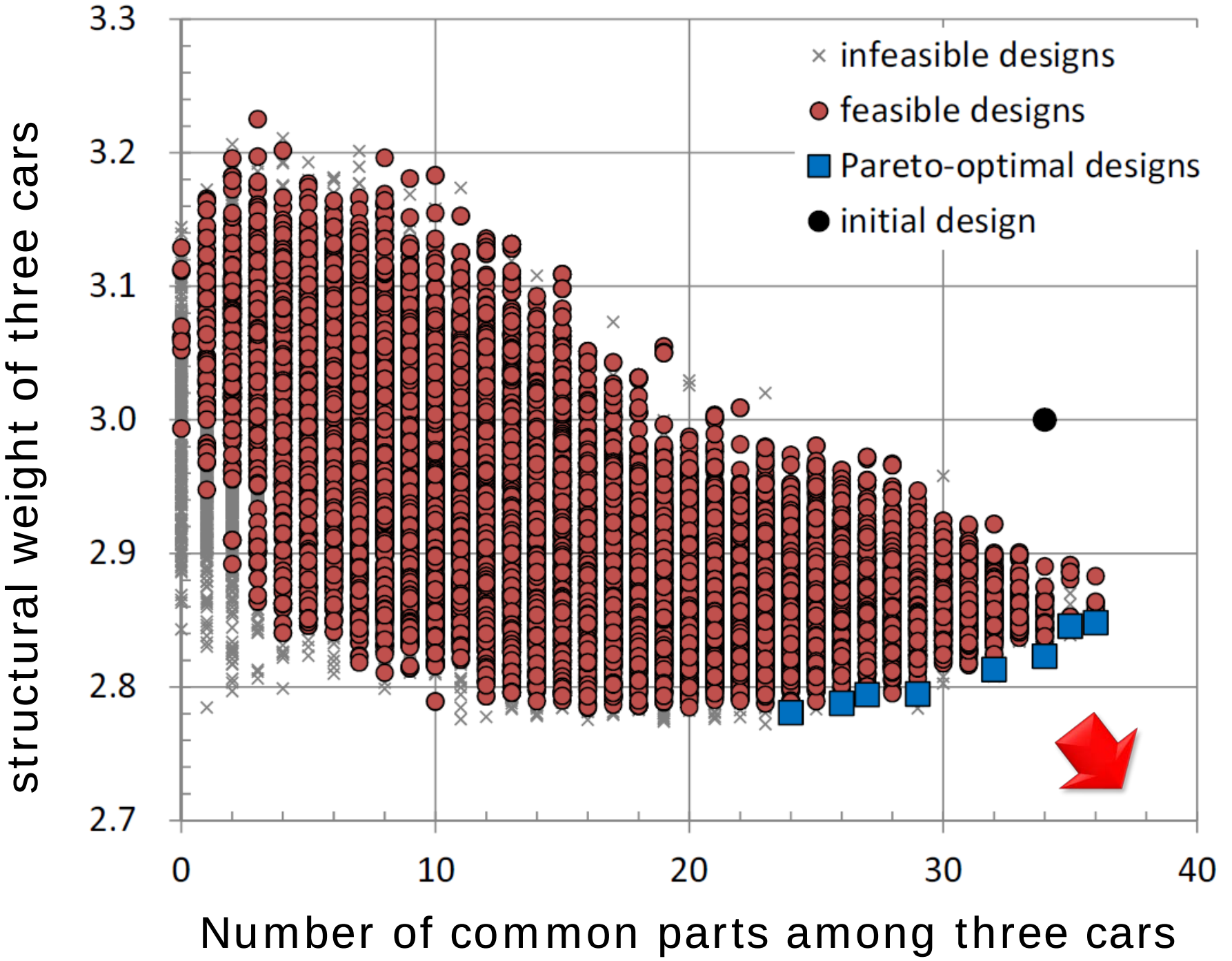
大山　聖
Akira Oyama
宇宙航空研究開発機構 宇宙科学研究所
Institute of Space and Astronautical Science, Japan Aerospace Exploration Agency
oyama@flab.isas.jaxa.jp

立川　智章
Tomoaki Tatsukawa
東京理科大学　情報工学科
Department of Information and Computer Technology, Tokyo University of Science
tatsukawa@rs.tus.ac.jp

Benchmark problem with features of actual engineering design problems is proposed

- The problem bases on a car-body structural development of Mazda motor company
- The problem is constructed with response surface approximation (RBF) of solutions obtained by multiobjective optimization.
- Design objectives are (1) minimization of car structure weight and (2) maximization of number of common thickness parts among three types of cars.
- Number of design variables is 222 (74x3)
- Number of design constraints is 42 (14x3)


Rate of individual violate constraints
100%
90%
80%
70%
60%
50%
40%
30%
20%
10%
0%
1
11
21
31
41
51
61
71
81
91
Generation
$N_g = 0$
$N_g = 1$
$5 > N_g \geq 2$
$10 > N_g \geq 5$
$15 > N_g \geq 10$
$20 > N_g \geq 15$
$N_g \geq 20$


infeasible designs
feasible designs
Pareto-optimal designs
initial design
structural weight of three cars
3.3
3.2
3.1
3.0
2.9
2.8
2.7
0
10
20
30
40
Number of common parts among three cars

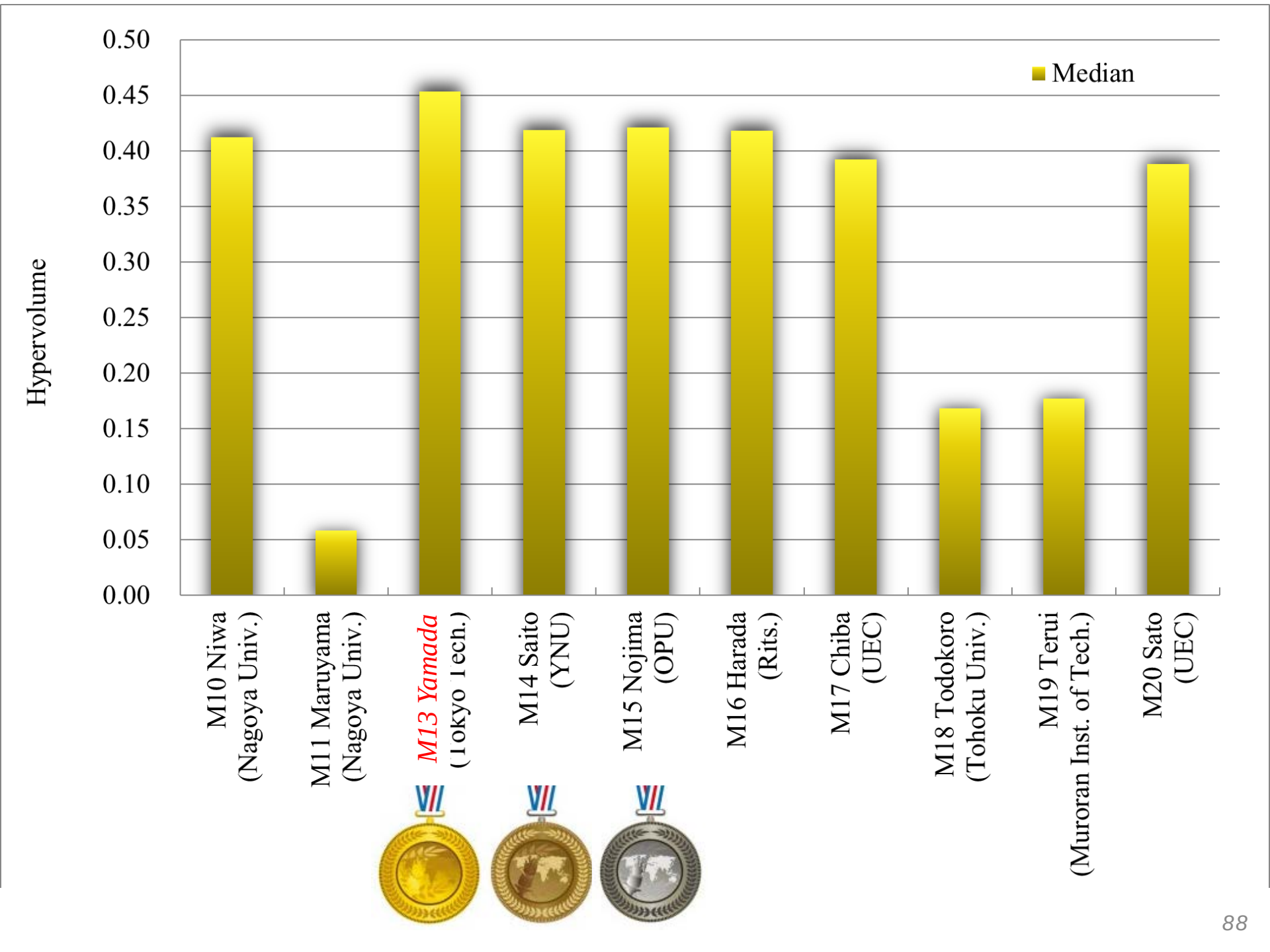
Evolutionary Computation Competition 2017

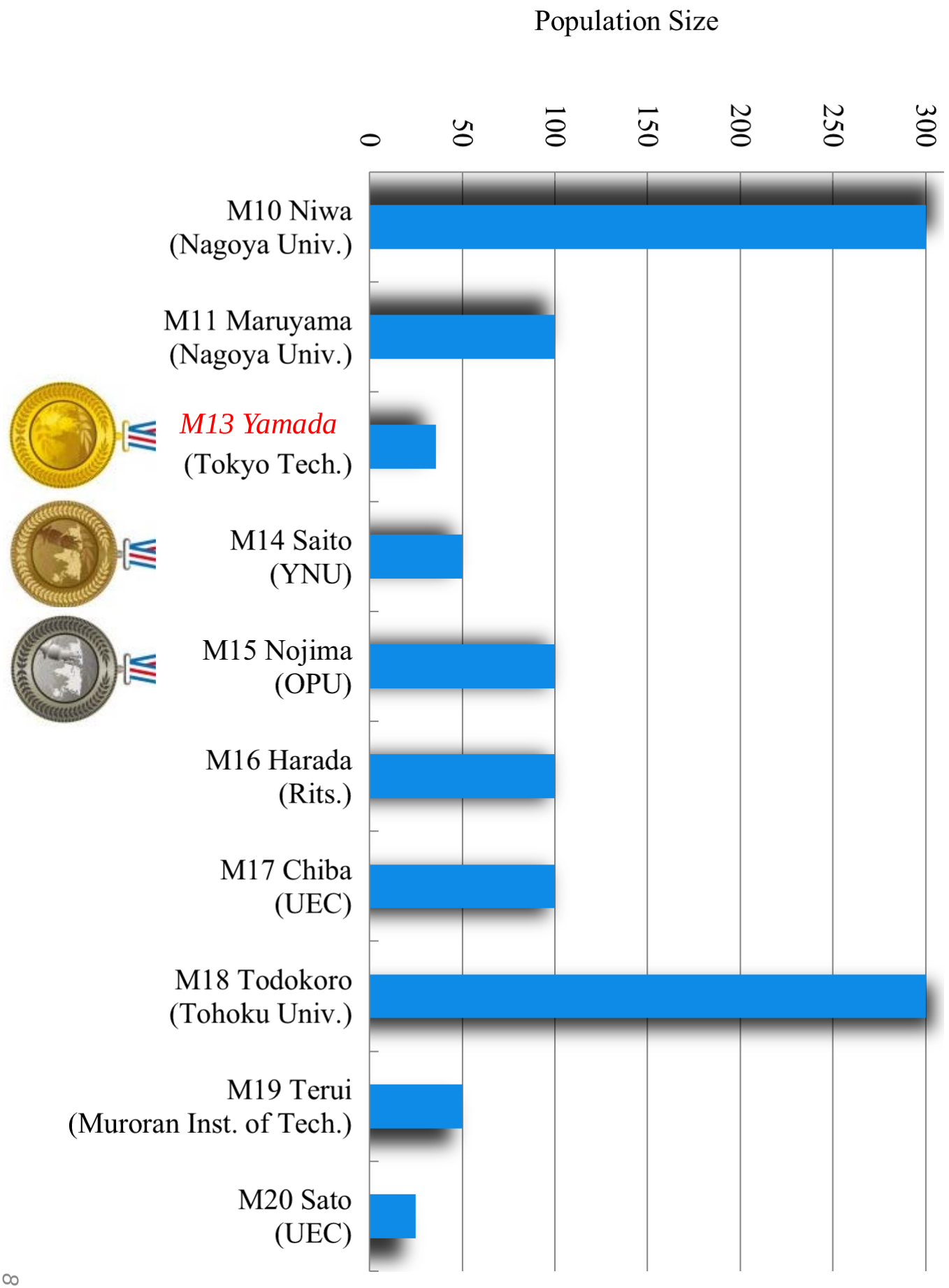
Category 1: Single-objective optimization

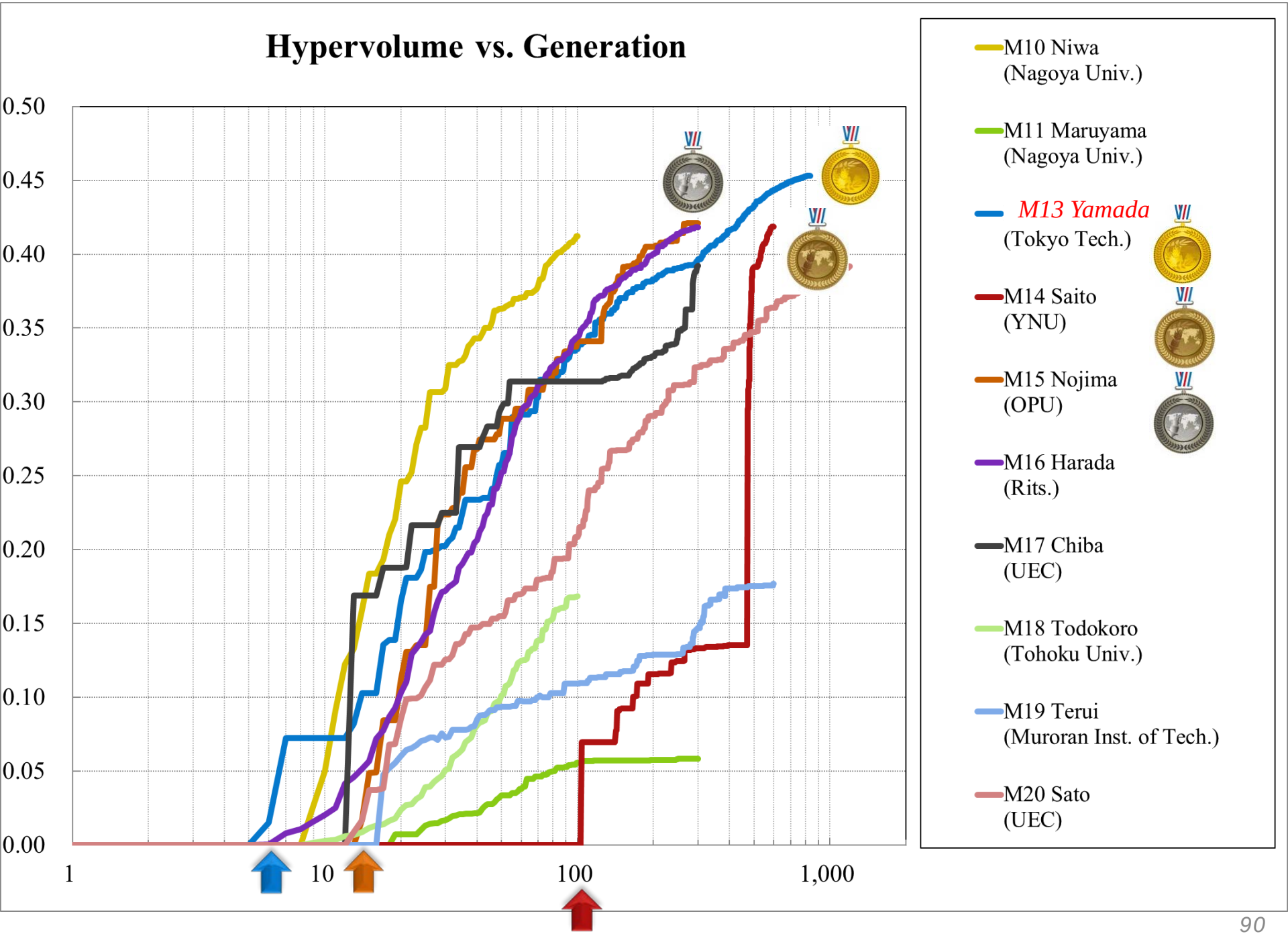
1	肖恒, 山下憂起, 横谷元, 畠中利治	大阪大学
2	丹羽健斗, 吉川大弘	名古屋大学
3	丸山功貴，吉川大弘	名古屋大学
4	開発拓也，渡邉真也	室蘭工業大学
5	大西圭	九州工業大学
6	青木 勇輔，加藤 遊馬，山田 大輝，小野 功	東京工業大学
7	内田絢斗, 斉藤翔汰, 白川真一	横浜国立大学
8	余俊, 高木英行	九州大学
9	串田淳一	広島市立大学

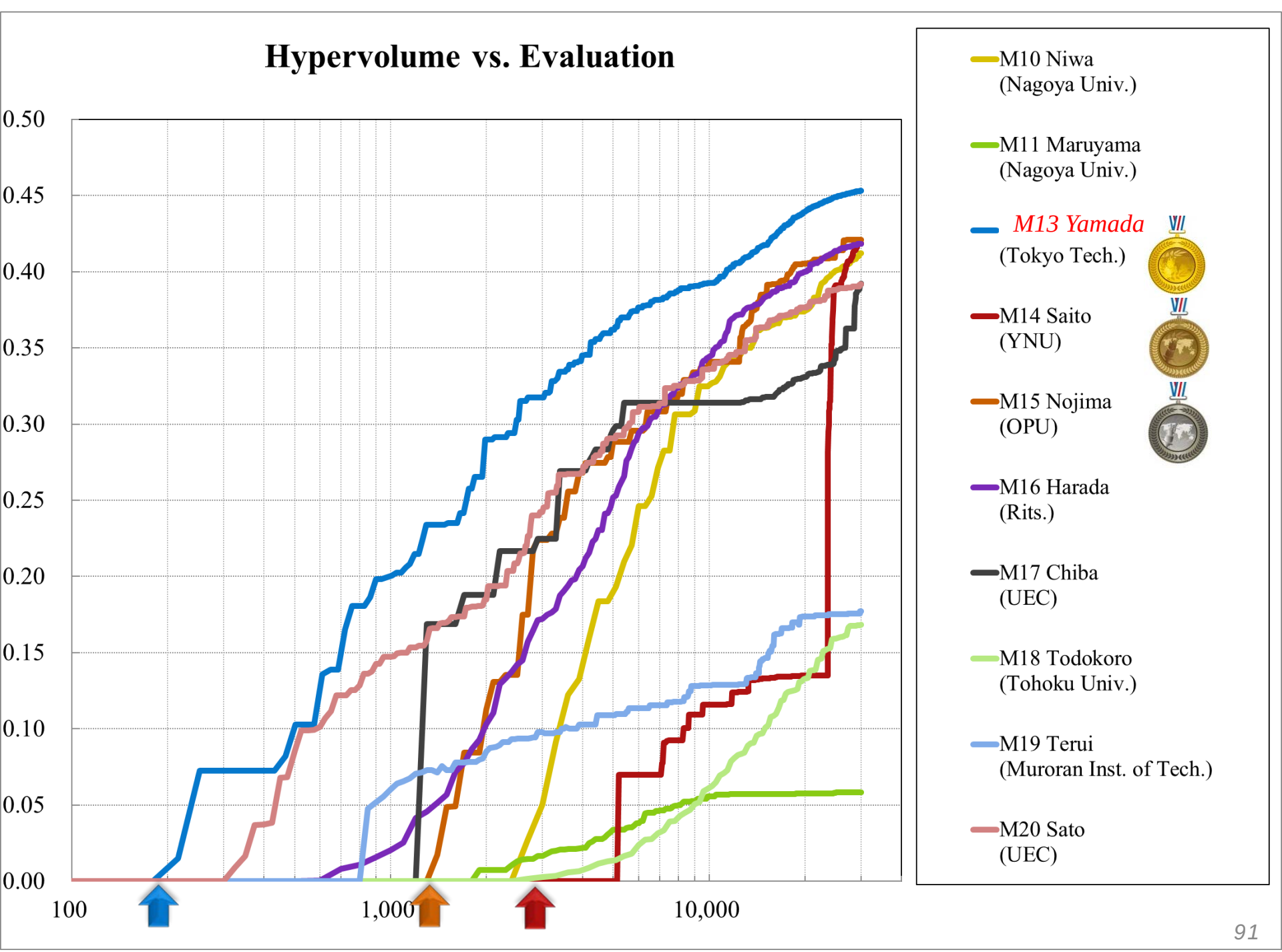
Category 2: Multi-objective optimization

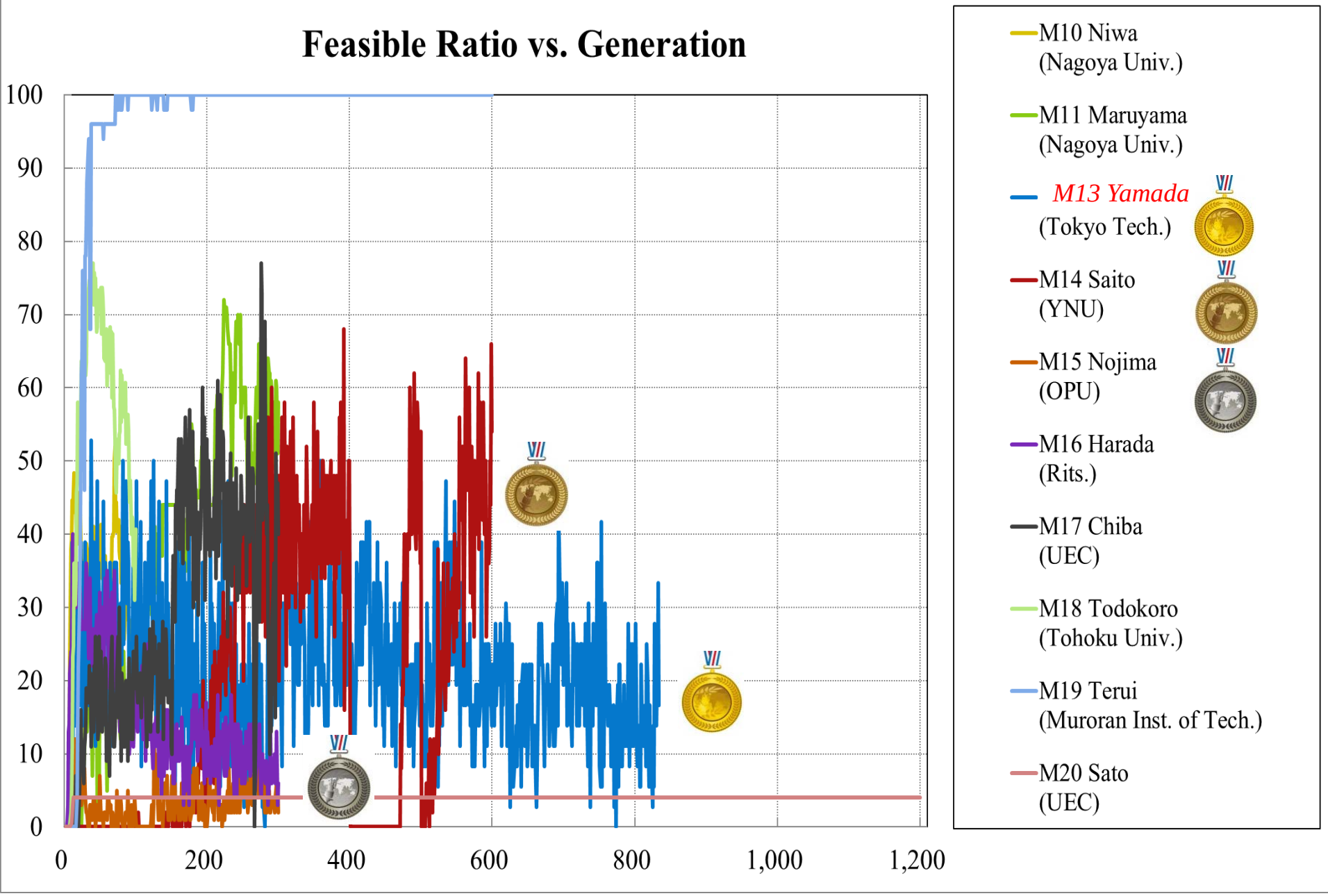
10	丹羽健斗, 吉川大弘	名古屋大学
11	丸山功貴，吉川大弘	名古屋大学
12	キャンセル	
13	山田 大輝，加藤 遊馬，青木 勇輔，小野 功	東京工業大学
14	斉藤翔汰, 内田絢斗, 白川真一	横浜国立大学
15	能島裕介	大阪府立大学
16	原田智広	立命館大学
17	住元剛志，千葉一永	電気通信大学
18	外処凌雲，下山幸治	東北大学
19	照井勇輔，渡邉真也	室蘭工業大学
20	佐藤寛之	電気通信大学

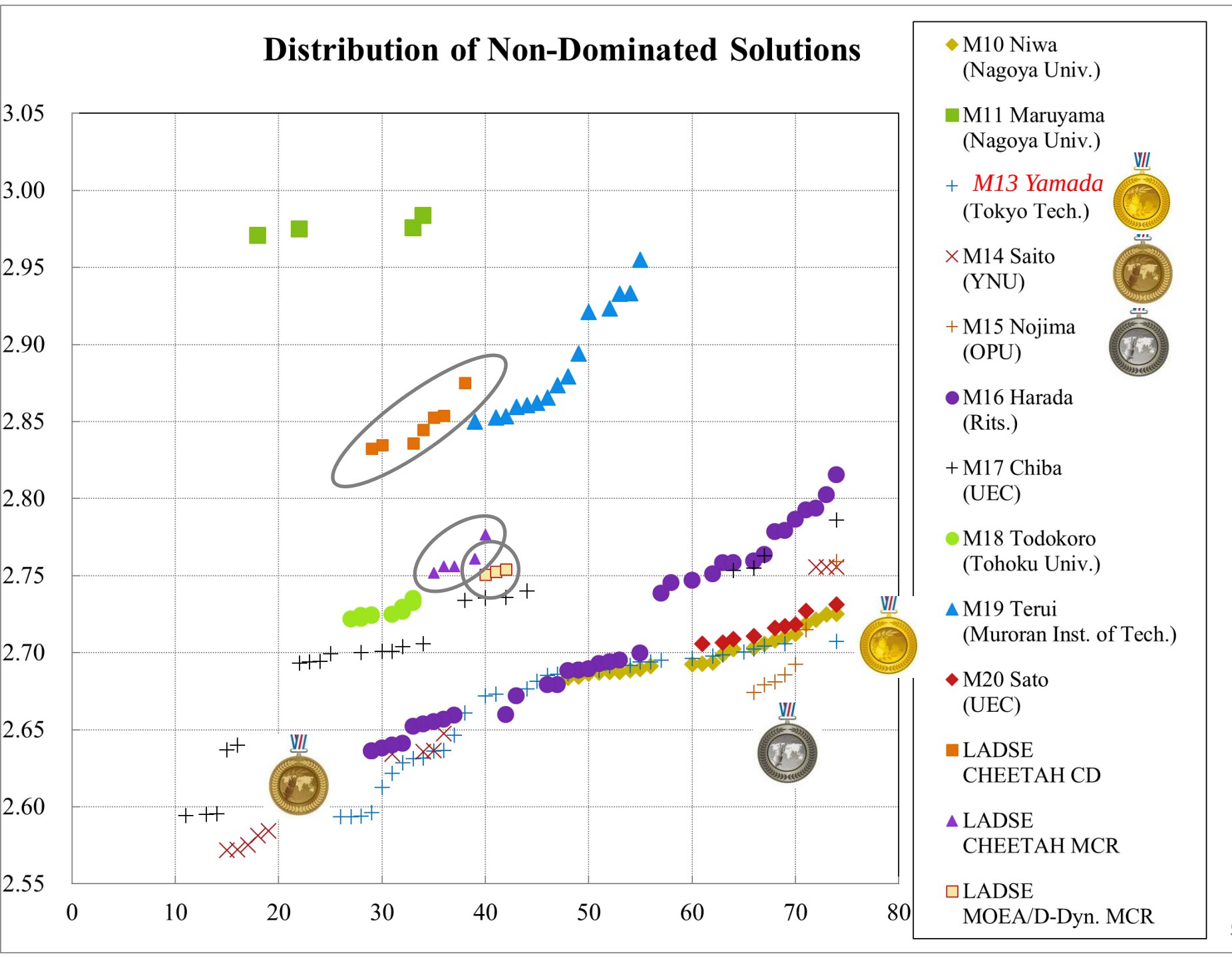

Hypervolume
0.50
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Median
M10 Niwa (Nagoya Univ.)
M11 Maruyama (Nagoya Univ.)
M13 Yamada (Tokyo Tech.)
M14 Saito (YNU)
M15 Nojima (OPU)
M16 Harada (Rits.)
M17 Chiba (UEC)
M18 Todokoro (Tohoku Univ.)
M19 Terui (Muroran Inst. of Tech.)
M20 Sato (UEC)


Population Size
0
50
100
150
200
250
300
M10 Niwa
(Nagoya Univ.)
M11 Maruyama
(Nagoya Univ.)
M13 Yamada
(Tokyo Tech.)
M14 Saito
(YNU)
M15 Nojima
(OPU)
M16 Harada
(Rits.)
M17 Chiba
(UEC)
M18 Todokoro
(Tohoku Univ.)
M19 Terui
(Muroran Inst. of Tech.)
M20 Sato
(UEC)


Hypervolume vs. Generation
0.50
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
1
10
100
1,000
M10 Niwa
(Nagoya Univ.)
M11 Maruyama
(Nagoya Univ.)
M13 Yamada
(Tokyo Tech.)
M14 Saito
(YNU)
M15 Nojima
(OPU)
M16 Harada
(Rits.)
M17 Chiba
(UEC)
M18 Todokoro
(Tohoku Univ.)
M19 Terui
(Muroran Inst. of Tech.)
M20 Sato
(UEC)


Hypervolume vs. Evaluation
0.50
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
100
1,000
10,000
M10 Niwa
(Nagoya Univ.)
M11 Maruyama
(Nagoya Univ.)
M13 Yamada
(Tokyo Tech.)
M14 Saito
(YNU)
M15 Nojima
(OPU)
M16 Harada
(Rits.)
M17 Chiba
(UEC)
M18 Todokoro
(Tohoku Univ.)
M19 Terui
(Muroran Inst. of Tech.)
M20 Sato
(UEC)


Feasible Ratio vs. Generation
100
90
80
70
60
50
40
30
20
10
0
0
200
400
600
800
1,000
1,200
M10 Niwa
(Nagoya Univ.)
M11 Maruyama
(Nagoya Univ.)
M13 Yamada
(Tokyo Tech.)
M14 Saito
(YNU)
M15 Nojima
(OPU)
M16 Harada
(Rits.)
M17 Chiba
(UEC)
M18 Todokoro
(Tohoku Univ.)
M19 Terui
(Muroran Inst. of Tech.)
M20 Sato
(UEC)


Distribution of Non-Dominated Solutions
3.05
3.00
2.95
2.90
2.85
2.80
2.75
2.70
2.65
2.60
2.55
0
10
20
30
40
50
60
70
80
M10 Niwa (Nagoya Univ.)
M11 Maruyama (Nagoya Univ.)
M13 Yamada (Tokyo Tech.)
M14 Saito (YNU)
M15 Nojima (OPU)
M16 Harada (Rits.)
M17 Chiba (UEC)
M18 Todokoro (Tohoku Univ.)
M19 Terui (Muroran Inst. of Tech.)
M20 Sato (UEC)
LADSE CHEETAH CD
LADSE CHEETAH MCR
LADSE MOEA/D-Dyn. MCR

Analysis of results

- Smaller population size may have advantage
- Algorithms strongly search for feasible region tends to convergence prematurely?
- HV does not guarantee uniform distribution
- Understanding the problem before optimization and Parameter tuning is important to get good Pareto-optimal designs (dilemma of real-world design optimization….)

制約条件取り扱い法
constraint handling techniques

constraint handling techniques

- Penalty approach
- Separatist approach
- Repair approach
- Hybrid approach

constraint handling techniques

- Penalty approach

Transform a constrained optimization problem into an unconstrained optimization problem

minimize $F(\vec{x}^i) = f(\vec{x}^i) + \alpha \cdot cv(\vec{x}^i)$

where $cv(\vec{x}^i)$ is constraint violation of $\vec{x}^i$

$cv(\vec{x}^i)$ is zero when all constraints are satisfied

if multiple constraints are involved

$$cv(\vec{x}^i) = a_1 cv_1(\vec{x}^i) + a_2 cv_2(\vec{x}^i) + \ldots$$

α and a_j need to be tuned.

constraint handling techniques

- separatist approach

  Handles objective function and constraints separately.

  1. Feasible solutions are better than infeasible solutions
  2. Feasible solutions are ranked based on objective function values
  3. Infeasible solutions are ranked based on constraint violation values

constraint handling techniques

- Repair approach

  converts infeasible solutions to feasible ones

  can be used only when repairing is easy

  Example) Thickness constraint in aerodynamic airfoil shape optimization problem. Here, thickness can be evaluated without CFD

constraint handling techniques

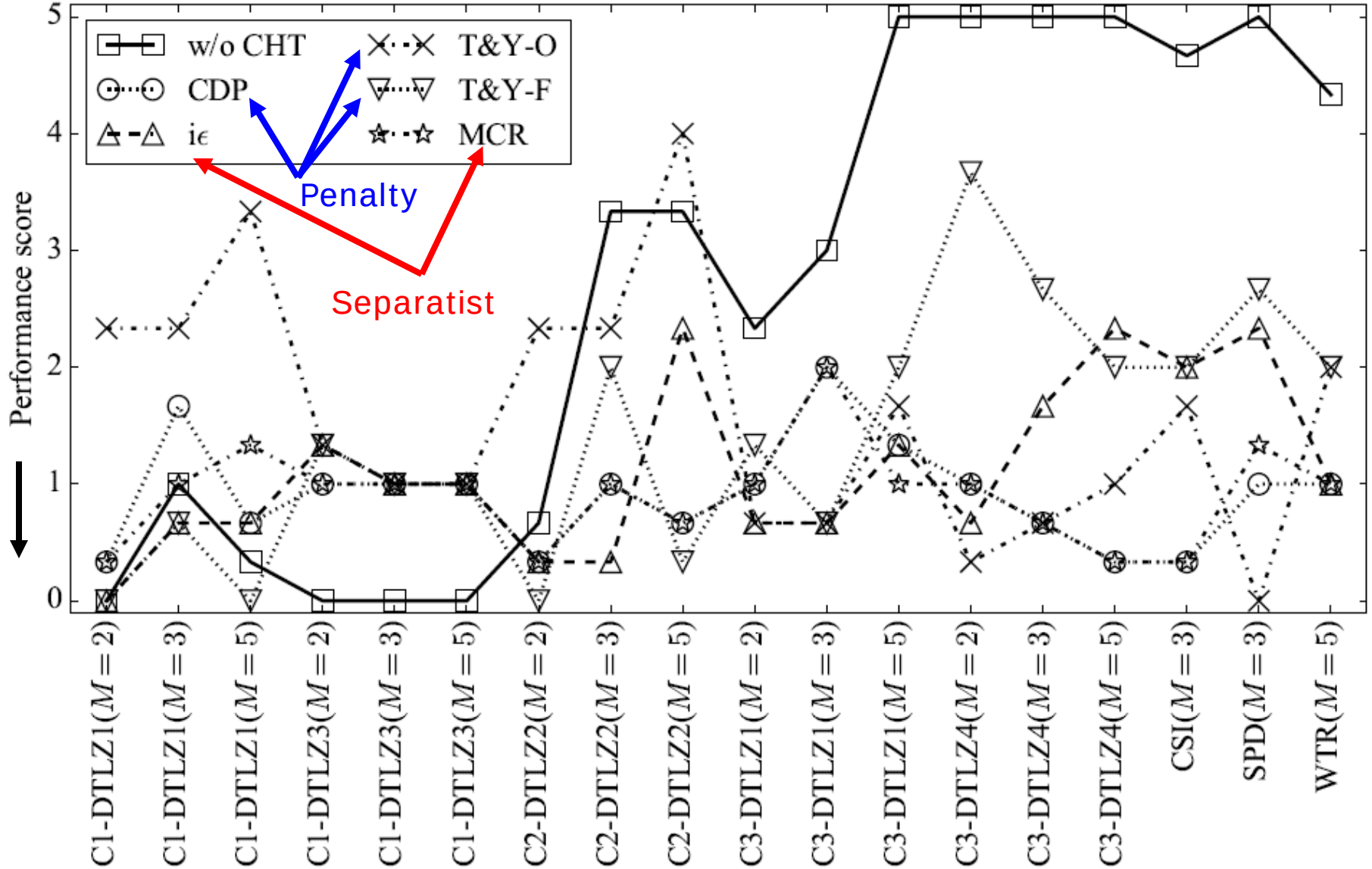
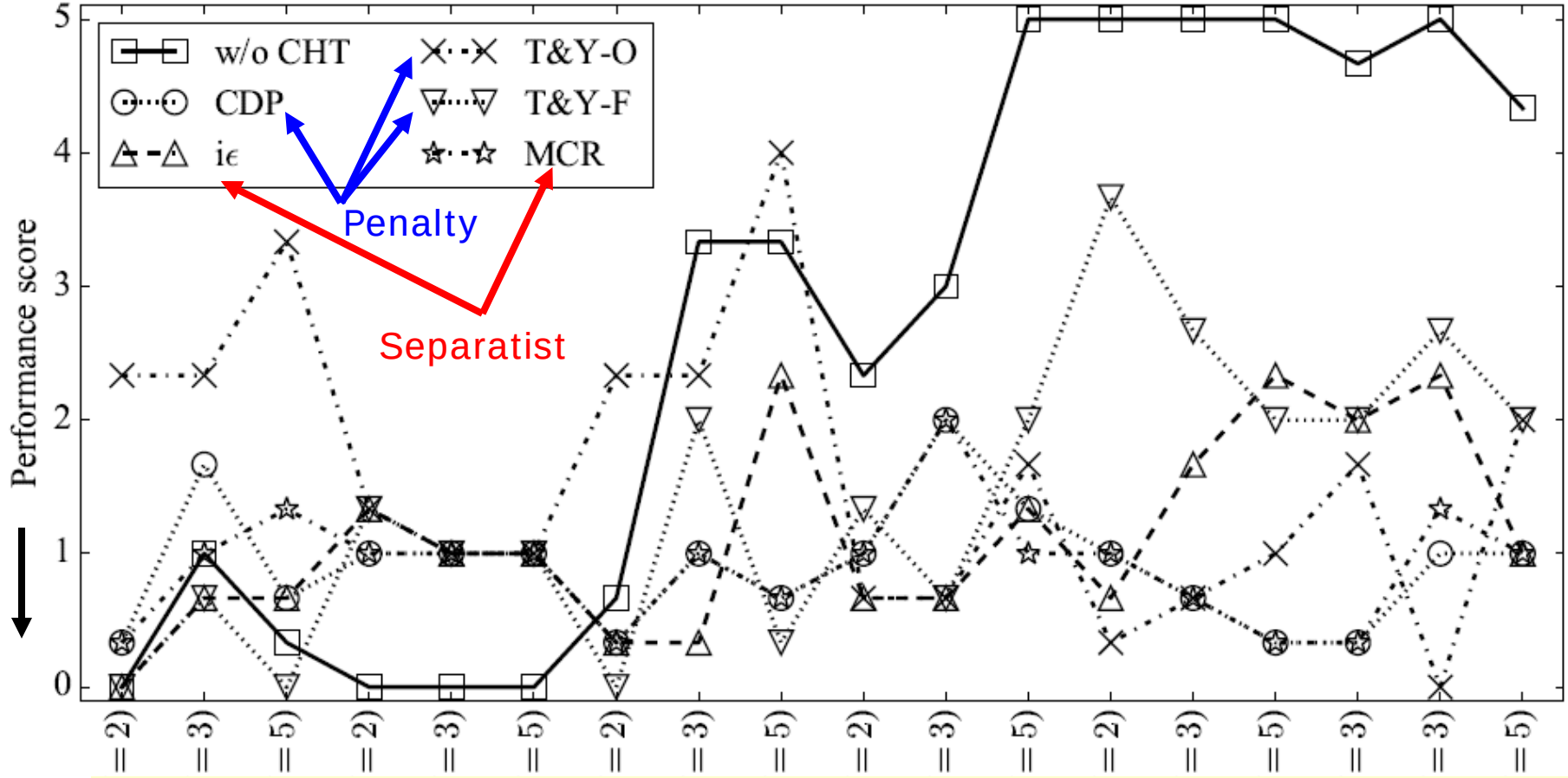


Fig. 3: Average performance score for all benchmark problems (lower is better)

constraint handling techniques

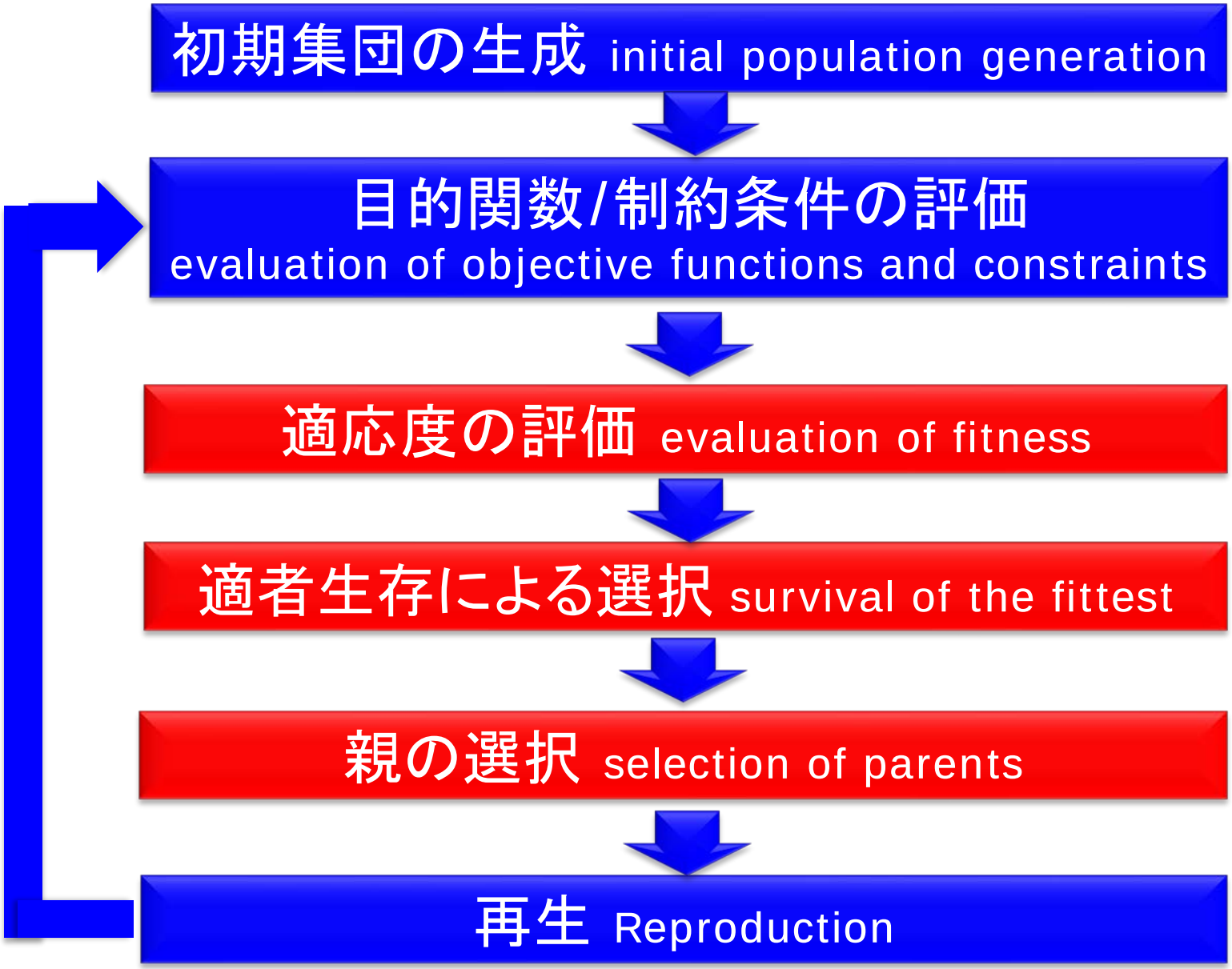


- Performance of CHT strongly depends on problem
- MCR performs better in average

Fig. 3: Average performance score for all benchmark problems (lower is better)

まとめ

Summary


初期集団の生成 initial population generation
目的関数/制約条件の評価
evaluation of objective functions and constraints
適応度の評価 evaluation of fitness
適者生存による選択 survival of the fittest
親の選択 selection of parents
再生 Reproduction

多目的進化アルゴリズムの利点・欠点

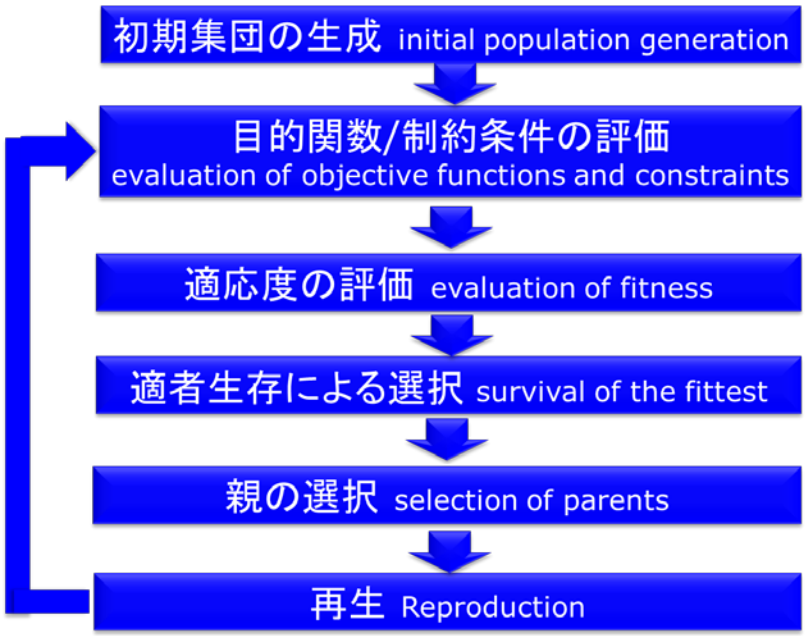
advantages/disadvantages of MOEA

- 目的関数分布が多峰的である場合でも，大域的探査が可能
  Global search even if the objective function is multimodal
- 並列化効率が高い
  Parallelization efficiency is very high
- 不連続最適化問題も解くことが可能
  Applicable to design problems with discontinuous objective function
- 適用が簡単（ブラックボックスを準備すればOK）
  Application is easy
- **多目的最適化問題の多数のパレート最適解を一度に得ることが可能**
  Find the Pareto-optimal solutions at a time
- 探査効率では勾配法などより劣る（たくさんの評価が必要）
  Less efficient compared with gradient-based method

最適化手法の比較 comparison of optimization methods

	勾配法 GM	焼き鈍し法 SA	単目的進化 アルゴリズム SOEA	多目的進化 アルゴリズム MOEA
単目的 最適化 問題	評価回数少ない 探査局所的	中間 探査中間	評価回数多い 大域的探査	N/A
多目的 最適化 問題	評価回数少ない 探査局所的 非劣解を 効率的に 得られない	中間 中間 非劣解を 効率的に 得られない	評価回数多い 大域的探査 非劣解を 効率的に 得られない	評価回数多い 大域的探査 非劣解を 効率的に 得られる

How to define population size and number of generations?



- No way to define best combination of population size N_P and number of generation N_G
- Total number of evaluations ($N_T = N_P \times N_G$) is usually restricted
- When N_G is larger, MOEA becomes more efficient but less robust
- From my (personal) experience, it is reasonable to set N_P and N_G equal value

参考文献 Reference on EC

最適デザインの概念 単行本 – 2008/10/7
松岡 由幸 (著), 宮田 悟志 (著)

多目的最適化と工学設計—しなやかシステム工学アプローチ 単行本 – 2008/1/1
中山 弘隆 (著), 岡部 達哉 (著), 荒川 雅生 (著), 尹 禮分 (著)
（販売終了）